

**INTEGRATING ERC-20 ALLOWANCE EDGES INTO
PAGERANK FOR ENHANCED ON-CHAIN WALLET
REPUTATION SCORING**

by

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Abstract

DeFi protocols require on-chain reputation scores that are computationally scalable, semantically interpretable, and aligned with meaningful financial outcomes ^{[32],[29]}. This dissertation introduces **EndorseRank**, an on-chain reputation score produced by applying PageRank to a directed endorsement graph built from latest ERC-20 allowance edges on Arbitrum One, and evaluates it against Adaptive Weighted PageRank (AWP), the transfer-graph baseline of Do et al.^[10], using five proxy families on a matched alignment cohort ($n = 5,521$).

EndorseRank runs $\approx 8.9\times$ faster than AWP on identical cohort inputs (2.1 s vs. 18.7 s PageRank wall time) and aligns more strongly with allowance proxies (mean Kendall $\tau = 0.355$ vs. -0.030 for AWP); AWP aligns more strongly with transfer proxies ($\tau = 0.534$ vs. 0.333 for EndorseRank). Inter-method rank correlation between EndorseRank and AWP ($\tau = 0.316$) indicates partially overlapping yet construct-distinct social reputation structures.

On an expanded wallet pool ($N = 31,612$), EndorseRank remains $\approx 10.4\times$ faster than AWP at $n = 10,000$ and $\approx 9.4\times$ at full pool size. Robustness checks (PageRank damping, top-20 token subgraph, sample-size sensitivity) leave EndorseRank allowance alignment stable at $\tau \approx 0.355$ without altering the allowance–transfer trade-off.

During empirical work, GMX V2 trading-success proxies were incorporated to extend domain validation on the same cohort, and **GF-PR** was added as a supplementary outcome-native ceiling reference built from the same GMX profit-and-loss stream. Among social methods, transfer proxies yield higher mean alignment than trading-success proxies on this chain; GF-PR achieves the highest

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

trading-success mean τ (0.583) by construction, illustrating construct overlap when ranking directly on outcome flow rather than social edges. GF-PR does not constitute a deployable social reputation score.

Keywords— DeFi, EndorseRank, Adaptive Weighted PageRank, On-Chain Reputation, GMX V2, Arbitrum One, Domain Alignment, Spearman’s rho, Kendall’s tau, GF-PR (outcome-native reference)

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Abbreviations and Notation

This section lists abbreviations and mathematical notation used throughout the dissertation. Symbols are defined again at first use in the main chapters when helpful for local reading.

Abbreviations

Table 1: Abbreviations used in this dissertation.

Abbreviation	Meaning
AMM	Automated Market Maker
AWP	Adaptive Weighted PageRank
BQ	BigQuery
CCS	ACM Conference on Computer and Communications Security
DeFi	Decentralized Finance
DEX	Decentralized Exchange
EIP	Ethereum Improvement Proposal
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
FC	Financial Cryptography and Data Security
GF-PR	GainFlow PageRank (outcome-native reference)
GMX	GMX perpetual swap protocol (V2 on Arbitrum One)
MEV	Maximal (Miner/Maximal) Extractable Value
PnL	Profit and Loss
PR	PageRank
RQ	Research Question
SoK	Systematization of Knowledge
SRQ	Supplementary Research Question
USENIX	USENIX Security Symposium
WWW	World Wide Web Conference

Mathematical notation

Table 2: Mathematical notation used in this dissertation.

Symbol	Meaning
G_{approve}	Directed endorsement graph from latest ERC-20 allowances (owner $\rightarrow$ spender)
G_{transfer}	Directed transfer graph with time-decayed edge weights (AWP input)
n	Alignment-cohort size (primary comparison: $n = 5,521$)
N	Expanded wallet-pool size (scaling: $N = 31,612$)
d	PageRank damping (teleportation) factor
k	AWP logistic time-decay steepness
t_0	AWP logistic time-decay midpoint (days)
Δt	Edge age in days for AWP decay
$\sigma(\Delta t)$	Logistic decay weight applied to transfer edges
ρ	Spearman’s rank correlation coefficient
τ	Kendall’s rank correlation coefficient
$ E $	Number of directed edges in a reputation graph
$\mathcal{W}$	Set of wallets in the evaluation cohort
$s(w)$	Reputation score of wallet w under a given method
$p(w)$	Validation-proxy value for wallet w

1 Introduction and Research Problem

Decentralized Finance (DeFi) has transformed how capital is allocated on public blockchains, enabling lending, borrowing, and leveraged trading without traditional financial intermediaries^{[32],[18]}. Yet most credit-like interactions remain highly collateralized because protocols cannot rely on off-chain identity systems or conventional credit histories^{[29],[45]}. In a pseudonymous environment, the operational challenge is to allocate capital safely: distinguishing wallets that manage leveraged exposure reliably from those that exhibit repeated losses, liquidations, or adversarial behavior.

This dissertation addresses a concrete research problem within that setting: whether an on-chain reputation score built from ERC-20 approve/permit allowances—implemented as **EndorseRank**—can reduce computational cost, represent endorsement semantics more faithfully, and yield a distinct yet credible alignment profile compared with transaction-based Adaptive Weighted PageRank (AWP) on **Arbitrum One**. The primary empirical comparison is **EndorseRank versus AWP** on an identical wallet sample, validated with five proxy families and GMX V2 perpetual swap outcomes following Do et al.^[10] rank-correlation framework.

In this study, **on-chain reputation score** computation refers to deriving a quantitative wallet score from observable on-chain behavior and relationships without off-chain identity^[10]. EndorseRank produces such a score from allowance-based endorsement edges. The remainder of this chapter situates the problem historically and economically, motivates allowance-based reputation, states the research problem and questions, defines key constructs, and outlines contributions, scope, and

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

the dissertation roadmap.

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains introduced programmable settlement and open participation without a central operator^{[27],[36]}. Smart contracts encode agreement terms as self-executing code, enabling financial applications that settle on-chain rather than through custodial intermediaries^{[6],[34]}. DeFi protocols compose these contracts into markets for lending, automated market making, and perpetual futures^{[32],[35]}. On Arbitrum One and similar Layer-2 networks, activity density is high enough that wallet-level behavioral signals—transfers, approvals, and protocol events—can be observed at scale.

Despite this transparency, DeFi credit markets remain structurally over-collateralized. Protocols typically require borrowers or traders to post collateral exceeding the value of their exposure because counterparties are pseudonymous addresses rather than identified legal persons^{[45],[18]}. Pseudonymity is a design feature of public ledgers: anyone may create addresses at negligible cost, and a single economic actor may control many wallets^[11]. Traditional credit scoring depends on identity-linked histories that do not transfer cleanly to this setting^[29]. Consequently, risk management defaults to collateral buffers, liquidation engines, and protocol-level parameters rather than individualized reputation.

Collateralization protects protocols but imposes capital inefficiency. Traders and borrowers must lock assets that could otherwise be deployed elsewhere; protocols forgo under-collateralized prod-

^[27]Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.

^[36]Wood, G. (2014). Ethereum: A secure decentralised generalised transaction ledger. [Ethereum Project]

^[6]Buterin, V. (2014). A next-generation smart contract and decentralized application platform. [Ethereum Foundation]

^[34]Szabo, N. (1997). Formalizing and securing relationships on public networks. [First Monday]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

ucts that traditional finance offers to borrowers with established credit. Researchers have therefore explored on-chain signals as substitutes for off-chain identity: transfer-graph centrality^[10], network risk propagation^[26], prospect-theory and blockchain-based credit models^[19], and machine-learning pipelines on lending protocols^[24]. These approaches share a premise: public ledger data encode economically meaningful relationships that can inform risk assessment without deanonymizing users.

Graph ranking methods are a natural fit for this premise. PageRank and its variants propagate influence through directed edges, ranking nodes by recursive endorsement from other nodes^{[30],[5],[25]}. Early web-trust systems such as EigenTrust and TrustRank applied similar ideas to peer-to-peer and web spam settings^{[21],[17]}. Do et al.^[10] adapt PageRank to Ethereum transfer graphs with value-weighted, time-decayed edges—Adaptive Weighted PageRank (AWP)—and validate scores against inbound transfer volume as a domain-intuition proxy for economic importance. That line of work establishes transfer-graph reputation as a credible baseline; it does not, however, model explicit spending authorization, which is a distinct on-chain act recorded by ERC-20 allowances^{[12],[13]}.

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- ^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]
- ^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]
- ^[19]Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. [IEEE Transactions on Network Science and Engineering]
- ^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]
- ^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]
- ^[5]Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. [Computer Networks and ISDN Systems]
- ^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]
- ^[21]Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. [Proceedings of the 12th International Conference on World Wide Web]
- ^[17]Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. [Proceedings of the Thirtieth International Conference on Very Large Data Bases]
- ^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]
- ^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]
- ^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

1.2 Motivation for on-chain reputation

Three practical pressures motivate on-chain reputation research in leveraged DeFi. First, **capital efficiency**: if protocols can identify wallets with historically reliable risk management, they may safely reduce collateral requirements or prioritize liquidations, improving utilization without abandoning on-chain settlement ^{[32],[45]}. Second, **auditability**: unlike opaque off-chain credit scores, graph-based scores derived from public events can be recomputed and inspected, addressing concerns about black-box decentralized credit scoring ^[29]. Third, **operational cost**: reputation systems that must refresh frequently—for example, before each leverage adjustment—cannot afford expensive historical transfer aggregation on every update.

Transfer-based AWP addresses the first two pressures but is costly on the third. Constructing G_{transfer} requires aggregating transfer history, applying logistic time-decay, and running PageRank on a dense edge set ^[10]. On the matched cohort studied here ($n = 5,521$ wallets),³² the AWP subgraph contains 137,087 edges and requires 18.711 s of PageRank wall-clock time. Allowance graphs offer a structurally different alternative. An ERC-20 approve or EIP-2612 permit records how much a token owner authorizes a spender to transfer ^{[12],[13]}. The latest allowance per owner–spender pair is a snapshot of current authorization intent: newer approvals override older ones, so temporal decay is unnecessary. The resulting endorsement graph G_{approve} is sparse—14,727 edges on the same cohort—and PageRank completes in 2.105 s, approximately $8.9\times$ faster than AWP.

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^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

³²The matched cohort is the intersection of wallets with at least three GMX V2 `PositionDecrease` closes in the observation window and non-zero connectivity in the reputation subgraphs. Runtime scaling uses a larger expanded wallet pool ($N = 31,612$).

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

Beyond efficiency, allowances carry distinct semantics. A transfer may reflect routing, market making, or passive receipt; an allowance is an explicit delegation of spending authority. Treating allowances as endorsement edges aligns PageRank’s citation metaphor ^[30] with a trust-like act that protocols already rely on for token interactions. Whether that semantic difference produces rankings that are merely different, or differently aligned with economically relevant outcomes, is an empirical question. This dissertation answers that question on Arbitrum One using GMX V2 perpetual trading outcomes as a dense validation frame ^[15], because under-collateralized lending default labels remain sparse on this chain.

1.3 Research problem

Existing Adaptive Weighted PageRank (AWP) methods infer trust from transfer volume and logistic time-decay over transaction history ^[10]. While economically interpretable, they (i) require expensive historical transfer aggregation, (ii) treat inbound transfer volume as a proxy for a “valuable wallet,” and (iii) do not model explicit spending authorization. EndorseRank instead uses the latest allowance per owner–spender pair as edge strength on a directed endorsement graph, yielding a sparser graph and a semantically distinct trust signal without temporal decay weighting—integrating ERC-20 allowance edges into PageRank for enhanced on-chain wallet reputation scoring.

The research problem is therefore comparative and evaluative: on an identical Arbitrum One wallet sample with GMX V2 perpetual activity, does EndorseRank produce wallet orderings that are construct-distinct from AWP, align credibly with domain-intuition proxies (including allowance-specific and trading-success families), and do so at substantially lower computational cost? The primary comparison is EndorseRank versus AWP; a supplementary outcome-native reference (GF-

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[15]GMX. (n.d.). GMX V2 documentation.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

PR) is introduced later to bound interpretation of trading-outcome proxies, not to replace either social method.

Table 1.1 summarizes the primary comparison. Table 1.3 summarizes hypotheses (H1–H3), research questions (RQ1–RQ4), supplementary question SRQ1, and Chapter 4 empirical claims (C1–C4, SC1).

Table 1.1: Primary comparison: EndorseRank versus AWP (matched wallet cohort, $n = 5,521$).

Method	Input graph	Role
EndorseRank	Latest ERC-20 allowances (G_{approve})	Primary construct: allowance delegation; target of this dissertation
AWP	Time-decayed transfers (G_{transfer})	Primary baseline: Do et al. transfer-graph reproduction

To avoid ambiguity, the key concepts used throughout this dissertation are defined as follows:

- **DeFi (Decentralized Finance):** Financial services—including lending, borrowing, and perpetual trading—implemented via smart contracts on public blockchains, typically without a central operator controlling custody or settlement ^{[32],[18]}. DeFi protocols compose open interfaces so that positions, liquidations, and token movements are observable on-chain.
- **Smart contract:** Self-executing program code deployed on a blockchain that enforces agreement terms without a trusted intermediary ^{[6],[34]}. In this dissertation, smart contracts are the source of ERC-20 allowance logs, transfer logs, and GMX V2 position events.
- **Wallet / Address:** A blockchain account that holds assets and interacts with smart contracts; the unit of analysis for reputation scoring ^{[6],[12]}. A wallet is identified by its address; no off-chain identity is assumed or recovered.
- **On-chain reputation score:** A numerical indicator derived from on-chain signals that rep-

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

^[6]Buterin, V. (2014). A next-generation smart contract and decentralized application platform. [Ethereum Foundation]

^[34]Szabo, N. (1997). Formalizing and securing relationships on public networks. [First Monday]

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

resents a wallet’s relative standing for financial interactions under pseudonymity ^[10]. Scores are ordinal for ranking purposes; absolute magnitudes are not interpreted as credit limits.

- **On-chain credit risk:** Observable adverse outcomes such as position liquidation or repeated trading losses on public blockchains ^[26]. This dissertation uses GMX V2 perpetual outcomes as domain-intuition proxies for risk-relevant behavior, not as loan-repayment ground truth.
- **GMX V2 perpetual swap position:** A collateralized long or short exposure on GMX V2, opened via **PositionIncrease** and closed via **PositionDecrease** events on Arbitrum One ^{[15],[32]}. Realized profit-and-loss from closes supplies trading-success and inverse-risk proxies.
- **PageRank:** A graph ranking method that propagates influence through directed edges under a damping factor ^{[30],[5],[25]}. **EndorseRank** applies PageRank to a latest-allowance endorsement graph: edge strength equals the current allowance per owner–spender pair, with no temporal decay. **Adaptive Weighted PageRank (AWP)** applies PageRank to transfer graphs with edge weights given by transfer value and logistic time-decay ^[10].
- **ERC-20 allowance / Approve / Permit:** Mechanisms recording how much a token owner permits a spender to transfer on the owner’s behalf ^{[12],[13]}. Approve is a standard on-chain call; EIP-2612 permit authorizes the same state change via signature. EndorseRank uses the

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[15]GMX. (n.d.). GMX V2 documentation.

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[5]Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. [Computer Networks and ISDN Systems]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

latest non-zero allowance snapshot per owner–spender pair.

- **Endorsement graph:** A directed graph in which edges represent explicit delegation of spending authority rather than mere asset transfers ^[10]. In EndorseRank, an edge $u \rightarrow v$ means wallet u currently authorizes wallet or contract v to spend tokens owned by u .
- **Domain alignment:** The degree to which a reputation ranking agrees with an observable proxy ranking, assessed by Spearman’s ρ and Kendall’s τ ^{[33],[22],[8],[10]}. Alignment is construct-specific: a method may align strongly with its primary graph construct and weakly with an orthogonal outcome family.

1.4 Supplementary extension (GF-PR)

During empirical analysis, **GF-PR** (GainFlow PageRank) was added as a supplementary outcome-native reference (Section 2.9). GF-PR builds a directed star graph from the same GMX profit-and-loss stream used to define validation proxies. It ranks wallets on synthetic POOL/SINK flow rather than social wallet-to-wallet edges, setting an upper bound on trading-outcome proxy alignment for diagnostic interpretation—not a deployable social reputation score.

Table 1.2: Supplementary extension added during empirical work (not part of the primary EndorseRank–AWP research question).

Method	Input graph	Role
GF-PR	GMX PnL (POOL→wallet, wallet→SINK)	star Supplementary ceiling reference; construct overlap on trading-outcome proxies expected

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

Table 1.3: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1 / RQ1	Do EndorseRank and AWP yield distinct wallet orderings?	§4.6; inter-method $\tau = 0.316$
H2 / RQ2	Which social method aligns more strongly with which proxy families?	Tables 4.7–4.12; Claims C2–C3
H3 / RQ3	Is EndorseRank faster than AWP at equal n ?	Tables 4.2–4.3; Claims C1, C4
RQ4	Transfer vs. trading-success validation frame for social methods	Table 4.7
SRQ1	What does GF-PR ceiling imply for trading-outcome proxy interpretation?	§4.7; Claim SC1
C1	EndorseRank vs. AWP runtime and graph size	Table 4.2
C2	Credible EndorseRank–AWP alignment on selected proxies	Tables 4.8–4.12
C3	Construct-specific winners (EndorseRank vs. AWP)	Table 4.7
C4	Efficiency–alignment trade-off for EndorseRank	Tables 4.2–4.3
SC1	Supplementary GF-PR construct-overlap diagnostic	§4.7

1.5 Research questions

This study tests three primary hypotheses: **H1**—EndorseRank and AWP yield distinct wallet orderings on the same cohort; **H2**—the two social methods differ in construct-specific proxy alignment; and **H3**—EndorseRank is computationally more efficient than AWP at equal n . The following research questions guide the empirical evaluation summarized in Table 1.3. RQ1–RQ4 form the primary EndorseRank–AWP comparison; SRQ1 addresses the supplementary GF-PR extension.

- RQ1.** Do EndorseRank and AWP produce meaningfully different wallet orderings on an identical Arbitrum wallet sample?
- RQ2.** Between EndorseRank and AWP, which social method aligns more strongly with each of the five proxy families (Spearman’s ρ , Kendall’s τ)?
- RQ3.** Is EndorseRank computationally more efficient than AWP on the same matched sample (runtime, memory, iterations)?
- RQ4.** For social reputation methods, which validation frame yields stronger rank alignment on this chain: Do et al. transfer proxies (in-degree/in-value) or trading-success proxies (from GMX V2 perpetual closes)?
- SRQ1.** What does the supplementary GF-PR outcome-native ceiling imply for interpreting trading-outcome proxy alignment of EndorseRank and AWP?

RQ1 tests whether allowance and transfer graphs induce distinct centrality structures, operationalized by inter-method Kendall τ between score vectors. RQ2 evaluates construct-specific alignment across transfer, allowance, inverse-risk, Sybil-adjusted stability, and trading-success proxy families. RQ3 compares PageRank wall-clock time, memory, and iterations on identical wallet sets, including scaling on an expanded wallet pool. RQ4 asks which validation frame—transfer centrality or trading success—is more informative for social reputation methods on this sample. SRQ1 uses GF-PR only to bound how high trading-outcome alignment can rise when ranking directly on

outcome flow, clarifying construct overlap ^[8].

1.6 Contributions

This dissertation makes four contributions to on-chain reputation research and DeFi risk assessment.

First, it formulates **EndorseRank**, an allowance-based endorsement graph model that applies PageRank to latest ERC-20 approve/permit edges. The model treats current spending authorization as endorsement strength, eliminates temporal decay by relying on allowance override semantics, and produces a sparse graph suitable for frequent refresh.

Second, it conducts a **five-proxy domain alignment evaluation** comparing EndorseRank and AWP on leveraged DeFi activity. Proxies span transfer centrality ^[10], allowance delegation, GMX V2 inverse risk and trading success ^[15], and Sybil-adjusted stability. Alignment is reported with Spearman’s ρ and Kendall’s τ ^{[33],[22]}, enabling construct-specific interpretation rather than a single global winner.

Third, it releases an **open reproducible evaluation pipeline** that links BigQuery extraction, graph construction, PageRank scoring, proxy computation, and LaTeX result tables to a machine-readable summary (Appendix 8.1). The pipeline supports independent verification of claims C1–C4 and SC1.

Fourth, as a supplementary contribution added during empirical work, it provides a **GF-PR ceiling analysis** that clarifies construct overlap on trading-outcome proxies for the matched wallet cohort ($n = 5,521$). GF-PR is not proposed as a deployable social reputation method; it bounds interpretation of high τ on outcome-native families (Claim SC1).

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[15]GMX. (n.d.). GMX V2 documentation.

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

Empirically, the primary comparison yields inter-method $\tau = 0.316$, EndorseRank allowance $\tau = 0.355$, AWP transfer $\tau = 0.534$, and an efficiency advantage of 2.105 s versus 18.711 s ($\approx 8.9\times$) on the matched cohort, with pool scaling of $\approx 10.4\times$ at $n = 10,000$ and $\approx 9.4\times$ at $N = 31,612$. These results support an efficiency–alignment trade-off rather than universal dominance of either social method.

1.7 Scope and limitations

This dissertation focuses on Arbitrum One and GMX V2 perpetual trading over the observation window 2025-12-01 to 2026-05-31. The primary empirical sample is a matched wallet cohort of $n = 5,521$ addresses with at least three GMX PositionDecrease events and non-zero reputation subgraph connectivity, comprising 365,488 decoded closes, 14,727 EndorseRank edges, and 137,087 AWP edges. Runtime scaling is reported on an expanded wallet pool ($N = 31,612$). Robustness checks examine damping sensitivity, top-token subgraphs, and sample-size sensitivity. Unsecured lending default labels are out of scope: under-collateralized credit events are too sparse on this chain for reliable ground-truth validation. Constructs and ethics are defined in Chapters 1 and 3. Among social methods, transfer proxies align more strongly than trading-success proxies on this sample (RQ4); this does not diminish EndorseRank’s allowance-specific validity or runtime advantage. GF-PR results are reported as supplementary diagnostics (SRQ1), not as a third peer reputation method. Extended baselines archived for future comparison are excluded from the main narrative (Appendix 8.1).

Generalization beyond Arbitrum One, GMX V2, and the six-month window is not claimed. Token-decimal normalization, price oracles, and permit-versus-approve coverage may affect edge weights. Adversarial manipulation of allowances or transfers remains a threat discussed in Chapter 5; Sybil-adjusted proxies partially address but do not eliminate that risk ^[11].

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

1.8 Dissertation roadmap

The dissertation proceeds as follows.

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics, and GMX V2 trading-success validation. It develops the theoretical framework—hyperlink propagation, domain alignment, and computational complexity—that motivates the primary EndorseRank versus AWP comparison, and situates GF-PR as a supplementary outcome-native ceiling.

Chapter 3 specifies data sources, sampling, graph construction for EndorseRank and AWP, the five-proxy evaluation design, computational benchmarks, robustness checks, reproducibility practices, and ethics. It defines the matched wallet cohort, the expanded wallet pool used for scaling, and the operational procedures that produce the Chapter 4 tables.

Chapter 4 reports empirical results: cohort characteristics, runtime and scaling benchmarks (Claim C1), robustness checks, social-method alignment across five proxy families (Claims C2–C3), rank divergence (H1/RQ1), and the supplementary GF-PR ceiling analysis (SRQ1/SC1). Primary claims C1–C4 and supplementary claim SC1 are stated explicitly.

Chapter 5 interprets findings by research question, develops theoretical and practical implications, formulates a Sybil attack cost model on allowance graphs, discusses regulatory and ethical considerations, and evaluates threats to validity and limitations.

Chapter 6 synthesizes the contribution to knowledge, summarizes findings against RQ1–RQ4 and SRQ1, and outlines future work on cross-protocol validation, adversarial robustness, hybrid methods, and live oracle integration.

2 Literature Review and Theoretical Framework

This chapter reviews the conceptual and technical foundations required to interpret EndorseRank and Adaptive Weighted PageRank (AWP). It begins with blockchain and DeFi primitives for readers who may not specialize in distributed ledgers, then develops graph ranking theory, reputation-system lineage, ERC-20 allowance semantics, GMX V2 trading-success validation, and rank-correlation statistics. The chapter closes with the theoretical framework—hyperlink propagation, domain alignment, and computational complexity—that motivates the primary EndorseRank versus AWP comparison in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A blockchain is a replicated append-only ledger in which batches of transactions (blocks) are linked by cryptographic hashes ^[27]. Participants agree on a canonical history through a consensus mechanism; once confirmed, entries are costly to rewrite. This property enables *trust-minimized* settlement: counterparties need not share a legal identity if they can verify the ledger state.

Bitcoin demonstrated peer-to-peer electronic cash ^[27]. Ethereum generalized the model to a world computer: accounts hold balances and code, and transactions invoke functions that update state according to the Ethereum Virtual Machine (EVM) ^{[6],[36],[2]}. Smart contracts are programs stored on-chain that execute deterministically when called ^[34]. Because execution is public and reproducible, financial logic—collateral checks, liquidations, fee accrual—can be encoded without a

^[27]Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.

^[6]Buterin, V. (2014). A next-generation smart contract and decentralized application platform. [Ethereum Foundation]

^[36]Wood, G. (2014). Ethereum: A secure decentralised generalised transaction ledger. [Ethereum Project]

^[2]Antonopoulos, A. M., & Wood, G. (2018). Mastering Ethereum: Building smart contracts and DApps. [O'Reilly Media]

^[34]Szabo, N. (1997). Formalizing and securing relationships on public networks. [First Monday]

central operator^[32].

Decentralized Finance (DeFi) refers to the ecosystem of such contracts offering exchange, lending, derivatives, and asset management^{[18],[35]}. Composability—the ability of one protocol to call another—creates powerful products but also systemic risk pathways, including flash-loan attacks and oracle manipulation^{[31],[9]}. For reputation research, the relevant implication is that *wallet-level histories* on public chains are rich, machine-readable, and incomplete: they reveal actions, not legal identities^[29].

2.1.1 Pseudonymity and the credit problem

Traditional credit scoring links legal persons to repayment histories. On public blockchains, the unit of analysis is an address (wallet). An address may represent an individual, a fund, a bot, or a smart contract. Multiple addresses may be controlled by one entity^[11]. Consequently, DeFi lending markets typically require over-collateralization: borrowers lock assets worth more than the loan, and liquidators seize collateral when health factors fall^{[3],[32]}. Over-collateralization is robust but capital-inefficient. On-chain reputation scores aim to summarize behavioral reliability so that protocols—or researchers studying them—can reason about risk without off-chain KYC

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[31]Qin, K., Zhou, L., Livshits, B., & Gervais, A. (2021). Attacking the DeFi ecosystem with flash loans for fun and profit. [Financial Cryptography and Data Security]

^[9]Daian, P., Goldfeder, S., Kell, T., Li, Y., Zhao, X., Bentov, I., Breidenbach, L., & Juels, A. (2020). Flash Boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. [2020 IEEE Symposium on Security and Privacy]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[3]Bartoletti, M., Chiang, J. H., & Lluch-Lafuente, A. (2021). SoK: Lending pools in decentralized finance. [Financial Cryptography and Data Security. FC 2021 International Workshops]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

[19],[24],[26].

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Ethereum mainnet provides security and liquidity but faces throughput and fee constraints under demand ^{[4],[14]}. Layer-2 (L2) protocols execute transactions off the base layer (or in a compressed form) and periodically post commitments to Ethereum, aiming to inherit security while reducing cost ^[45]. Optimistic rollups such as Arbitrum assume correctness unless challenged during a dispute window ^[20].

Arbitrum One is an EVM-compatible optimistic rollup widely used for DeFi activity, including GMX V2 perpetual markets ^[15]. For this dissertation, Arbitrum One is the empirical setting because (i) ERC-20 Approval and Transfer logs are available in public BigQuery datasets, (ii) GMX V2 emits wallet-attributed PositionDecrease events with realized PnL, and (iii) fee levels permit dense retail and professional trading activity within a six-month window. Results should not be assumed to transfer automatically to other L2s or to Ethereum mainnet without re-validation, but the methodological template—allowance graph versus transfer graph, proxy families, rank correlation—is chain-agnostic wherever equivalent logs exist.

^[19]Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. [IEEE Transactions on Network Science and Engineering]

^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[4]Bonneau, J., Miller, A., Clark, J., Narayanan, A., Kroll, J. A., & Felten, E. W. (2015). SoK: Research perspectives and challenges for Bitcoin and cryptocurrencies. [2015 IEEE Symposium on Security and Privacy]

^[14]Gervais, A., Karame, G. O., Wüst, K., Glykantzis, V., Ritzdorf, H., & Capkun, S. (2016). On the security and performance of proof of work blockchains. [Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[20]Kalodner, H., Goldfeder, S., Chen, X., Weinberg, S. M., & Felten, E. W. (2018). Arbitrum: Scalable, private smart contracts. [27th USENIX Security Symposium]

^[15]GMX. (n.d.). GMX V2 documentation.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens: `transfer`, `approve`, `transferFrom`, `allowance`, and related events ^[12]. When an owner calls `approve(spender, amount)`, the token contract records that `spender` may transfer up to `amount` from the owner. The spender later calls `transferFrom` to move tokens. EIP-2612 permit allows allowance updates via off-chain signatures settled in a single transaction ^[13], improving user experience without changing the fundamental authorization semantics.

Figure 2.1 illustrates the lifecycle. Critically, an allowance is an *explicit authorization*, not a completed transfer. Owners may set allowances to routers, vaults, or trading contracts they trust to act within limits. Revocation (approving zero) and replacement (a new non-zero allowance) update the live authorization state. EndorseRank therefore uses the *latest* allowance per owner–spender pair as edge strength: older approvals are superseded rather than decayed continuously.

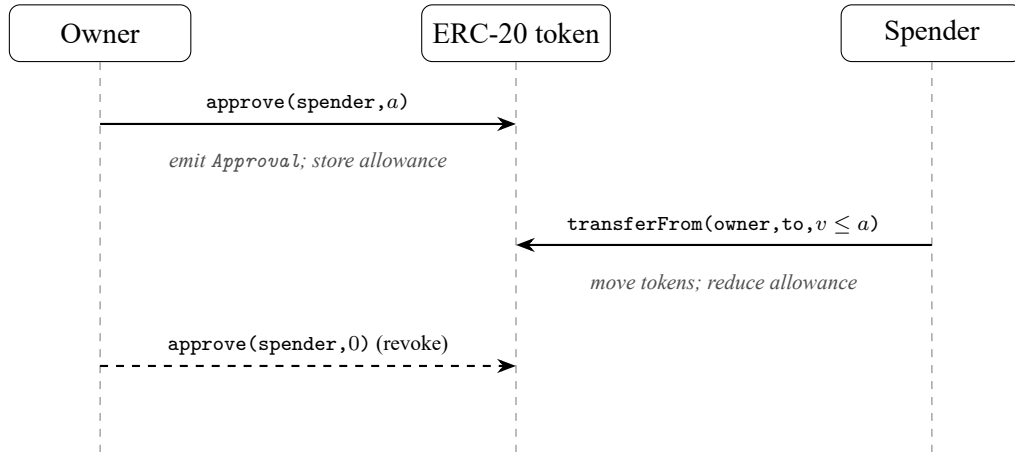


Figure 2.1: ERC-20 allowance lifecycle: approve (or permit), optional `transferFrom`, and revoke. EndorseRank edges reflect the latest non-historical allowance snapshot per owner–spender pair.

From a measurement perspective, allowances are sparser than transfers: many transfers occur under a single approval, and many wallets never approve third-party spenders. This sparsity is both a

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

computational advantage and a semantic claim—EndorseRank measures authorization networks, not payment networks.

2.4 Graph-based on-chain reputation

PageRank and variants rank nodes by propagated influence through directed edges ^{[30],[5]}. Do et al.^[10] apply Adaptive Weighted PageRank (AWP) to Ethereum transaction graphs, weighting edges by transfer value and logistic time-decay. They validate scores against **in-degree** and **in-value**—count and sum of inbound transfers—as domain-intuition proxies for economic importance. Lin et al.^[26] show that transaction connectivity alone can miss risk-relevant structure unless risk signals propagate through the network, motivating refined graph semantics beyond vanilla PageRank. Broader DeFi credit-scoring literature includes prospect-theory and blockchain-based models ^[19] and machine-learning pipelines on lending protocols ^[24]; Packin and Lev-Aretz^[29] emphasize auditability constraints under pseudonymity. EndorseRank stays within interpretable graph propagation rather than black-box predictive scoring.

Graph-based approaches are attractive because they (i) use only public data, (ii) produce continuous scores suitable for ranking, and (iii) admit complexity analysis in terms of edge counts. They are limited by Sybil attacks ^[11], wash trading, and the gap between graph centrality and economic default risk. This dissertation does not claim that PageRank scores are credit scores in the regulatory

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[5]Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. [Computer Networks and ISDN Systems]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[19]Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. [IEEE Transactions on Network Science and Engineering]

^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

sense; it claims that allowance-based and transfer-based rankings can be compared rigorously on shared proxies.

2.5 PageRank formalism

Let $G = (V, E)$ be a directed graph with $|V| = n$ nodes. Write $w_{ij} \geq 0$ for the weight of edge $i \rightarrow j$ (influence from i toward j in the endorsement or transfer sense used here). Let W be the weighted adjacency matrix and define the out-strength $d_i^{\text{out}} = \sum_j w_{ij}$. The row-stochastic transition matrix P is

$$P_{ij} = \begin{cases} w_{ij}/d_i^{\text{out}} & \text{if } d_i^{\text{out}} > 0, \\ 1/n & \text{if } d_i^{\text{out}} = 0 \end{cases} \quad (2.1)$$

(dangling nodes distribute mass uniformly). The Google matrix with damping $d \in (0, 1)$ is

$$G_{\text{PR}} = d P^\top + \frac{1-d}{n} \mathbf{1}\mathbf{1}^\top, \quad (2.2)$$

and the PageRank vector π satisfies $\pi = G_{\text{PR}} \pi$ with $\pi \geq 0$ and $\sum_i \pi_i = 1$ ^{[30],[25]}. Equivalently,

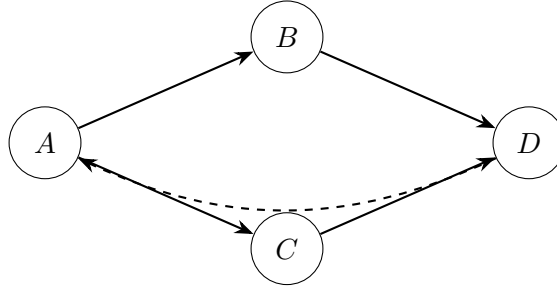
$$\pi_j = d \sum_{i:(i \rightarrow j) \in E} \pi_i \frac{w_{ij}}{d_i^{\text{out}}} + \frac{1-d}{n}. \quad (2.3)$$

2.5.1 Random-surfer interpretation

Equation (2.3) admits a probabilistic reading: a surfer follows outgoing edges with probability d and teleports uniformly with probability $1 - d$ (Figure 2.2). The stationary distribution ranks nodes by long-run visit frequency. Damping prevents sinks from trapping all mass and guarantees primitivity of G_{PR} under mild conditions, so the Perron vector is unique ^[25].

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google's PageRank and beyond: The science of search engine rankings. [Princeton University Press]



solid: follow edge (prob. d); dashed: teleport (prob. $1 - d$)

Figure 2.2: Random-surfer intuition for PageRank. Solid arrows are graph transitions; dashed arrows represent teleportation that mixes the chain.

2.5.2 Power iteration and complexity

In practice π is computed by power iteration:

$$\pi^{(t+1)} = G_{\text{PR}} \pi^{(t)}, \quad (2.4)$$

until $\|\pi^{(t+1)} - \pi^{(t)}\|_1 < \varepsilon$ or a maximum iteration count is reached. Each iteration costs $O(|E|)$ arithmetic operations for sparse graphs. Therefore, methods that induce fewer edges—such as latest-allowance snapshots versus full transfer histories—reduce per-iteration cost and often memory. This complexity observation grounds hypothesis H3.

Default parameters in this dissertation use damping $d = 0.85$ following common practice ^[30], with convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of 300 iterations as study-specific solver settings.³⁶ Robustness checks vary $d \in \{0.75, 0.85, 0.95\}$.

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

³⁶Tolerance and iteration caps are implementation choices for reproducible timing; they are not claimed as part of the classical PageRank formulation.

2.6 Reputation systems lineage

PageRank is one member of a family of spectral and propagation-based reputation mechanisms. Kleinberg’s HITS algorithm distinguishes hubs and authorities in hyperlinked corpora ^[23]. EigenTrust aggregates local trust scores in peer-to-peer networks with normalization to limit malicious inflation ^[21]. TrustRank propagates trust from a seed set of known-good nodes to combat web spam ^[17]. These systems share a theme: reputation is relational and should be hard to forge without costly endorsements from already-reputable nodes.

On-chain, Sybil attacks remain central ^[11]. Creating wallets is cheap; creating wallets that receive allowances or transfers from diverse counterparties is costlier. Sybil-adjusted stability proxies in this dissertation (inbound counterparty ratio, tenure, active months) partially capture diversity of interaction but do not eliminate adversarial manipulation. Chapter 5 sketches a cost model for allowance-graph Sybils.

Machine-learning credit models on DeFi features ^[24] can outperform simple graph scores on labeled defaults when labels exist, but they trade interpretability for accuracy. EndorseRank and AWP prioritize transparent propagation rules suitable for audit ^[29].

^[23]Kleinberg, J. M. (1999). Authoritative sources in a hyperlinked environment. [Journal of the ACM]

^[21]Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. [Proceedings of the 12th International Conference on World Wide Web]

^[17]Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. [Proceedings of the Thirtieth International Conference on Very Large Data Bases]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

2.7 ERC-20 allowances as endorsement

ERC-20 approve and EIP-2612 permit record explicit spending authorization ^{[12],[13]}. EndorseRank treats the latest allowance per owner–spender pair as edge strength on a directed endorsement graph and runs PageRank on that snapshot, eliminating historical time-decay because newer approvals override older ones.

Formally, for each token and owner–spender pair, let $a_{o \rightarrow s}$ be the decimal-adjusted latest allowance. Edges are aggregated across tokens by summing allowances into a single weight $w_{o \rightarrow s}$. The resulting graph $G_{\text{approve}} = (V, E_{\text{approve}}, w)$ is typically much sparser than the transfer graph over the same wallets and window. PageRank on G_{approve} yields EndorseRank scores s_{ER} .

The endorsement interpretation is intentional but imperfect. Users approve routers for convenience, not always as a social judgment of the router’s “trustworthiness.” Contract addresses and EOAs coexist. Nevertheless, allowances are a clearer authorization signal than passive receipt of transfers, which may reflect market making, airdrops, or routing without any endorsement intent ^[10].

2.8 Trading success validation frame (GMX V2)

On Arbitrum One, dense wallet-attributed **PositionDecrease** events with realized PnL (**basePnlUsd**) provide trading-success proxies: close-success count, realized gain proxy, and close success rate ^{[15],[32]}. Under-collateralized lending default labels are sparse on this chain; GMX V2 perpetual closes offer dense, wallet-level realized outcomes for domain-intuition validation. These proxies parallel Do et al.’s in-degree/in-value logic but target collateralized perpetual trading rather than raw inflows.

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[15]GMX. (n.d.). GMX V2 documentation.

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

Perpetual markets allow leveraged long and short exposure with funding and liquidation rules ^[35]. A profitable close is not equivalent to loan repayment, but it is an observable outcome that risk systems may care about when screening active traders. This dissertation treats GMX outcomes as *domain-intuition proxies*, not regulatory ground truth.

2.9 Supplementary extension: outcome-native reference (GF-PR)

During the empirical phase, **GF-PR** (GainFlow PageRank) was added as a supplementary diagnostic to bound trading-outcome proxy alignment. GF-PR builds a directed star graph from the same GMX PositionDecrease stream: a synthetic profit pool node sends edges to wallets with positive realized PnL; wallets send edges to a loss sink for negative closes ($|\text{PnL}| + 0.1$) and liquidations ($|\text{PnL}| + 1.0$). GF-PR has no wallet-to-wallet social edges; it serves as an *outcome-native ceiling reference* (SRQ1), not a peer to EndorseRank or AWP. High alignment with trading-success proxies is expected by construction and must be interpreted as construct overlap, not superior social reputation.

2.10 Rank correlation theory

Reputation scores and validation proxies are continuous variables without a universal classification threshold. Following Do et al.^[10] and classical psychometrics ^[8], this dissertation evaluates monotonic and pairwise rank agreement.

Spearman’s rank correlation ^[33] is Pearson correlation on ranks. For paired ranks (R_i, S_i) of n items,

$$\rho = 1 - \frac{6 \sum_{i=1}^n (R_i - S_i)^2}{n(n^2 - 1)} \quad (2.5)$$

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

in the absence of ties (with tie-aware variants used in software). Spearman’s ρ captures monotonic association.

Kendall’s τ ^[22] counts concordant and discordant pairs. For all pairs (i, j) with $i < j$,

$$\tau = \frac{C - D}{\binom{n}{2}}, \quad (2.6)$$

where C (D) is the number of concordant (discordant) pairs, with tie corrections in the τ -b variant. Kendall’s τ is often more interpretable for pairwise ordering agreement and is the primary family-level summary in Chapter 4.

2.10.1 Interpretation bands

There is no universal cutoff for “strong” τ in blockchain ranking studies. This dissertation uses operational bands for narrative clarity: $|\tau| < 0.10$ negligible noise; 0.10 – 0.35 weak-to-moderate; ≥ 0.35 substantively meaningful in comparable PageRank/blockchain studies. Values are compared *relatively* across methods on the same cohort rather than against an absolute clinical threshold.

2.11 Theoretical framework

This study is grounded in hyperlink-citation propagation, construct (domain) alignment, and computational complexity to compare EndorseRank and AWP across reputation structure (H1), trading-success alignment (H2), and computational efficiency (H3).

Hyperlink–citation semantics. The original PageRank algorithm ^[30] treats inbound hyperlinks as citations. Do et al. ^[10] apply this logic to Ethereum transfers with time-decayed edge weights.

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

Figure 3.2 (Chapter 3) plots the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))} \quad (2.7)$$

with $k = 0.01$ and $t_0 = 180$ days as used for AWP in this study. EndorseRank extends citation-based propagation to ERC-20 approve/permit edges because allowances are explicit, on-chain delegations of spending authority ^{[12],[13]}. Unlike transfers, which may reflect trading or routing without trust intent, an allowance records durable authorization until revoked. Graph-based risk models ^[26] further show that network structure carries information beyond raw activity counts, supporting a directed endorsement graph rather than transfer volume alone.

Domain alignment framework. In measurement theory, construct alignment assesses how well an indicator tracks an observable outcome ^[8]. For rank-based reputation scoring, Spearman’s ρ and Kendall’s τ evaluate monotonic and pairwise ordering agreement without assuming a classification threshold ^[10]. Close-success count from GMX V2 and realized gain proxy serve as observable trading-success measures on Arbitrum One, satisfying the domain-intuition requirement without claiming external loan-repayment ground truth.

Computational complexity. PageRank’s per-iteration cost is $O(|E|)$ ^[30]. Approve/permit graphs are sparser than transfer graphs (fewer edges $|E|$), yielding lower per-iteration cost, faster convergence, and reduced memory—the basis for H3.

Figure 2.3 presents the conceptual model mapping the primary EndorseRank–AWP comparison to H1–H3 and RQ1–RQ4. GF-PR (dashed) is a supplementary outcome-native ceiling for SRQ1,

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

added during empirical work.

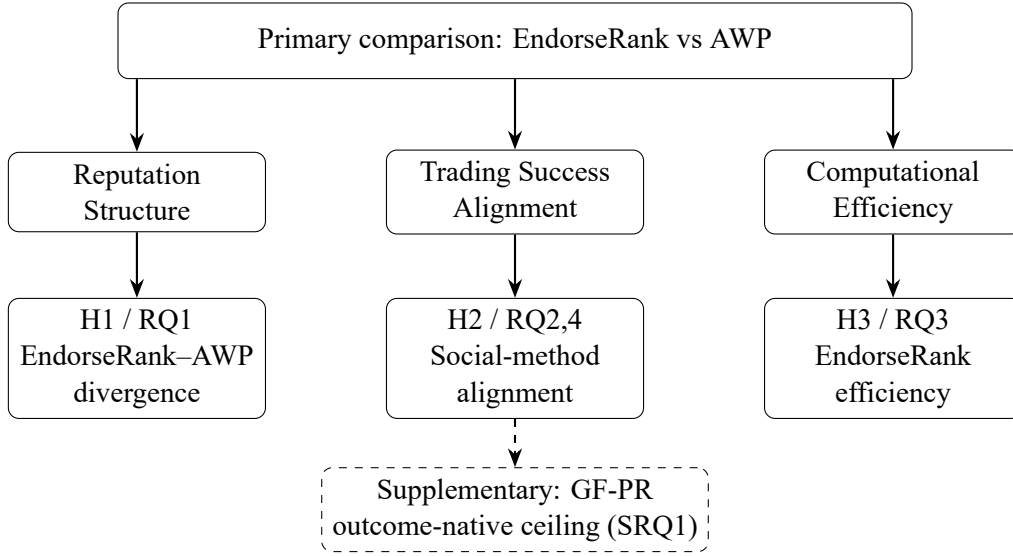


Figure 2.3: Conceptual model: primary EndorseRank vs. AWP comparison mapped to reputation structure (H1/RQ1), social-method alignment (H2/RQ2–4), and computational efficiency (H3/RQ3). GF-PR (dashed) is a supplementary outcome-native ceiling reference (SRQ1), not a third peer social method.

Supplementary role of GF-PR. H1 and H3 concern EndorseRank and AWP only. H2 compares social-method alignment between EndorseRank and AWP across five proxy families (RQ2, RQ4). GF-PR bounds the maximum trading-outcome proxy alignment achievable when ranking directly on outcome flow, clarifying construct overlap for SRQ1 without entering the primary hypothesis tests.

2.12 Reader’s guide to terminology

The definitions in Chapter 1 establish formal constructs; the following guide clarifies how to interpret results and avoid common misreadings:

- **Primary methods:** EndorseRank (allowance graph) and AWP (transfer graph). Both are social PageRank methods on wallet-to-wallet edges; intended as deployable reputation signals.
- **Supplementary reference:** GF-PR on GMX PnL flow to/from synthetic POOL/SINK nodes;

outcome-native ceiling for SRQ1, not a deployable social score.

- **Primary graph construct:** The graph a method is *designed* to rank (e.g., G_{approve} for EndorseRank). Distinct from validation proxies applied after scoring.
- **Validation proxy:** Observable wallet metric (e.g., in-degree, GMX close-success count) used to test whether rankings align with an external construct ^[8].
- **Construct overlap:** When a method’s input graph and a validation proxy share the same data source (GF-PR vs. trading-outcome proxies), inflating τ without implying social reputation.
- **Alignment bands (Kendall τ):** $|\tau| < 0.10$ negligible noise; 0.10–0.35 weak-to-moderate; ≥ 0.35 substantively meaningful in comparable PageRank/blockchain studies; values are compared *relatively* across methods on the same cohort.
- **Matched wallet cohort:** The primary evaluation sample of $n = 5,521$ wallets with at least three GMX closes and non-zero reputation-subgraph connectivity.
- **Expanded wallet pool:** A larger address set ($N = 31,612$) used for runtime scaling beyond the matched cohort.

2.13 Domain alignment metrics

Following Do et al.^[10] and Cronbach & Meehl^[8], construct alignment for rank-based scores uses Spearman’s ρ and Kendall’s τ between reputation scores and proxy rankings. This dissertation evaluates **five proxy families** (transfer, allowance, inverse risk, Sybil-adjusted stability, trading success); see Chapter 3 for operational definitions and formulas. Family-level mean τ summarizes each construct; per-proxy tables report ρ and τ for transparency.

The literature reviewed here supports three design choices: (i) PageRank as an interpretable propagation backbone; (ii) allowances as endorsement edges distinct from transfers; and (iii) multi-family rank correlation as validation in the absence of loan-default labels. Chapter 3 turns these choices

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

into an operational pipeline on Arbitrum One.

2.14 Worked example: PageRank on a four-node endorsement graph

To make Equation (2.3) concrete, consider four wallets $\{A, B, C, D\}$ with latest-allowance edges

$$A \rightarrow B (2), \quad A \rightarrow C (1), \quad B \rightarrow D (3), \quad C \rightarrow D (1),$$

and damping $d = 0.85$. Out-strengths are $d_A^{\text{out}} = 3$, $d_B^{\text{out}} = 3$, $d_C^{\text{out}} = 1$, $d_D^{\text{out}} = 0$ (dangling). After one power-iteration step from the uniform vector $\pi^{(0)} = \mathbf{1}/4$, mass flows preferentially toward D because both B and C endorse D , while A splits mass between B and C . Subsequent iterations mix in teleportation mass $(1 - d)/4 = 0.0375$ at every node. The stationary ranking places D highest, then B and C , then A —illustrating that *receiving* endorsements from nodes that themselves receive endorsements raises score, exactly the citation logic of PageRank^{[30],[25]}.

If the same wallets are connected by many small transfers instead of few allowances, the transition matrix changes and the ranking can reorder even though the node set is identical. That is the microcosm of H1: EndorseRank and AWP need not agree.

2.15 DeFi risk taxonomy and where reputation fits

DeFi risk is multi-layered^{[35],[31],[45]}. Smart-contract risk concerns bugs and upgrade keys. Oracle risk concerns manipulated or stale prices^[1]. Market risk concerns volatility and liquidation

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[31]Qin, K., Zhou, L., Livshits, B., & Gervais, A. (2021). Attacking the DeFi ecosystem with flash loans for fun and profit. [Financial Cryptography and Data Security]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[1]Angeris, G., & Chitra, T. (2020). Improved price oracles: Constant function market makers. [Proceedings of the 2nd ACM Conference on Advances in Financial Technologies]

cascades^[54]. Governance risk concerns token-weighted capture. *Counterparty* or *behavioral* risk—the focus of this dissertation—concerns whether a wallet’s historical interactions suggest reliable management of leveraged exposure.

On-chain reputation scores address only the behavioral layer, and only partially. They do not replace collateral, audits, or oracle design. Their role is to summarize relational structure (who authorizes whom; who pays whom) into an ordinal signal that can be audited and recomputed^[29]. EndorseRank targets the authorization facet of behavioral risk; AWP targets the payment-flow facet. Neither claims to predict exploits or governance attacks.

Wash trading and MEV further complicate behavioral measurement^{[58],[9],[38]}. Transfer graphs are especially exposed to artificial volume; allowance graphs are exposed to approval farming. Multi-family validation and Sybil-oriented proxies are partial mitigations, not complete defenses.

2.16 Perpetual markets as a validation laboratory

Perpetual futures (“perps”) are derivative contracts without expiry, popularized in crypto markets and implemented on-chain by protocols such as GMX^{[15],[18]}. Traders post collateral, choose long or short exposure, and pay or receive funding. Positions may be reduced or closed voluntarily, or liquidated when collateral is insufficient. Each close emits an on-chain event with realized PnL attributable to a wallet.

For reputation research, perps offer three advantages over sparse lending defaults on Arbitrum:

^[54]Perez, D., Werner, S. M., Xu, J., & Livshits, B. (2021). Liquidations: DeFi on a knife-edge. [Financial Cryptography and Data Security]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[58]Victor, F., & Weintraud, A. M. (2021). Detecting and quantifying wash trading on decentralized cryptocurrency exchanges. [Proceedings of the Web Conference 2021]

^[9]Daian, P., Goldfeder, S., Kell, T., Li, Y., Zhao, X., Bentov, I., Breidenbach, L., & Juels, A. (2020). Flash Boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. [2020 IEEE Symposium on Security and Privacy]

^[38]Zhou, L., Qin, K., Torres, C. F., Le, D. V., & Gervais, A. (2021). High-frequency trading on decentralized on-chain exchanges. [2021 IEEE Symposium on Security and Privacy]

^[15]GMX. (n.d.). GMX V2 documentation.

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

(i) high event density (365,488 closes in the study window); (ii) wallet-level attribution; and (iii) signed outcomes (profit, loss, liquidation) that can be turned into oriented proxies. The disadvantage is construct distance from unsecured credit: a profitable trader is not necessarily a good borrower. This dissertation therefore treats GMX outcomes as domain-intuition proxies and reports transfer proxies alongside them (RQ4).

2.17 Related measurement traditions

Beyond blockchain, reputation systems have been studied in e-commerce ^[55], peer-to-peer networks ^[21], and web spam ^[17]. Common lessons include: (i) reputation is relational; (ii) cheap identities enable Sybil attacks ^[11]; (iii) seed sets and normalization limit inflation; and (iv) evaluation requires external criteria, not only internal consistency. EndorseRank inherits (i)–(iii) via PageRank damping and endorsement costs, and addresses (iv) via multi-family rank correlation ^{[8],[33],[22]}.

Personalized and topic-sensitive PageRank variants ^{[46],[47]} could, in future work, bias teleportation toward protocol-approved contracts or known-good wallets, analogous to TrustRank seeds. This dissertation uses uniform teleportation for comparability with Do et al.^[10].

^[55]Resnick, P., & Zeckhauser, R. (2002). Trust among strangers in Internet transactions: Empirical analysis of eBay’s reputation system. [The Economics of the Internet and E-commerce]

^[21]Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. [Proceedings of the 12th International Conference on World Wide Web]

^[17]Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. [Proceedings of the Thirtieth International Conference on Very Large Data Bases]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

^[46]Haveliwala, T. H. (2003). Topic-sensitive PageRank: A context-sensitive ranking algorithm for web search. [IEEE Transactions on Knowledge and Data Engineering]

^[47]Jeh, G., & Widom, J. (2003). Scaling personalized web search. [Proceedings of the 12th International Conference on World Wide Web]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

2.18 Accounts, gas, and why logs are the measurement surface

Ethereum-style accounts are either externally owned accounts (EOAs), controlled by private keys, or contract accounts, controlled by code ^{[36],[2]}. Every state-changing interaction consumes gas and emits logs when contracts are written to do so. Reputation research that aims for auditability therefore privileges *logs* over off-chain indexes: logs are part of the canonical chain history and can be re-derived by any full node or public warehouse ^[28].

Gas costs shape the graphs we observe. On L2s, lower fees increase transfer frequency and approval churn, densifying G_{transfer} more than G_{approve} because payments are routine while approvals are intermittent setup actions. This economic asymmetry is one reason EndorseRank graphs remain sparse even when transfer graphs explode (Chapter 4).

2.19 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) are a well-known user-experience hazard: a compromised spender can drain tokens until the owner revokes. Security analyses of honeypots, phishing approvals, and malicious spenders ^{[57],[52]} underscore that allowances are not neutral metadata—they are capabilities. EndorseRank interprets inbound allowances as endorsement of a spender’s capability, which is precisely why the construct is semantically distinct from inbound transfers. A wallet may receive many transfers without ever being trusted to spend on others’ behalf.

From a measurement standpoint, unlimited approvals create heavy-tailed edge weights. Decimal adjustment and optional winsorization can moderate extremes; this dissertation sums decimal-

^[36]Wood, G. (2014). Ethereum: A secure decentralised generalised transaction ledger. [Ethereum Project]

^[2]Antonopoulos, A. M., & Wood, G. (2018). Mastering Ethereum: Building smart contracts and DApps. [O’Reilly Media]

^[28]Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). Bitcoin and cryptocurrency technologies. [Princeton University Press]

^[57]Torres, C. F., Steichen, M., & State, R. (2019). The art of the scam: Demystifying honeypots in Ethereum smart contracts. [28th USENIX Security Symposium]

^[52]Li, W., Bu, J., Li, X., & Chen, X. (2022). Security analysis of DeFi: Vulnerabilities, attacks and advances. [2022 IEEE International Conference on Blockchain]

adjusted latest allowances and relies on PageRank’s normalization by out-strength so that a single huge approval does not automatically dominate unless the owner has few outgoing endorsements.

2.20 AWP in depth: what Do et al. measure

Do et al.^[10] propose Adaptive Weighted PageRank on Ethereum transfers as a social-credit signal. Edges are transfers; weights combine value and recency through a logistic decay. Validation uses inbound degree and inbound value—proxies for “how much economic attention a wallet receives.” The framework is attractive because it is fully on-chain, rank-based, and free of off-chain identity. EndorseRank preserves the PageRank-and-rank-correlation template but replaces the edge semantics. The empirical question is not whether AWP is “wrong,” but whether allowance edges yield a different, useful, and cheaper signal. Chapter 4 answers in the affirmative on cost and allowance alignment, and in the negative on universal dominance: AWP remains stronger on transfer and several outcome-adjacent families.

2.21 Comparison of graph reputation methods

Table 2.1 situates EndorseRank among related methods discussed in this chapter.

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank (web)	Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank	Links from seed goods	Static / seeded	Spam demotion
AWP	Token transfers	Logistic decay	In-degree / in-value
RiskProp	Risk-bearing links	Propagation of risk	Account risk labels
EndorseRank	Latest allowances	Snapshot (override)	Allowance proxies; secondary families
GF-PR (supp.)	PnL flow to POOL/SINK	Outcome-native	Trading-success (overlap)

The table highlights that EndorseRank’s novelty is not the use of PageRank per se, but the choice

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

of latest-allowance endorsement edges as the primary social construct for DeFi wallets, evaluated under a multi-family alignment protocol.

2.22 Open problems motivating this dissertation

Several open problems in on-chain reputation remain unresolved in the literature and motivate the present study:

1. **Edge semantics:** Which on-chain relations best encode trust or authorization under pseudonymity ^[29]?
2. **Validation without defaults:** How can methods be compared when loan-default labels are scarce ^{[10],[8]}?
3. **Computational cost:** Can reputation scores refresh at interactive latencies as graphs grow ^[25]?
4. **Construct overlap:** How can evaluation avoid mistaking shared inputs for predictive power (SRQ1)?
5. **Adversarial robustness:** What is the cost of forging reputation under cheap identities ^[11]?

EndorseRank addresses (1)–(3) directly, contributes a diagnostic for (4) via GF-PR, and analyzes (5) at a model level in Chapter 5. Full empirical adversarial evaluation remains future work.

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

3 Research Methodology

This chapter specifies the empirical design used to compare EndorseRank and Adaptive Weighted PageRank (AWP) as social reputation scores on Arbitrum One. The design covers notation, data sources, a three-phase collection strategy, pipeline architecture, event decoding, graph construction algorithms, validation proxies, statistical tests, computational benchmarks, scaling and robustness extensions, reproducibility, and ethics. The primary comparison is between EndorseRank on the ERC-20 allowance graph and AWP on the time-decayed transfer graph ^{[10],[30]}. GainFlow PageRank (GF-PR) appears only as a supplementary outcome-native ceiling reference and is not treated as a peer social method.

The methodological stance follows construct (domain) alignment: reputation scores are continuous rankings, and validation uses observable wallet-level proxies rather than binary default labels ^{[8],[10]}. In measurement terms, each ranking method is an indicator of an unobserved reputation construct; proxies are alternative indicators chosen for domain intuition (inbound economic activity, received endorsements, risk-avoiding trading behavior, activity stability, and trading success). Alignment statistics ask whether the ordering induced by a method co-varies with the ordering induced by each proxy on the same wallets. This approach is appropriate when ground-truth loan defaults are sparse and when the scientific claim is comparative—EndorseRank versus AWP—rather than absolute prediction of an external credit bureau label.

All procedures operate on publicly available blockchain logs within a fixed observation window, with deterministic seeds and an open evaluation pipeline so that reported tables can be regenerated from cached parquet. The chapter is organized to move from symbols and data acquisition, through

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

graph and proxy formalisms, to evaluation statistics, computational protocol, and research ethics. Readers primarily interested in empirical outcomes may skim Sections 3.1–3.5 and focus on Sections 3.6–3.10; readers implementing a replication should follow the full pipeline description and Section 3.13.

- **Design principles:** Four principles govern every operational choice. First, *same-seed comparability*: EndorseRank and AWP are always scored on identical wallet sets so that differences cannot be attributed to sampling inconsistency. Second, *pre-registration by configuration*: observation window, topic hashes, damping, decay parameters, minimum closes, and benchmark seeds are fixed in configuration before interpreting results. Third, *separation of construction and timing*: graph construction is performed once per cohort; computational benchmarks time only the PageRank solve. Fourth, *primary versus supplementary scope*: claims C1–C4 and hypotheses H1–H3 concern EndorseRank and AWP; GF-PR enters only SRQ1 as an outcome-native ceiling.

3.1 Notation

Table 3.1 summarizes the operational symbols used throughout this chapter and in Chapters 4–5 (a compact front-matter summary appears in Table 2). Vectors are bold; sets are calligraphic when helpful. Wallet addresses are treated as opaque identifiers and are never linked to off-chain identities.

Unless otherwise stated, PageRank is the weighted formulation of Page et al.^[30]: at iteration t ,

$$\mathbf{r}^{(t+1)} = (1 - d) \boldsymbol{\pi} + d P^\top \mathbf{r}^{(t)} + d (\mathbf{r}^{(t)\top} \mathbf{1}_{\text{dang}}) \boldsymbol{\pi}, \quad (3.1)$$

where P is the row-stochastic transition matrix induced by positive edge weights, $\boldsymbol{\pi}$ is the uniform teleportation distribution over nodes, and $\mathbf{1}_{\text{dang}}$ marks dangling nodes (zero out-strength). Explicitly,

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched wallet cohort ($ \mathcal{W} = n = 5,521$)
$\mathcal{W}_{\text{pool}}$	Expanded wallet pool ($ \mathcal{W}_{\text{pool}} = N = 31,612$)
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph
$G_{\text{GF-PR}}$	Supplementary GF-PR star graph
$w_A(u, v)$	Latest allowance weight, owner $u \rightarrow$ spender v
$w_T(u, v)$	Time-decayed transfer weight, sender $u \rightarrow$ receiver v
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_P$	Proxy value vector for proxy P
d	PageRank damping factor (default 0.85)
τ_{tol}	PageRank convergence tolerance (10^{-8})
I_{max}	Maximum PageRank iterations (300)
$\sigma(\Delta t)$	Logistic time-decay on transfer age
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before observation end
$\varepsilon_{\text{loss}}, \varepsilon_{\text{liq}}$	GF-PR sink additives (0.1, 1.0)
ρ	Spearman’s rank correlation
τ	Kendall’s rank correlation
T_{obs}	Observation end timestamp (2026-05-31 UTC)

if $w(u, v)$ denotes the weight of the directed edge $u \rightarrow v$ and $s(u) = \sum_v w(u, v)$ is the out-strength, then

$$P_{u,v} = \begin{cases} w(u, v)/s(u) & \text{if } s(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Dangling mass is redistributed uniformly through the teleportation term, which is the standard treatment in sparse web and transaction graphs. Iteration stops when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \tau_{\text{tol}}$ or after I_{max} iterations. Scores are renormalized to sum to one over reachable nodes. Dense ranks for alignment are assigned by sorting scores descending (rank 1 = highest score), with wallet address as a deterministic tie-break when scores are equal.

The damping factor d interpolates between pure graph propagation ($d \rightarrow 1$) and uniform scores ($d \rightarrow 0$). The default $d = 0.85$ follows the classical web setting and the AWP baseline of Do et al.^[10]. Robustness sweeps vary d while holding edge weights fixed (Section 3.12). Tolerance $\tau_{\text{tol}} = 10^{-8}$ and $I_{\text{max}} = 300$ are intentionally strict relative to interactive analytics settings, so that runtime differences reflect graph structure rather than early stopping.

3.2 Data sources and sampling

This section fixes the empirical universe: which chain, which time window, which public tables, which event types, and which wallets enter the primary sample. Downstream graph construction and proxy formulas never invent events outside this universe.

- **Primary chain:** All on-chain events are drawn from Arbitrum One (chain ID 42161), an Ethereum Layer-2 rollup with dense DeFi activity and public log availability^[32]. Restricting to a single chain avoids cross-chain address aliasing—the same private key controlling addresses on multiple chains would otherwise appear as unrelated nodes—and keeps gas-cost

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

and latency regimes comparable across wallets. Arbitrum One is also the deployment venue for GMX V2’s high-volume perpetual markets, which supply wallet-attributed realized PnL at a density that lending defaults do not match within scope.

- **Observation window:** The study window is 2025-12-01 00:00:00 UTC through 2026-05-31 23:59:59 UTC (six calendar months). Exclusive end timestamps in SQL use 2026-06-01 00:00:00 UTC so that the inclusive calendar end is 2026-05-31. This window is long enough to accumulate multiple GMX closes per active trader—the minimum-close filter requires at least three—while remaining tractable for monthly-batched reputation extracts under BigQuery byte budgets. AWP time-decay is anchored at observation end $T_{\text{obs}} = 2026-05-31\ 23:59:59\ \text{UTC}$, so $\Delta t = 0$ for events at the window’s close and $\Delta t \approx 181$ days for events at the window’s open.
- **BigQuery source:** Event logs are read from the public dataset

`bigquery-public-data.goog_blockchain_arbitrum_one_us.logs.`

Each row corresponds to a contract log entry with block timestamp, emitting address, topics, and data. Queries filter on `block_timestamp`, contract address (when applicable), and event topic hashes. The public dataset removes the need for a private archive node while remaining auditable: any researcher with Google Cloud access can re-issue the same SQL against the same table. SQL templates, dry-run cost guardrails, and acquisition metadata are documented in Appendix 8.1.

- **Reputation events:** EndorseRank uses ERC-20 Approval logs ^[12], including allowances established via EIP-2612 permit flows that ultimately emit the same Approval event ^[13]. AWP uses ERC-20 Transfer logs. Both streams are wallet-filtered to the cohort under study: a log is retained if the owner or spender (approvals) or the sender or recipient (transfers) lies in the seed wallet set. Filtering at query time reduces scanned payload relative to full-chain

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

extracts and focuses the reputation subgraph on economically relevant neighbors of the cohort. Extracts run in six monthly batches to respect per-query byte caps and to allow resumable checkpoints if a month fails.

- **GMX events:** Trading-success and risk proxies are derived from GMX V2 `PositionDecrease` events emitted by the GMX `EventEmitter` at

`0xC8ee91A54287DB53897056e12D9819156D3822Fb`

^[15]. Traders are identified via the event `account` field. Realized profit and loss use `basePnlUsd`; position size changes use `sizeDeltaUsd` when size-weighted robustness variants are computed; liquidations are flagged from order-type metadata in the decoded payload. GMX is used because it provides dense, wallet-level realized outcomes on collateralized perpetual trading—a domain-intuition validation frame analogous to Do et al.’s use of inbound transfer activity, but targeted at trading success rather than passive inflows ^{[10],[32]}. Lending-protocol default labels are excluded because under-collateralized lending volume on Arbitrum within scope is insufficient for stable wallet-level labels.

- **Matched cohort definition:** The primary alignment sample is the intersection of (i) wallets with at least three GMX `PositionDecrease` closes in the observation window and (ii) wallets with non-zero connectivity in the reputation subgraph induced by approval or transfer edges. The minimum-close threshold $m = 3$ excludes one-off traders whose success rates are dominated by sampling noise, while remaining inclusive of active retail participants. The connectivity requirement excludes wallets that appear in GMX events but have no approval or transfer edges touching the reputation extract, for whom social PageRank scores would be uniformly zero and uninformative. After BigQuery extraction and preprocessing, this matched wallet cohort has size $n = 5,521$. Both `EndorseRank` and `AWP` are scored on the same seed

^[15]GMX. (n.d.). GMX V2 documentation.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

set so that differences in alignment and runtime are attributable to ranking logic and graph structure, not to inconsistent sampling.

- **What is not in scope:** The study does not ingest mempool data, off-chain order books, centralized-exchange account maps, or cross-chain bridges. It does not attempt entity resolution across multiple addresses controlled by one actor. It does not use proprietary GMX indexer APIs as the primary source of truth; the public log table is authoritative, and decoding is performed in the open pipeline.

3.3 Three-phase collection strategy

Data acquisition and evaluation proceed in three phases that separate expensive remote scans from local, repeatable analysis. The separation is intentional: once parquet artifacts exist, alignment tables, benchmarks, and robustness sweeps can be regenerated without further cloud cost, which is essential for iterative dissertation writing and for third-party replication.

- **Phase 1 (GMX validation stream):** Extract and decode all GMX `PositionDecrease` logs from the `EventEmitter` over the full observation window. The extract is not wallet-filtered a priori because the cohort is defined from this stream: every decrease in the window is a candidate label. After decoding, wallets with fewer than three closes are dropped from the eligibility set. This stream supplies trading-success, inverse-risk, and liquidation-family proxies, and it defines the matched wallet cohort used for primary alignment ($n = 5,521$ after intersecting with reputation connectivity in Phase 2).
- **Phase 2 (reputation streams):** For the matched cohort, extract wallet-filtered ERC-20 `Approval` and `Transfer` events in monthly batches. Each month is an independent query with its own dry-run estimate and checkpoint, so a failure in one month does not invalidate prior months. Preprocess approvals to latest allowances per owner–spender–token: given multiple approvals for the same triple, only the last timestamp in the window survives, and a final zero allowance removes the edge. Preprocess transfers to decimal-adjusted event tables with

timestamps, preserving sender, recipient, value, token, and block time. These tables are the sole inputs to EndorseRank and AWP graph construction for the primary comparison.

- **Phase 3 (local evaluation):** From cached parquet, compute reputation ranks, five primary proxy families, Spearman and Kendall alignment, computational benchmarks, and scaling/robustness extensions. Supplemental active wallets for the expanded pool are drawn from reputation-subgraph addresses (and optional supplemental parquet) without requiring a full-chain re-scan for every robustness experiment. No additional BigQuery scans are required once the Phase 1 and Phase 2 extracts and any supplemental active-wallet parquet are on disk.
- **Cost guardrails:** Every remote query begins with a mandatory `--dry-run` that estimates bytes processed before extraction. Each query is aborted if the estimate exceeds a 150 GiB per-query cap.¹² Operators must pass an explicit confirmation flag before paid scans proceed. Extraction metadata—wallet counts, row counts, bytes processed, timestamps, and output paths—is logged in the pipeline manifest (Appendix 8.1). These controls implement responsible use of public cloud resources and make the acquisition cost of the study auditable.

3.4 Pipeline architecture

Figure 3.1 summarizes the end-to-end flow from public logs to dissertation tables. Remote stages write immutable parquet; local stages are idempotent given those inputs and configuration in `margin_config.yaml`.

Operationally, Phase 1 scripts write GMX raw logs and decoded events; Phase 2 scripts write monthly approval/transfer shards and consolidated latest-allowance and transfer-event tables; Phase 3 scripts read those tables, build graphs, run PageRank, compute proxies, and export tables under `results/tables/`. The same local path supports fixture-mode dry runs for continuous integration without cloud credentials.

¹²The per-query byte cap is an operational policy of this study’s acquisition pipeline, not a property of BigQuery itself.

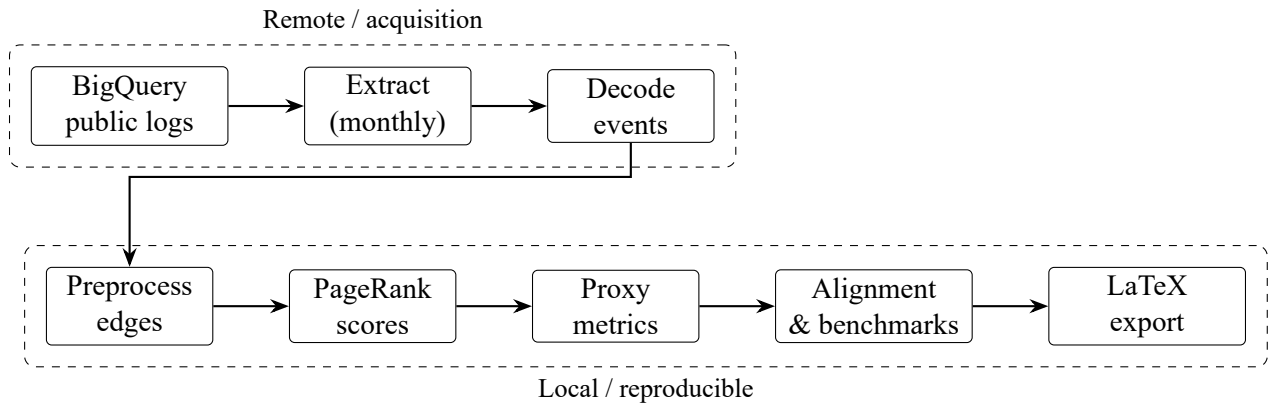


Figure 3.1: Evaluation pipeline: public Arbitrum logs are extracted and decoded, reputation edges and GMX outcomes are preprocessed, EndorseRank and AWP ranks are computed, proxies and alignment statistics are evaluated, and LaTeX result fragments are exported.

3.5 Event decoding

Blockchain logs store events as topics and a data payload rather than as named fields. Decoding maps each log row to a typed record used by graph construction and proxy computation. Incorrect decoding would silently corrupt both reputation graphs and validation labels; this section therefore records the exact topic hashes and field mappings used in the pipeline.

- **Log anatomy:** An Ethereum-compatible log contains an emitting contract address, an ordered list of topics, and an ABI-encoded data blob. Topic 0 is the keccak-256 hash of the event signature string, which uniquely identifies the event type among well-known standards. Indexed parameters occupy subsequent topics as 32-byte words (addresses are left-padded); non-indexed parameters occupy the data field in ABI order^[12]. BigQuery exposes these fields as columns; the pipeline applies topic filters in SQL and performs ABI decoding in Python for fields that require numeric interpretation.
- **ERC-20 Approval:** Topic 0 is

0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925,

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

the hash of `Approval(address,address,uint256)`. Topics 1 and 2 hold the owner and spender addresses (left-padded); the allowance amount is ABI-decoded from the data field and scaled by the token’s decimals. Unlimited approvals (`type(uint256).max`) are retained as large finite weights after decimal adjustment; they contribute strong endorsement edges, which is intentional: an unlimited approval is a strong on-chain authorization. EIP-2612 `permit` does not introduce a separate reputation event: successful permits emit the same `Approval` log, so `EndorseRank` observes the resulting allowance snapshot regardless of whether authorization was set by an on-chain `approve` call or an off-chain signature ^[13].

- **ERC-20 Transfer:** Topic 0 is

`0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef,`

the hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed; value is decimal-adjusted. Mint and burn patterns that use the zero address are retained if they touch cohort wallets, but they are rare relative to wallet-to-wallet transfers in the filtered extract. Self-transfers are retained in the event table but excluded from Sybil-stability inbound-counterparty calculations (Section 3.9), because a wallet transferring to itself does not evidence an external counterparty relationship.

- **GMX PositionDecrease:** GMX V2 emits structured events through `EventEmitter` rather than a minimal ERC-20-style signature ^[15]. Extraction filters on the emitter address and the verified `EventLog1` topic 0, then decodes key–value fields from the event payload to recover `account`, `basePnlUsd`, `sizeDeltaUsd`, and liquidation indicators. Only decreases (closes, partial closes, and liquidations) enter the validation frame; increases open or enlarge positions and do not realize PnL for the success proxies used here. Liquidation flags distinguish forced closes from voluntary ones, which matters for both inverse-risk proxies and GF-PR sink weights.

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

^[15]GMX. (n.d.). GMX V2 documentation.

- **Address normalization and units:** All addresses are lower-cased before joins so that checksummed and non-checksummed hex forms collide correctly. Token amounts are stored as floating-point decimal-adjusted values for weighting; ranking uses relative magnitudes within a cohort rather than absolute USD conversion for allowance and transfer edges. GMX PnL fields are already denominated in USD units in the protocol’s event schema and are used directly for trading-success and risk proxies.

3.6 Graph construction

EndorseRank and AWP share the PageRank solver in Equation (3.1) but differ in edge semantics. Both methods induce a directed, weighted graph on wallet nodes, restrict attention to the subgraph touching the seed wallet set (edges with at least one endpoint in the cohort, including one-hop neighbors), and emit a score for every cohort wallet (missing nodes receive score 0 before ranking). Subgraph restriction is important for scalability and for interpretability: the score of a cohort wallet may receive mass from non-cohort neighbors that endorse or transfer to it, but the evaluation reports scores only on $\mathcal{W}$.

The conceptual distinction is endorsement versus flow. An allowance is an explicit, revocable authorization to spend; a transfer is a completed movement of value that may reflect payment, routing, market-making, or wash activity without a trust commitment^{[10],[12]}. EndorseRank therefore treats the latest allowance snapshot as a citation-like endorsement graph in the sense of Page et al.^[30]. AWP treats historical transfers as time-decayed citations, down-weighting older flows so that recent economic activity dominates.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

3.6.1 EndorseRank allowance graph

Let $\mathcal{A}$ be the set of latest allowance records after preprocessing: for each token and owner–spender pair, only the chronologically last `Approval` in the observation window is retained (zero allowances remove the edge). Across tokens, weights are summed:

$$w_A(u, v) = \sum_{\text{token}} a^*(u, v, \text{token}), \quad (3.3)$$

where a^* is the latest decimal-adjusted allowance. The directed edge $u \rightarrow v$ encodes owner u endorsing spender v . PageRank damping is $d = 0.85$. No temporal decay is applied: allowance override semantics replace time-based weighting, because a newer approval supersedes an older one for the same pair^[12].

Algorithm 1 formalizes construction and scoring.

Algorithm 1 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$, seed wallets $\mathcal{W}$, damping d , tolerance τ_{tol} , max iterations I_{max}

Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$

```

1:  $E \leftarrow \emptyset$ 
2: for each  $(u, v)$  in unique owner–spender pairs in  $\mathcal{A}$  do
3:    $w \leftarrow \sum_{\text{token}} a^*(u, v, \text{token})$ 
4:   if  $w > 0$  then
5:     add edge  $(u \rightarrow v, w)$  to  $E$ 
6:   end if
7: end for
8:  $E \leftarrow$  edges in  $E$  incident to at least one wallet in  $\mathcal{W}$ 
9:  $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, d, \tau_{\text{tol}}, I_{\text{max}})$ 
10: return  $\mathbf{s}_{\text{ER}}[w] \leftarrow \mathbf{s}[w]$  for  $w \in \mathcal{W}$  (else 0)

```

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

3.6.2 AWP transfer graph

AWP follows Do et al.^[10]: each transfer of value v at age Δt days before observation end T_{obs} contributes a decayed weight

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))}, \quad k = 0.01, \quad t_0 = 180 \text{ days.} \quad (3.4)$$

Edge weights aggregate over transfers:

$$w_T(u, v) = \sum_{e: u \rightarrow v} v_e \sigma(\Delta t_e). \quad (3.5)$$

Damping is again $d = 0.85$. Figure 3.2 plots $\sigma(\Delta t)$ over the relevant age range: recent transfers retain nearly full weight, while transfers older than the midpoint t_0 are smoothly down-weighted.

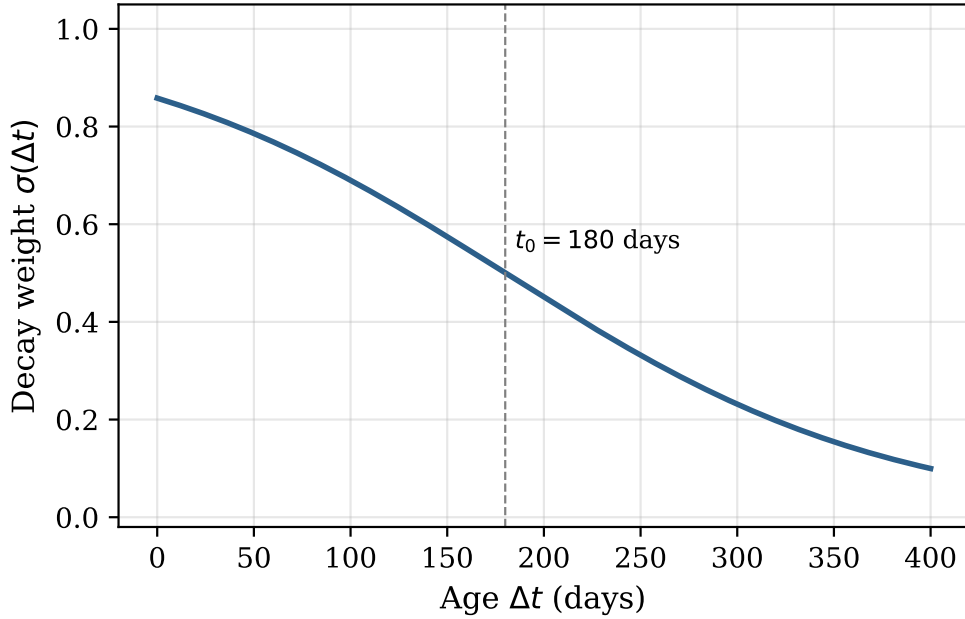


Figure 3.2: Logistic time-decay $\sigma(\Delta t)$ used by AWP ($k = 0.01$, $t_0 = 180$ days). Horizontal axis is event age in days before 2026-05-31 UTC; vertical axis is the multiplicative weight applied to transfer value.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

Algorithm 2 formalizes construction and scoring.

Algorithm 2 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$, seed wallets $\mathcal{W}$, observation end T_{obs} , parameters $k, t_0, d, \tau_{\text{tol}}, I_{\text{max}}$

Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$

```

1:  $E \leftarrow \emptyset$ 
2: for each transfer  $e = (u, v, v_e, t_e)$  in  $\mathcal{T}$  do
3:    $\Delta t \leftarrow (T_{\text{obs}} - t_e)$  in days
4:    $w_e \leftarrow v_e \cdot \sigma(\Delta t)$  with  $\sigma$  as in Equation (3.4)
5:   accumulate  $w_e$  on edge  $(u \rightarrow v)$ 
6: end for
7: remove non-positive edges; retain edges incident to  $\mathcal{W}$ 
8:  $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, d, \tau_{\text{tol}}, I_{\text{max}})$ 
9: return  $\mathbf{s}_{\text{AWP}}[w] \leftarrow \mathbf{s}[w]$  for  $w \in \mathcal{W}$  (else 0)

```

Because G_{approve} records durable authorizations and G_{transfer} records historical value flows, the two graphs differ in density and temporal sensitivity. Approvals are relatively rare, high-intent events; transfers are frequent and often mechanical. On the matched cohort, the approval subgraph is substantially sparser than the transfer subgraph (Chapter 4), which is the structural basis for the computational-efficiency hypothesis (H3): per-iteration PageRank cost scales with $|E|$, so fewer edges imply faster solves and lower memory for the adjacency structures in Equation (3.2).

- **Implementation notes shared by both methods:** Edge lists are stored as triples (`from_node, to_node, weight`) in memory. Duplicate edges are aggregated by summing weights before the solve. Non-positive weights are dropped. The solver builds a sparse compressed-row transition matrix, iterates Equation (3.1), and returns a dictionary from node identifier to score. Cohort scores are aligned to $\mathcal{W}$ with missing entries set to zero, then converted to dense ranks for correlation.

3.7 Supplementary GF-PR graph (extension)

GF-PR is an outcome-native reference built only from GMX `PositionDecrease` events. It is evaluated solely for supplementary question SRQ1 (construct-overlap diagnostic) and is not a pri-

mary peer to EndorseRank or AWP. The method exists to answer a methodological question: if a ranking is allowed to read the same PnL stream that defines trading-success proxies, how high can alignment rise? That ceiling clarifies how much of a social method’s trading-success τ might be attributed to genuine social signal versus residual correlation with outcomes.

The graph is a directed star with two synthetic nodes, POOL and SINK. For each close of wallet w with realized PnL x (field `basePnLUsd`):

- if the close is a liquidation, add weight $|x| + \varepsilon_{\text{liq}}$ on $w \rightarrow \text{SINK}$ with $\varepsilon_{\text{liq}} = 1.0$;
- else if $x > 0$, add weight x on $\text{POOL} \rightarrow w$;
- else if $x < 0$, add weight $|x| + \varepsilon_{\text{loss}}$ on $w \rightarrow \text{SINK}$ with $\varepsilon_{\text{loss}} = 0.1$.

The additives $\varepsilon_{\text{loss}}$ and ε_{liq} break ties when PnL is exactly zero after absolute value and ensure that liquidations contribute strictly more sink mass than ordinary losses of equal magnitude. There are no wallet-to-wallet edges: GF-PR does not encode social endorsement or transfer structure. PageRank with $d = 0.85$ propagates mass from POOL toward profitable wallets and from loss-making wallets toward SINK. High alignment with trading-success proxies is expected by construction and must be interpreted as construct overlap, not as superior social reputation ^[8].

3.8 Five-proxy evaluation design

Primary evaluation compares EndorseRank and AWP only. For each method M (EndorseRank or AWP) and each proxy P , the pipeline computes scores $\mathbf{s}_M$ and proxy values $\mathbf{p}_P$ on the matched cohort, then reports Spearman’s ρ and Kendall’s τ between the induced rankings (Section 3.10). All proxies are oriented so that higher values imply higher expected reputation before correlation, following the domain-alignment approach of Do et al.^[10]. Orientation is part of construct validity: without it, a method that correctly ranks safer traders could appear negatively aligned with a loss-magnitude proxy.

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

Five families structure the primary comparison. They were chosen to span (i) baselines from prior work, (ii) within-construct checks for each social graph, and (iii) outcome-oriented trading and risk measures that are not direct inputs to EndorseRank or AWP:

1. **Transfer** (Do et al. baseline): inbound transfer activity.
2. **Allowance**: inbound approval activity (wallet as spender).
3. **Inverse risk** (GMX PnL): loss avoidance and non-loss close behavior.
4. **Sybil-adjusted stability**: counterparty diversity and activity tenure.
5. **Trading success**: GMX close outcomes and realized gains.

Transfer and allowance families are *graph-adjacent*: they are computed from the same event classes that build G_{transfer} and G_{approve} , but they are simple local statistics (degree and value sums) rather than global PageRank scores. Strong alignment of AWP with transfer proxies, or of EndorseRank with allowance proxies, is therefore evidence of within-construct coherence, not of independent predictive power. Inverse-risk, Sybil-stability, and trading-success families are the more demanding tests for social reputation as a signal of trading behavior and account quality.

Family-level mean τ summarizes construct alignment for the EndorseRank–AWP comparison; the evaluation pipeline records which social method aligns more strongly per family (Appendix 8.1). A sixth liquidation family is computed in the codebase for completeness and is mentioned where relevant, but main-text claims use the five families above.

Supplementary evaluation includes GF-PR only for SRQ1, to diagnose construct overlap on trading-outcome proxies. Lending-protocol default labels remain out of scope. Packin and Lev-Aretz^[29] caution that decentralized credit scores can become opaque; the proxy design counters opacity by publishing explicit formulas (Section 3.9) and by separating social methods from outcome-native references.

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

3.9 Operational definitions of proxies

Let $\mathcal{W}$ be the matched cohort. For GMX-derived proxies, only wallets with at least $m = 3$ closes in the decoded event table are eligible; ineligible wallets receive zeros after left-join to $\mathcal{W}$. Notation below matches `proxy_metrics.py` in the open evaluation pipeline.

3.9.1 Transfer family

Let $\mathcal{T}_{\text{in}}(w)$ be inbound transfers to wallet w (any sender). Define

$$\text{in_degree}(w) = |\{ u : \exists \text{ transfer } u \rightarrow w \}|, \quad (3.6)$$

$$\text{in_value}(w) = \sum_{e: \cdot \rightarrow w} v_e. \quad (3.7)$$

These are the domain-intuition baselines of Do et al.^[10]: economically important addresses tend to receive transfers from more counterparties and in larger aggregate value. Unlike PageRank, `in_degree` and `in_value` are strictly local: they ignore the reputation of senders. Comparing AWP (global, time-decayed) to these local baselines tests whether propagation and recency weighting add ordering information beyond raw inflows.

3.9.2 Allowance family

Let $\mathcal{A}_{\text{in}}(w)$ be latest allowances in which w is the spender (owner $\rightarrow w$). Define

$$\text{in_approve_degree}(w) = |\{ u : a^*(u, w, \cdot) > 0 \}|, \quad (3.8)$$

$$\text{in_approve_value}(w) = \sum_{u, \text{token}} a^*(u, w, \text{token}). \quad (3.9)$$

These proxies measure endorsement received, the construct `EndorseRank` is designed to rank. Strong alignment of `EndorseRank` with this family is expected and is interpreted as within-

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

construct validity rather than independent trading skill. As with transfer baselines, the allowance proxies are local: they count and sum inbound approvals without propagating through multi-hop endorsement chains. EndorseRank’s incremental value, if any, appears when it aligns with non-allowance families while remaining coherent with allowance baselines.

3.9.3 Inverse-risk family

Let $\mathcal{C}(w)$ be the set of non-missing GMX closes for wallet w , with PnL x_c and liquidation flag $\ell_c \in \{0, 1\}$. A loss close is a non-liquidation close with $x_c < 0$. Define

$$\text{loss_avoidance}(w) = - \sum_{c \in \mathcal{C}(w)} \min(x_c, 0), \quad (3.10)$$

$$\text{non_loss_close_rate}(w) = 1 - \frac{|\{c \in \mathcal{C}(w) : \ell_c = 0 \wedge x_c < 0\}|}{|\mathcal{C}(w)|}, \quad (3.11)$$

$$\text{worst_close_pnl_score}(w) = - \min_{c \in \mathcal{C}(w)} x_c. \quad (3.12)$$

Equation (3.10) sums absolute loss magnitudes (higher is better after the sign flip). Equation (3.11) is the fraction of closes that are not ordinary losses. Equation (3.12) rewards wallets whose worst close is less severe (again after sign flip). Liquidation closes are included in the worst-close minimum but are excluded from the ordinary-loss count in Equation (3.11), so that forced liquidations are not double-counted as ordinary losses. Inverse-risk proxies are not inputs to EndorseRank or AWP; alignment here is evidence that social structure co-varies with risk-avoiding trading behavior.

3.9.4 Sybil-adjusted stability family

Self-transfers ($u = w$) are excluded from inbound counters. Let $N_{\text{tx}}(w)$ be the number of inbound transfers to w from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Let $t_{\min}(w)$ and $t_{\max}(w)$ be the first and last timestamps at which w appears as sender or recipient in the transfer table, and let $M(w)$ be the number of distinct calendar months of activity (inbound

months and full-span months, taking the maximum). Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \quad (3.13)$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad (\text{seconds to days}), \quad (3.14)$$

$$\text{active_months}(w) = M(w). \quad (3.15)$$

High counterparty ratio penalizes wallets that receive many transfers from few addresses (a simple Sybil/wash pattern). Tenure and active months capture persistence of on-chain activity within the observation window. These measures are imperfect Sybil defenses—coordinated adversaries can still diversify counterparties—but they operationalize the intuition that durable, multi-counterparty activity is more reputation-relevant than bursty self-dealing ^[29].

3.9.5 Trading-success family

A successful close is a non-liquidation close with $x_c > 0$. Define

$$\text{close_success_count}(w) = |\{c \in \mathcal{C}(w) : \ell_c = 0 \wedge x_c > 0\}|, \quad (3.16)$$

$$\text{realized_gain_proxy}(w) = \sum_{c \in \mathcal{C}(w)} \max(x_c, 0), \quad (3.17)$$

$$\text{close_success_rate}(w) = \frac{\text{close_success_count}(w)}{|\mathcal{C}(w)|}. \quad (3.18)$$

The primary definition of `realized_gain_proxy` is the raw sum of positive PnL. A size-weighted variant multiplies each positive PnL by $\max(|s_c|, 1)$, where s_c is the close’s `sizeDeltaUsd`, and is used only in robustness checks (Section 3.12). These proxies operationalize trading success on GMX V2 perpetual markets ^{[15],[32]}. They are outcome measures, not social-graph features: neither

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[15]GMX. (n.d.). GMX V2 documentation.

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

EndorseRank nor AWP reads `basePnlUsd` when building edges. Consequently, non-trivial alignment with trading success is the strongest form of domain evidence available in this design short of external credit labels. Count, sum, and rate capture complementary aspects of success—frequency, magnitude, and consistency—so family-level mean τ is not dominated by a single operationalization.

3.9.6 Liquidation family (ancillary)

For completeness, the pipeline also computes

$$\text{liquidation_free_rate}(w) = 1 - \frac{\sum_c \ell_c}{|\mathcal{C}(w)|}, \quad (3.19)$$

$$\text{zero_liquidation_flag}(w) = \mathbf{1} \left[\sum_c \ell_c = 0 \right], \quad (3.20)$$

$$\text{liquidation_free_closes}(w) = \sum_c (1 - \ell_c). \quad (3.21)$$

These metrics support sensitivity analyses and GF-PR diagnostics but are not part of the five-family primary claims.

3.10 Statistical tests

Reputation scores and validation proxies are continuous rank variables, not binary default labels. Following Do et al.^[10] and the construct-validity tradition of Cronbach and Meehl^[8], alignment is measured with rank correlations that are invariant to monotonic transforms of either margin.

- **Spearman’s ρ :** Let R_i and S_i be the dense ranks of score and proxy for wallet $i = 1, \dots, n$.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

Spearman’s coefficient is the Pearson correlation of these ranks:

$$\rho = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2}}. \quad (3.22)$$

Spearman captures monotonic agreement: if higher reputation consistently co-occurs with higher proxy values, ρ is positive even when the relationship is nonlinear in the raw scores.

- **Kendall’s τ :** Kendall’s coefficient counts concordant and discordant pairs. For all pairs $i < j$,

$$\tau = \frac{C - D}{\binom{n}{2}}, \quad (3.23)$$

where C is the number of pairs ordered the same way by both rankings and D is the number ordered differently (ties handled by the implementation’s standard tie correction). Kendall’s τ is more interpretable as a pairwise ordering probability and is less sensitive to large rank displacements than Spearman in some heavy-tailed settings.

- **Why both:** Spearman summarizes global monotonic association; Kendall emphasizes pairwise order fidelity. Reporting both follows Do et al.^[10] and allows readers to verify that family-level conclusions are not an artifact of a single rank statistic. In blockchain reputation settings, score distributions are often heavy-tailed: a few hubs receive large mass while many wallets share near-zero scores. Rank correlations remain well-defined in that regime, whereas Pearson correlation on raw scores would be dominated by hubs.
- **Family aggregation:** Family-level summaries use mean τ across proxies within a family. For example, the transfer-family mean is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. Method comparisons on the same cohort and proxies are relative, not absolute claims about universal reputation. Chapter 2 notes interpretive bands used in related PageRank studies ($|\tau| < 0.10$ negligible; 0.10–0.35 weak-to-moderate; ≥ 0.35 substantively meaningful); these bands guide discussion but do not replace the primary EndorseRank-versus-AWP contrast on identical inputs.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

- **Orientation and missingness:** Proxies are pre-oriented so that larger values indicate better reputation. Wallets absent from a proxy source receive zeros after alignment to $\mathcal{W}$, which is conservative for sparse activity and is applied identically to both methods. No wallet is dropped from one method but retained in the other: the correlation sample is always the full matched cohort for primary tables.
- **What is not tested:** The study does not claim causal identification (reputation does not “cause” trading success in an interventional sense), does not fit predictive classifiers with train/test splits, and does not report p -values as the primary decision criterion. The scientific target is comparative construct alignment and computational cost under a fixed, transparent protocol.

3.11 Computational benchmark (Claim C1)

Claim C1 concerns computational efficiency of EndorseRank versus AWP on identical inputs. The benchmark protocol isolates PageRank solve cost from one-time graph construction, because construction cost is dominated by I/O and preprocessing that would be amortized in a deployed scorer that updates incrementally, whereas the iterative solve is the recurring computational core of both methods.

- **Timed quantity:** For each method, edge lists are built once per cohort outside the timed loop. The timed section runs weighted PageRank only, with damping $d = 0.85$, tolerance $\tau_{\text{tol}} = 10^{-8}$, and at most $I_{\text{max}} = 300$ iterations. These numerical settings match the configuration used for alignment scores, so runtime and alignment refer to the same mathematical object.
- **Repetitions and metrics:** Each configuration is repeated for five independent runs to average out operating-system noise. Per run, the pipeline records wall-clock solve time (`time.perf_counter`), peak memory via `tracemalloc`, and iterations to convergence. Reported statistics are the mean runtime, runtime standard deviation, mean iterations, maximum peak memory across runs, and edge/node counts of the pre-built graph. Edge counts are reported because

they are the leading structural explanation for runtime differences under $O(|E|)$ per-iteration complexity ^[30].

- **Cohorts:** The primary benchmark uses the matched wallet cohort ($n = 5,521$). Scaling benchmarks use deterministic subsamples of the expanded wallet pool (Section 3.12). GF-PR runtime may be reported parenthetically in Chapter 4 as supplementary context only and is not part of Claim C1.
- **Fairness of comparison:** Both methods use the same solver, tolerance, iteration cap, damping (unless a robustness sweep varies d), and seed wallet set. Differences therefore reflect graph density and weight structure—allowance snapshots versus time-decayed transfers—rather than implementation asymmetry. Hardware and software versions are recorded in the evaluation manifest so that absolute seconds can be interpreted relative to the experimental machine; speedup ratios (AWP time divided by EndorseRank time) are the preferred comparative summary.

3.12 Scaling and robustness extensions

Beyond the matched cohort, the study evaluates runtime scaling on a larger address set and stress-tests alignment under parameter and sample perturbations. These extensions support H3 (efficiency at larger n) and the stability of family-level conclusions. They are not alternative primary samples: unless otherwise noted, alignment claims in Chapter 4 refer to the matched cohort, and scaling claims refer to the expanded pool.

3.12.1 Expanded wallet pool and runtime scaling

The expanded wallet pool $\mathcal{W}_{\text{pool}}$ unions the matched GMX cohort with supplemental active addresses derived from the reputation subgraph (and, when present, supplemental active-wallet parquet), yielding $N = |\mathcal{W}_{\text{pool}}| = 31,612$ addresses in this study. Supplemental addresses increase

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

graph size for timing experiments without requiring every address to carry GMX labels. Runtime benchmarks at intermediate sizes use deterministic SHA256 subsampling with seed string `benchmark-tier2-v1`: wallets are ordered by

`SHA256(seed || address)` with seed `benchmark-tier2-v1`

and the first n addresses form the subsample. The seed string is a technical identifier in configuration only; the dissertation narrative refers to the expanded wallet pool and robustness checks, not to numbered evaluation stages. Reported stages include $n = 10,000$ and the full pool $n = N = 31,612$, always on the same approval and transfer parquet. In-cohort scaling stages with $n \leq 5,521$ are reported in Appendix 8.1. Deterministic hashing avoids the non-reproducibility of pseudorandom draws across library versions.

3.12.2 Damping sensitivity

PageRank damping is swept over $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort, holding graphs and proxies fixed. The default $d = 0.85$ matches the classical web PageRank setting^[30] and Do et al.^[10]. Lower d increases teleportation and flattens scores; higher d emphasizes multi-hop structure and can slow convergence. The sweep tests whether allowance- and transfer-family alignment are artifacts of a single teleportation weight.

3.12.3 Top-token subgraph

A restricted subgraph retains only the top-20 ERC-20 tokens by combined transfer and allowance volume within the observation window. EndorseRank and AWP are recomputed on this subgraph to assess sensitivity to long-tail tokens and possible spam assets. If family-level conclusions reverse when low-volume tokens are removed, the primary result would be fragile to token composition; stability supports interpreting the full-token graphs as representative of major economic activity.

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

3.12.4 Sample-size sensitivity

Alignment statistics are recomputed on staged subsamples drawn from the expanded wallet pool using the same deterministic seed, to verify that family-level mean τ and method orderings are not driven by a single cohort size. Because rank correlations have sampling variability, modest movement in τ across sizes is expected; the robustness question is whether EndorseRank and AWP preserve their relative ordering on key families.

3.12.5 GMX proxy definition variants

For trading-success robustness, realized_gain_proxy is computed under both the primary raw-sum definition (Equation (3.17)) and a size-weighted definition that multiplies positive PnL by $\max(|s_c|, 1)$ for close size s_c (sizeDeltaUsd). Size-weighting up-weights large positions and can change the identity of top traders. Primary tables use the raw-sum definition; size-weighted results appear only in robustness tables.

3.13 Reproducibility

All empirical tables in Chapter 4 are produced by the open evaluation pipeline under A-Skill-Programs/margin_rank. After Phase 1–2 parquet artifacts are present, the dissertation evaluation is:

```
pythonscripts/run_dissertation_eval.py--real[--robustness]--export-latex
```

The `--real` flag selects cached BigQuery-derived inputs rather than synthetic fixtures; `--robustness` enables damping, top-token, sample-size, and GMX-proxy variant sweeps; `--export-latex` writes table fragments to 2-Dissertation-Draft/results/tables/. Configuration (observation window, damping, decay parameters, topic hashes, benchmark seeds) lives in `config/margin_config.yaml`. Fixture mode (`--fixtures`) supports offline continuous integration without cloud credentials.

Reproducible elements include: deterministic wallet subsampling via SHA256 seeds; fixed PageRank tolerance and iteration caps; explicit proxy formulas in Section 3.9; and logged extraction manifests. Peer reviewers can regenerate alignment and benchmark tables from the same parquet without re-querying BigQuery.

3.14 Sampling considerations and missingness

The matched cohort is a purposeful sample, not a random sample of all Arbitrum addresses. Eligibility requires GMX activity and reputation-subgraph connectivity, so results speak to *active perpetual traders with observable approval or transfer structure*. Idle wallets, pure holders, and addresses that interact only through non-ERC-20 mechanisms are out of scope by design.

Missingness arises in several forms. Tokens with unknown decimals may be dropped or defaulted during preprocessing; such decisions are logged in the pipeline. Approvals that never emit standard ERC-20 logs (non-standard tokens) are invisible to EndorseRank. Transfers that occur off-chain or through custodial systems never enter AWP. These missingness patterns bias both methods toward on-chain, standard-token activity—acceptable for an on-chain reputation study, but relevant when generalizing to the full user population of a protocol.

Supplemental addresses in the expanded wallet pool ($N = 31,612$) increase coverage for runtime tests but do not automatically carry GMX labels. Alignment claims therefore remain anchored to the matched cohort ($n = 5,521$), while scaling claims use the pool.

3.15 Implementation notes

PageRank is implemented with sparse adjacency lists and power iteration in Python, with optional NumPy vectorization for score updates. Graphs are built once per configuration; timed benchmarks exclude I/O and preprocessing. Randomness enters only through optional fixture generation; real-data evaluation is deterministic given parquet inputs and configuration seeds. Floating-point differences across machines are expected to be negligible relative to the reported τ precision (three

decimal places).

Figure generation (`generate_dissertation_figures.py`) reads the same artifacts and writes vector PDFs under `Figures/generated/`. Dissertations built from stale figures or tables should regenerate both before submission.

3.16 Ethics

This study uses exclusively publicly available blockchain data and does not intervene with human participants. Ethics clearance and data-handling procedures were approved by the institution prior to empirical data collection; the exemption or clearance confirmation is retained as part of the research record.

- **Privacy and anonymity:** Arbitrum One addresses are pseudonymous identifiers, not inherently linked to real-world identities ^[29]. Pseudonymity is not anonymity: determined adversaries can sometimes link addresses to persons through exchange deposits, public profiles, or clustering heuristics. This study deliberately refrains from such linkage. Collection is restricted to on-chain events—Approval, Transfer, and GMX PositionDecrease—with no scraping or integration of off-chain personal data such as IP addresses, exchange KYC records, social-media handles, or deanonymization graphs. No attempt was made to cluster addresses into real-world entities, to publish address-level rankings that could reasonably identify individuals, or to combine wallet histories with external identity providers. Reported results are cohort-level aggregates (correlations, runtimes, family means), not leaderboards of named users. Where examples are discussed in prose, they refer to abstract roles (e.g., “high-allowance spenders”) rather than concrete addresses.
- **Data minimization:** Only fields required for graph construction and proxies are retained in processed tables (addresses, token amounts, timestamps, GMX PnL fields, liquidation flags).

Raw logs are stored for auditability of the research pipeline but are not redistributed as a

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

deanonymization aid. Query filters are wallet-scoped where possible to avoid indiscriminate full-chain dumps beyond the study design.

- **Data usage and compliance:** All queries targeted public datasets under `bigquery-public-data.google.blockchain.arbitrum_one_us` in accordance with Google Cloud terms of service and rate limits. Dry-run byte estimates and per-query caps limit unnecessary scanning. No transactions were broadcast, no smart contracts were called in ways that could disrupt network operations, and no privileged or non-public indexer access was used.
- **Fairness and transparency:** Graph reputation systems can systematically undervalue new entrants, low-volume participants, or wallets that avoid public approvals ^[29]. EndorseRank and AWP were therefore audited for such structural biases: allowance graphs favor addresses that receive spending authority; transfer graphs favor addresses embedded in value-flow networks. Limitations—including cold-start effects, Sybil susceptibility of approval graphs, and construct overlap for GF-PR—are discussed in Chapter 5. All code paths, parameter choices, and evaluation metrics are documented in the open pipeline (Appendix 8.1) to enable peer review and independent replication.
- **Responsible disclosure and impact:** Although EndorseRank may inform risk assessment in DeFi contexts, it is not financial advice, credit advice, or a consumer reporting product. Human oversight remains necessary for lending, trading, and compliance decisions. Potential adversarial manipulation (for example, Sybil attacks on G_{approve} or wash transfers on G_{transfer}) is acknowledged and discussed in Chapter 5. The study reports aggregated cohort-level findings rather than wallet-level rankings intended for targeting individual users. Any future operational deployment would require additional governance, appeal mechanisms, and fairness monitoring beyond the scope of this dissertation.
- **Summary of methodological commitments:** The design fixes the chain, window, cohort rules, graph formulas, proxy definitions, and statistical tests before interpreting results. Pri-

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

mary claims compare only EndorseRank and AWP on the matched wallet cohort; the expanded pool and robustness sweeps test efficiency and stability; GF-PR bounds outcome-native alignment for SRQ1. Chapter 4 reports the empirical outcomes of this protocol.

4 Implementation and Empirical Results

This chapter reports the empirical evaluation of EndorseRank against Adaptive Weighted PageRank (AWP) on Arbitrum One. The analysis proceeds from cohort construction and descriptive distributions, through computational benchmarks and robustness checks, to social-method alignment across five proxy families, inter-method rank divergence, and a supplementary GainFlow PageRank (GF-PR) ceiling analysis. Primary claims C1–C4 address the EndorseRank–AWP comparison; supplementary claim SC1 interprets GF-PR as an outcome-native diagnostic rather than a deployable social reputation method.

All primary results use an identical matched cohort of $n = 5,521$ wallets that satisfy both a GMX trading-activity filter and non-zero connectivity in the reputation subgraphs. Runtime scaling is evaluated on an expanded wallet pool of $N = 31,612$ addresses. Robustness checks vary PageRank damping, restrict the reputation graph to high-volume tokens, and recompute alignment at staged subsample sizes drawn from the expanded pool. Methodological definitions appear in Chapter 3; the open evaluation pipeline is documented in Appendix 8.1. Claims are summarized in Section 4.8 and mapped to hypotheses and research questions in Table 1.3.

4.1 Dataset

The matched cohort comprises $n = 5,521$ wallets with at least three GMX V2 PositionDecrease closes and non-zero reputation-subgraph connectivity, drawn from six months of Arbitrum One BigQuery logs spanning 2025-12-01 through 2026-05-31. EndorseRank scores are computed on the latest-allowance endorsement graph G_{approve} ; AWP scores are computed on the time-decayed transfer graph G_{transfer} . Five proxy families—transfer, allowance, inverse risk, Sybil-adjusted stability, and trading success—are evaluated against both social methods. GF-PR scores are computed

from the same cohort for supplementary analysis only (Section 4.7).

Table 4.1 summarizes cohort size, event counts, and subgraph edge counts. The endorsement graph is substantially sparser than the transfer graph (14,727 vs. 137,087 edges), a structural difference that underpins both the runtime advantage reported in Section 4.2 and the construct-specific alignment patterns in Sections 4.4–4.5.

Table 4.1: Dataset characteristics for the matched cohort ($n = 5,521$ wallets with ≥ 3 GMX closes and non-zero reputation-subgraph connectivity).

Statistic	Value
Matched wallets	5,521
GMX PositionDecrease events (decoded)	365,488
EndorseRank edges (subgraph)	14,727
AWP edges (subgraph)	137,087
Observation window	2025-12-01 – 2026-05-31
Min. GMX closes per wallet	3

The edge-count asymmetry is not an artifact of incomplete extraction: both graphs are built from the same wallet seed set and the same observation window. EndorseRank retains only the latest decimal-adjusted allowance per owner–spender pair, so repeated approvals collapse to a single directed edge. AWP aggregates transfer history with logistic time-decay, producing many more directed edges among the same wallets. Consequently, differences in scores and alignment are attributable to ranking logic and graph semantics, not to inconsistent sampling.

The matched-cohort design deliberately intersects two filters. The GMX activity filter (≥ 3 closes) ensures that trading-success and inverse-risk proxies are defined for every wallet, avoiding undefined rates and unstable count statistics for one-off traders. The reputation-subgraph connectivity filter ensures that both EndorseRank and AWP produce non-degenerate scores: wallets with no approval edges and no transfer edges cannot be ranked meaningfully by either social method. The intersection yields $n = 5,521$ wallets and 365,488 decoded PositionDecrease events over the six-month window. This sample is large enough for stable Kendall τ estimates at the family level, yet small enough that full PageRank solves remain interactive on commodity hardware—a practical

advantage when iterating on evaluation design.

The observation window (2025-12-01 – 2026-05-31) is long enough to capture multiple market regimes within a single perpetual-futures venue, but short enough that latest-allowance snapshots remain contemporaneous with transfer histories used by AWP. Longer windows would increase AWP’s historical edge accumulation and could further widen the runtime gap; shorter windows would reduce proxy stability for tenure and active-month measures. The chosen window therefore balances recency, activity volume, and comparability between snapshot-based and history-based graphs.

4.1.1 GMX trading activity in the matched cohort

Before comparing reputation rankings, it is useful to characterize the trading-outcome distributions that define the inverse-risk and trading-success proxy families. Figure 4.1 shows the distribution of GMX close counts per wallet in the matched cohort. The minimum of three closes is enforced by construction; the right tail indicates a minority of highly active traders whose close counts substantially exceed the cohort median. This heterogeneity matters for interpretation: proxies based on close-success *counts* can favor high-activity wallets, whereas rate-based proxies normalize by activity and may yield weaker rank correlations when success rates concentrate near the center of the distribution.

Figure 4.2 displays the distribution of realized profit-and-loss (PnL) summaries used to construct inverse-risk and trading-success proxies. The mass around zero and the presence of both profitable and loss-making tails indicate that the cohort is not dominated by uniformly successful traders. Social reputation methods that align only weakly with PnL-derived proxies are therefore not necessarily failing a validation test; they may be measuring a construct—authorization or transfer centrality—that is only partially related to leveraged trading outcomes over a six-month window. Figure 4.3 shows close success rates. Relative to count-based proxies, success rates compress much of the activity-driven variation and concentrate mass in a narrower band. Weak alignment of social methods with success-rate proxies (Section 4.5) should be read in this light: when many

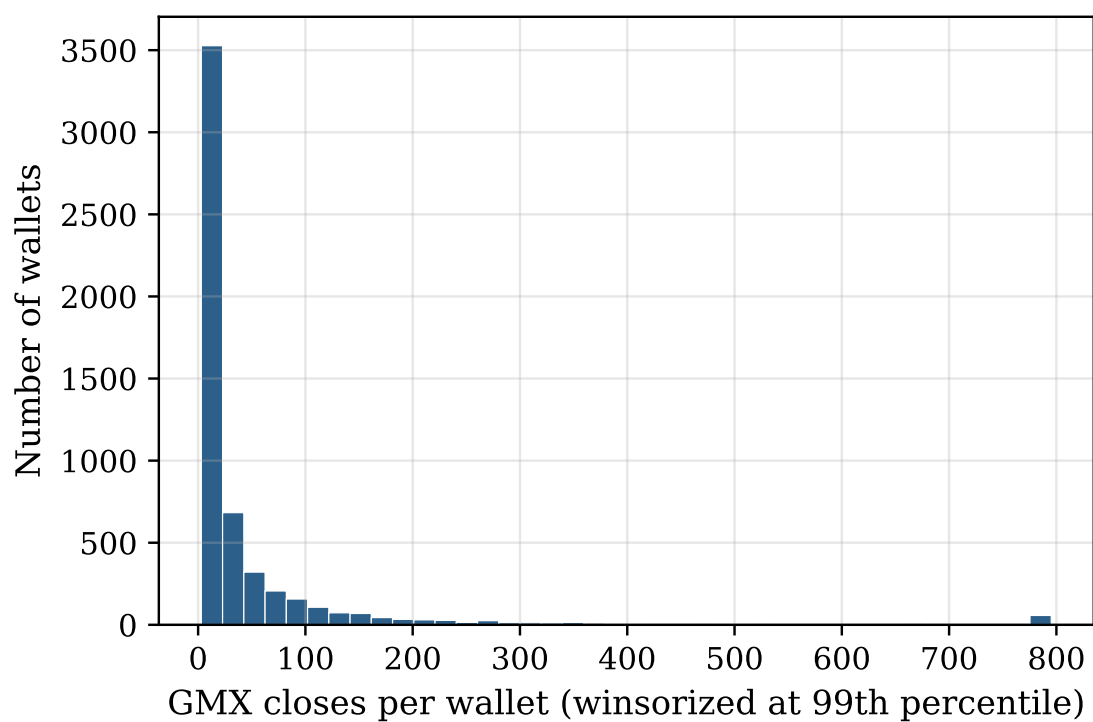


Figure 4.1: Distribution of GMX V2 PositionDecrease close counts per wallet in the matched cohort ($n = 5,521$). The cohort filter requires at least three closes; the long right tail reflects a minority of highly active traders.

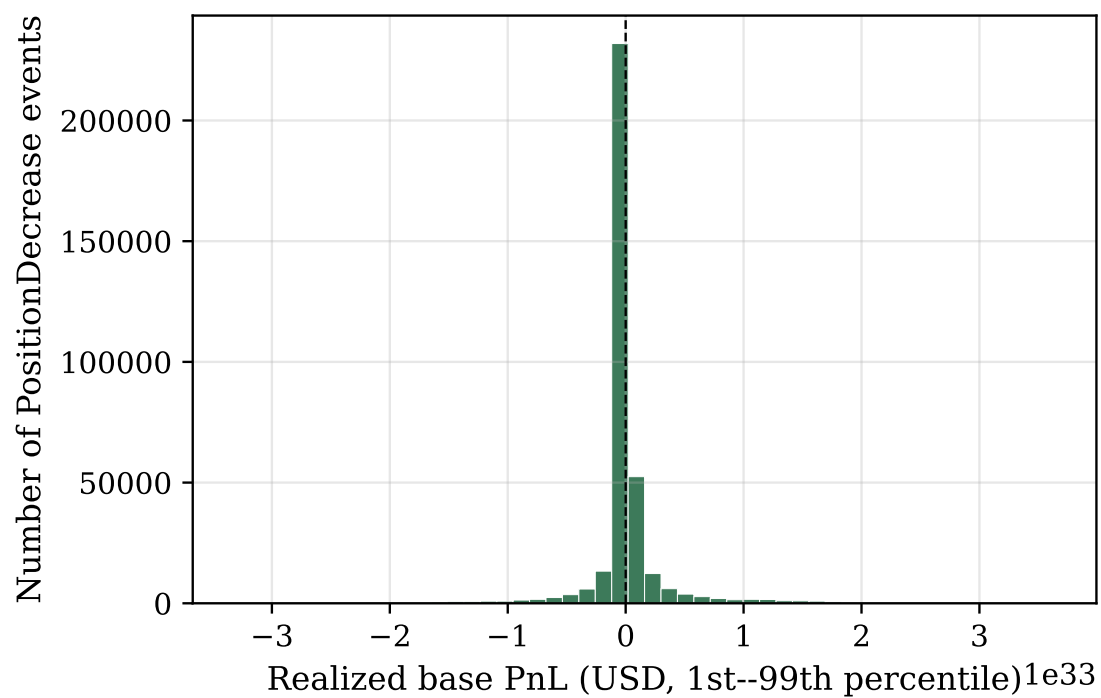


Figure 4.2: Distribution of GMX realized PnL summaries in the matched cohort. Heterogeneous profit and loss outcomes motivate treating trading-success proxies as one validation axis among several, rather than as a sole ground truth for social reputation.

wallets share similar rates, rank correlation has limited room to rise even if reputation scores are informative on other axes.

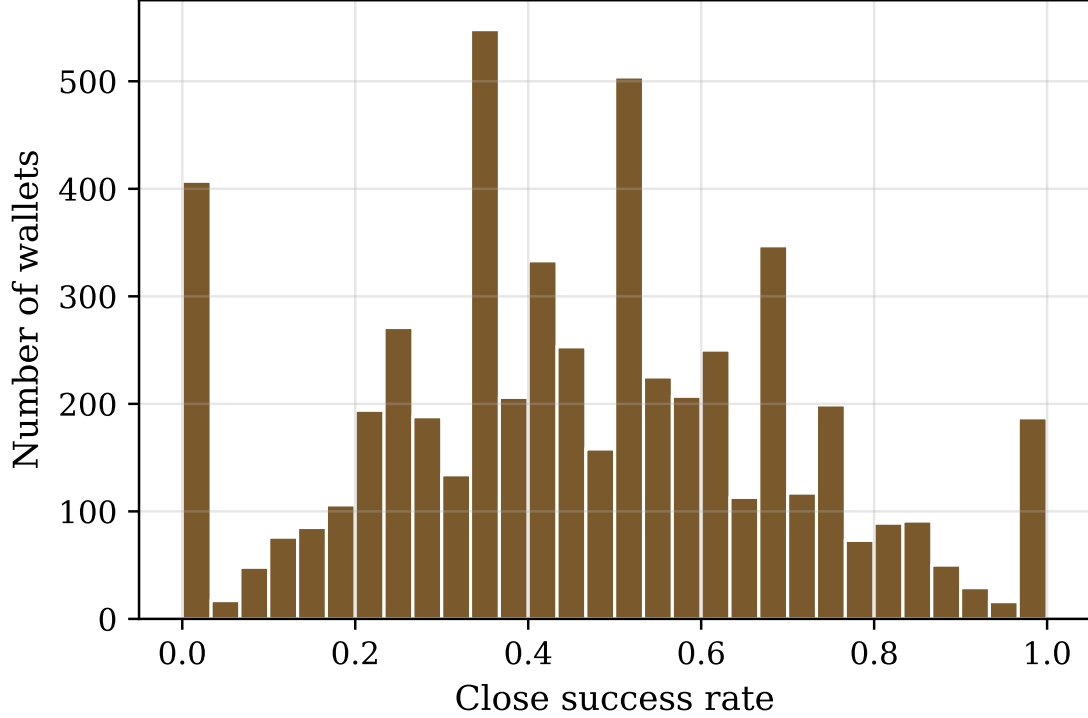


Figure 4.3: Distribution of GMX close success rates in the matched cohort. Concentration of rates reduces the discriminative power of rate-based trading-success proxies relative to count- or value-based proxies.

Taken together, the descriptive figures establish three interpretive constraints for later sections. First, the matched cohort is active enough for stable rank statistics ($n = 5,521$; 365,488 decoded closes) but heterogeneous in trading intensity and outcomes. Second, endorsement and transfer graphs differ by nearly an order of magnitude in edge count, so efficiency and alignment claims must be evaluated jointly rather than in isolation. Third, trading-outcome proxies are informative validation axes, not interchangeable substitutes for social-graph constructs; Section 4.7 returns to this point when GF-PR inflates trading-family alignment through construct overlap.

A fourth constraint follows from the joint reading of Figures 4.1–4.3. Count-based and value-based proxies inherit the heavy-tailed activity structure of perpetual trading, so social methods that

correlate with activity—especially transfer activity—can appear to predict trading success partly because active wallets trade more often and accumulate more closes. Rate-based proxies partially correct for this, but they also compress signal. Throughout Section 4.5, family means average across count-, value-, and rate-style proxies; readers should therefore inspect per-proxy rows when adjudicating whether a method aligns with skill-like outcomes or with activity volume.

4.2 EndorseRank vs. AWP benchmark (Claim C1)

Claim C1 asserts that EndorseRank is computationally cheaper than AWP on identical inputs. Table 4.2 reports PageRank solve time, peak memory, iteration count, and edge count on the full matched cohort ($n = 5,521$). Timing excludes one-time graph construction and averages three runs at convergence tolerance 10^{-8} .¹

Table 4.2: Computational benchmark on matched wallet sample ($n = 5521$).

Method	Runtime (s)	Peak mem. (MB)	Iters	Edges
EndorseRank	2.105	4.6	37.0	14727
AWP	18.711	38.2	100.0	137087
GF-PR	1.461	3.2	89.0	10338

EndorseRank completes in 2.105 s versus 18.711 s for AWP, a wall-clock speedup of approximately $8.9\times$. Peak memory follows the same pattern (4.6 MB vs. 38.2 MB). The dominant driver is graph sparsity: EndorseRank operates on 14,727 edges, whereas AWP operates on 137,087 edges—roughly $9.3\times$ more edges. Because PageRank’s per-iteration cost scales with $|E|$, the sparser endorsement graph reduces both work per iteration and the number of iterations required for convergence (EndorseRank ≈ 37 iterations; AWP ≈ 100). The iteration gap is consistent with a better-conditioned, lower-density adjacency structure under latest-allowance semantics: allowance overrides replace historical accumulation, so the operator does not encode long-tailed transfer histories that slow mixing.

¹Excluding graph construction isolates the recurring PageRank solve cost that would dominate incremental score refresh in a deployed scorer.

GF-PR runtime (1.461 s) is reported in the same table for context only. GF-PR’s star graph has 10,338 edges and converges in 89 iterations; it is not a peer social method and is not used to support Claim C1. The primary efficiency comparison is EndorseRank versus AWP on wallet-to-wallet social graphs.

Memory footprints reinforce the same structural story. EndorseRank’s peak memory (4.6 MB) is an order of magnitude below AWP’s (38.2 MB) under identical profiling. Although absolute megabytes are modest at $n = 5,521$, the ratio matters for deployment at larger n : transfer-graph adjacency, weight vectors, and residual buffers all scale with $|E|$. A method that stores and iterates over roughly $9\times$ fewer edges inherits proportional savings in cache pressure and allocator traffic, not only in arithmetic operations. Iteration counts (37 vs. 100) compound this effect: EndorseRank performs fewer passes over a smaller edge list.

From a systems perspective, Claim C1 is therefore multi-dimensional. Wall-clock time is the headline metric for interactive refresh; memory and iterations are the mechanisms that make the headline durable as cohorts grow. A method that were faster only because of an implementation accident—for example, an unfairly optimized sparse kernel applied to one graph but not the other—would be less persuasive. Here both methods use the same PageRank solver, tolerance, and damping default; only the edge lists differ. The efficiency gap is a property of the endorsement representation.

4.2.1 Runtime scaling on the expanded wallet pool

The runtime advantage is not merely a constant-factor curiosity on a single cohort size. If EndorseRank’s edge set remains substantially smaller than AWP’s as the wallet pool grows, the speedup should persist under staged scaling. Table 4.3 reports runtime scaling on an expanded wallet pool of $N = 31,612$ addresses: the GMX-qualified matched cohort unioned with supplemental active wallets drawn from the reputation subgraph. Deterministic SHA256 subsampling yields staged cohorts at $n = 10,000$ and at full pool size. In-cohort scaling for $n \leq 5,521$ is reported in Appendix 8.1 (Table 8.1).

At $n = 10,000$, EndorseRank runs in 1.18 s versus 12.28 s for AWP (speedup $\approx 10.4\times$). At full

Table 4.3: Runtime scaling on expanded wallet pool ($N = 31612$; SHA256 seed benchmark-tie r2-v1). Stages subsample the extraction wallet set; ER speedup = AWP/ER runtime ratio.

n	ER (s)	AWP (s)	Speedup	ER $ E $	AWP $ E $
10000	1.178	12.277	$10.4\times$	7067	72158
31612	2.286	21.402	$9.4\times$	14727	137087

pool size ($N = 31,612$), EndorseRank runs in 2.29 s versus 21.40 s for AWP (speedup $\approx 9.4\times$). Edge counts at the full pool remain 14,727 for EndorseRank and 137,087 for AWP—identical to the matched-cohort edge totals in Table 4.1—because the reputation subgraph edges among the evaluation wallets are fixed by the extraction; expanding the node set without adding new edges among scored wallets does not inflate $|E|$ for either method in this configuration. The speedup therefore remains in the 9–10 $\times$ band across both staged sizes.

Figure 4.4 visualizes the scaling relationship. The curves confirm that AWP’s wall-clock cost grows more steeply with n , while EndorseRank remains in a low-second regime even at $N = 31,612$. For protocols that refresh reputation scores frequently—for example, after each batch of new approvals or transfers—an order-of-magnitude reduction in PageRank solve time is operationally material: it lowers the cost of continuous monitoring and makes more frequent re-ranking feasible under the same compute budget.

Two caveats qualify the efficiency claim. First, the benchmark times PageRank solve on pre-built edge lists; end-to-end pipelines that include BigQuery extraction and edge construction incur additional costs shared or partially shared across methods. Second, memory footprints reported here are modest because the cohort is wallet-filtered; full-chain transfer graphs would increase AWP’s absolute cost further and would likely widen, not narrow, the relative gap. Within the scope of this dissertation—matched wallets with reputation-subgraph connectivity—Claim C1 is supported at both the matched cohort and the expanded pool.

A third caveat concerns what “faster” does and does not buy. EndorseRank’s speedup does not automatically imply superior risk discrimination. Efficiency is valuable only insofar as the resulting scores remain credible on the constructs a protocol cares about. Section 4.4 and Claim C4

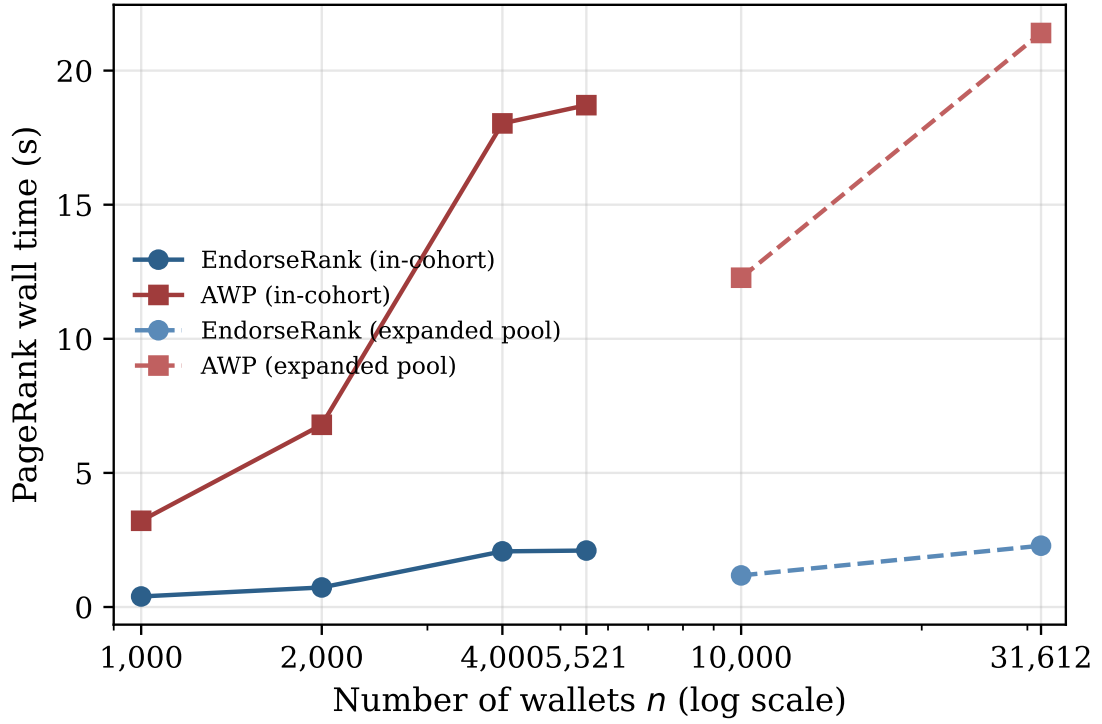


Figure 4.4: PageRank wall-clock runtime versus cohort size (log x -axis). Solid lines: in-cohort scaling on the matched GMX sample ($n \leq 5,521$). Dashed lines: expanded wallet pool ($N = 31,612$). EndorseRank remains approximately 9–10 $\times$ faster than AWP at $n = 10,000$ ($\approx 10.4\times$) and at full pool size ($\approx 9.4\times$).

therefore pair the runtime results with alignment results: EndorseRank is not merely cheaper; it is cheaper while preserving strong allowance alignment and moderate transfer alignment. Protocols that need transfer-hub detection above all else may still prefer AWP and pay the runtime premium; protocols that need authorization-aware scores at low latency have a clear efficiency rationale for EndorseRank.

4.3 Robustness checks

Efficiency and alignment results are useful only if they are not artifacts of a single damping choice, a single token mix, or a single sample size. This section reports three robustness checks: PageRank damping sensitivity, restriction to the top-20 ERC-20 token subgraph, and sample-size sensitivity on the expanded wallet pool.

4.3.1 Damping sensitivity

Table 4.4 reports mean Kendall τ for allowance and transfer families under damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort, together with corresponding runtimes.

Table 4.4: Robustness: mean Kendall τ under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on matched alignment cohort ($n = 5521$).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)
0.75	0.355	-0.031	0.333	0.531	3.436	21.797
0.85	0.355	-0.030	0.333	0.534	2.913	21.383
0.95	0.355	-0.030	0.333	0.536	2.320	21.077

EndorseRank allowance mean τ is stable at approximately 0.355 across all three damping values. Transfer-family τ for EndorseRank is likewise unchanged at 0.333. AWP transfer τ varies only from 0.531 to 0.536 (at most ≈ 0.005), and AWP allowance τ remains near zero (-0.031 to -0.030). The construct-specific pattern—EndorseRank leading on allowance, AWP leading on transfer—is therefore not an artifact of the conventional $d = 0.85$ setting.

Figure 4.5 plots these relationships. Flat allowance and transfer curves for EndorseRank indicate that the endorsement operator’s ranking is insensitive to moderate changes in the teleportation probability. Runtime decreases slightly as d increases for both methods, consistent with faster mixing when less mass is retained on the graph operator, but the relative speedup remains large throughout the sweep.

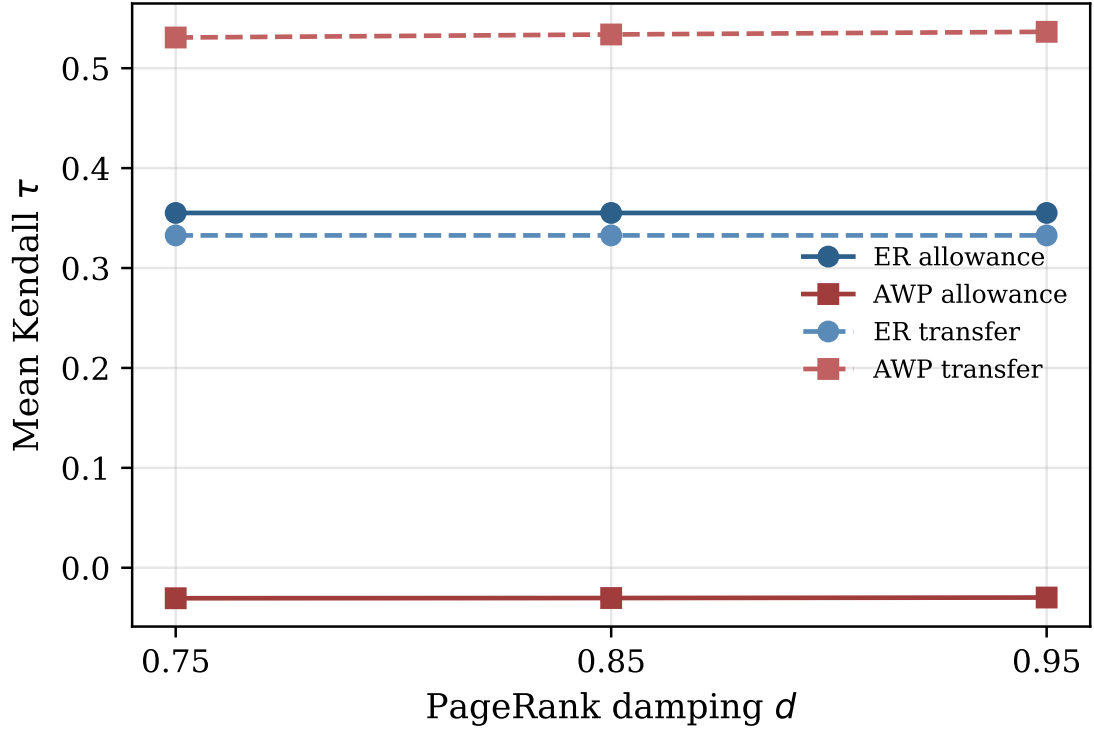


Figure 4.5: Damping sensitivity of mean Kendall τ for allowance and transfer families ($d \in \{0.75, 0.85, 0.95\}$). EndorseRank allowance alignment remains ≈ 0.355 across the sweep; AWP transfer alignment varies by at most ≈ 0.005 .

The damping result has a methodological implication. Because allowance edges encode a latest snapshot rather than a long historical path structure, changing how much mass teleports does not reorder wallets enough to alter family-level alignment. Protocols adopting EndorseRank can therefore treat $d = 0.85$ as a stable default without extensive hyperparameter search for the allowance construct, at least within the range examined here.

4.3.2 Top-token subgraph

Table 4.5 reports family-level mean τ when reputation graphs are restricted to the top-20 ERC-20 tokens by transfer and allowance volume.

Table 4.5: Robustness: mean Kendall τ on top-20 ERC-20 token subgraph (matched cohort).

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX
EndorseRank	0.331	0.355	0.092	0.036	0.234	0.098
AWP	0.532	-0.031	0.119	0.108	0.283	0.185

Relative to the full-token matched-cohort results in Table 4.7, family-level means change only marginally. EndorseRank allowance remains 0.355; transfer is 0.331 (vs. 0.333 on the full graph). AWP transfer is 0.532 (vs. 0.534), and AWP allowance remains near zero (-0.031). Trading-success and Sybil-adjusted stability means are likewise within a few thousandths of the full-graph values. The primary alignment pattern is therefore not driven by long-tail tokens with sparse, noisy edges. High-volume tokens—the economically salient core of ERC-20 activity—suffice to reproduce the EndorseRank–AWP contrast.

4.3.3 Sample-size sensitivity

Table 4.6 summarizes allowance and transfer τ at staged subsample sizes drawn from the expanded wallet pool ($N = 31,612$).

Table 4.6: Robustness: alignment and runtime vs. subsample size from wallet pool ($N = 31612$; seed `benchmark-tier2-v1`).

n	ER all.	AWP all.	ER tr.	AWP tr.	Speedup
10000	0.555	0.016	0.429	0.789	8.960
31612	0.533	0.021	0.443	0.797	9.930

At $n = 10,000$ and $N = 31,612$, EndorseRank continues to dominate on allowance and AWP on transfer, with EndorseRank speedups of approximately $9.0\times$ and $9.9\times$, respectively. Absolute τ values differ from the matched-cohort figures because the expanded pool includes supplemental

active wallets that are not GMX-matched; proxies and score supports therefore change with the sample definition. The relevant robustness claim is ordinal and comparative: the construct-specific winners and the runtime advantage persist under pool expansion and staged subsampling. In-cohort scaling detail for the GMX-matched sample alone appears in Appendix 8.1.

Collectively, the robustness checks support reading Claims C1–C4 as stable properties of the EndorseRank–AWP comparison on this chain and window, rather than as fragile point estimates at a single configuration. Robustness does not mean invariance of every decimal place. Absolute τ values shift when the sample definition changes from the GMX-matched cohort to the expanded pool, and runtimes shift with damping and n . What remains stable is the comparative structure: EndorseRank is faster; EndorseRank wins on allowance; AWP wins on transfer; allowance alignment under EndorseRank is insensitive to damping in $\{0.75, 0.85, 0.95\}$. That comparative structure is what the claims assert, and it is what the robustness section is designed to protect.

4.4 Social-method alignment: EndorseRank vs. AWP (Claims C2–C3)

Claims C2 and C3 address whether EndorseRank and AWP achieve credible domain alignment and whether winners are construct-specific. Table 4.7 reports mean Kendall τ by proxy family for EndorseRank, AWP, and the supplementary GF-PR reference. Rows 1–2 constitute the primary social-method comparison; row 3 is interpreted only in Section 4.7.

Table 4.7: Mean Kendall τ by proxy family across 3 PageRank methods ($n = 5521$). Bold = column maximum. Runtime = mean PageRank wall time (s, 3 repeats). EndorseRank and AWP are social reputation graphs; GF-PR is an outcome-native GMX PnL star baseline (construct overlap on GMX proxies).

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX	Runtime (s)
EndorseRank	0.333	0.355	0.093	0.035	0.233	0.096	2.105
AWP	0.534	-0.030	0.113	0.107	0.286	0.179	18.711
GF-PR	0.246	-0.041	0.132	0.376	0.093	0.583	1.461

Among social methods, EndorseRank leads on allowance ($\tau = 0.355$), while AWP leads on transfer ($\tau = 0.534$), Sybil-adjusted stability ($\tau = 0.286$), and trading success ($\tau = 0.179$). EndorseRank’s

transfer alignment is moderate ($\tau = 0.333$)—credible as a secondary axis, but below AWP. On inverse-risk proxies, both social methods are weak (EndorseRank 0.035; AWP 0.107), indicating that neither allowance nor transfer centrality strongly tracks loss-avoidance rankings on this cohort. GF-PR’s high inverse-risk (0.376) and trading-success (0.583) means reflect construct overlap with GMX PnL inputs and are not evidence of superior social reputation.

Figure 4.6 provides a compact visual summary of family-level alignment. The heatmap makes the construct-specific pattern immediate: EndorseRank’s strongest cell is allowance; AWP’s strongest social-method cells are transfer and, secondarily, Sybil-adjusted stability and trading success; GF-PR dominates outcome-native families. The visual separation between social-method cells and GF-PR outcome cells is the empirical signature of Claim C3 and the caution embodied in SC1.

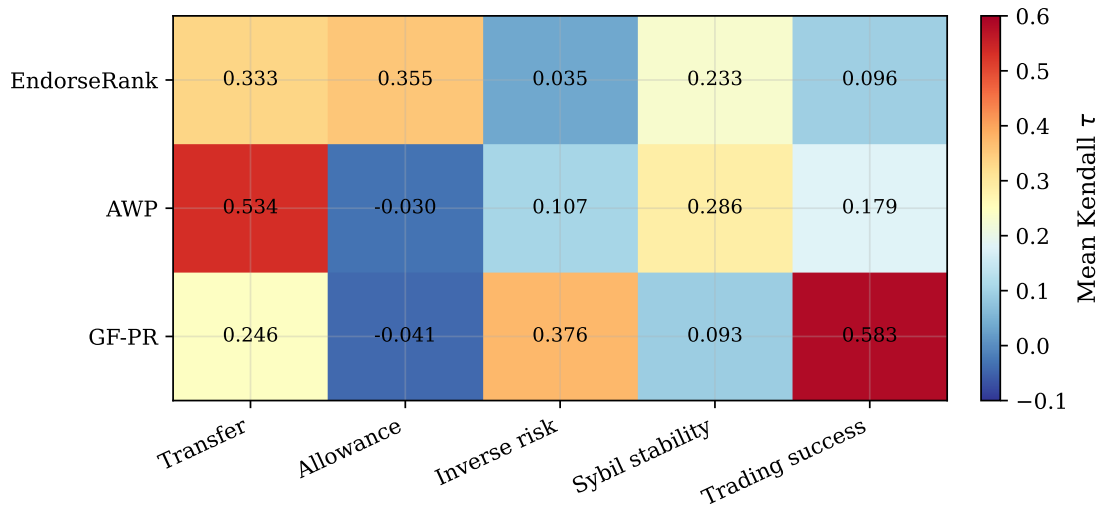


Figure 4.6: Family-level mean Kendall τ heatmap for EndorseRank, AWP, and supplementary GF-PR on the matched cohort ($n = 5,521$). EndorseRank peaks on allowance; AWP peaks on transfer among social methods; GF-PR peaks on trading-outcome families via construct overlap.

Claim C2 is supported insofar as both social methods achieve non-negligible alignment on at least one primary construct: EndorseRank on allowance ($\tau \approx 0.35$, at the lower edge of the substantively meaningful band defined in Chapter 2), and AWP on transfer ($\tau = 0.534$). Claim C3 is supported by the crossed pattern of family winners. No single social method dominates all families; alignment is construct-specific, as predicted when endorsement and transfer encode different

wallet relationships.

Reading Table 4.7 requires care on two points. First, family means average heterogeneous proxies; a method can lead on a family mean while trailing on an individual proxy, or vice versa. Section 4.5 therefore reports per-proxy ρ and τ so that readers can audit the means. Second, primary-construct correlations (EndorseRank with allowance; AWP with transfer) are necessary but not sufficient for a reputation method’s usefulness. They confirm that PageRank recovers the graph signal it was given. External usefulness depends on secondary axes—whether endorsement rankings also relate to transfer activity, trading outcomes, or stability—and on whether those secondary relationships are strong enough for a deployment context. EndorseRank’s moderate transfer $\tau = 0.333$ is encouraging as a secondary axis; its weak trading-success $\tau = 0.096$ is a limitation for protocols that hoped allowance endorsement would double as a trading-skill score.

The exported summary also includes a liquidation-related column and the GF-PR row. Liquidation-family means are low for both social methods and are not the focus of Claims C2–C3; they are retained for completeness with the evaluation pipeline. GF-PR’s outcome-family dominance is discussed only in Section 4.7. The primary narrative of this section is the EndorseRank–AWP contrast across transfer, allowance, inverse risk, Sybil-adjusted stability, and trading success.

Per-proxy detail for each family appears in Section 4.5. Inter-method divergence between EndorseRank and AWP score vectors is quantified in Section 4.6.

4.5 Proxy family detail

The following subsections report Spearman ρ and Kendall τ for each proxy within family. Each family is interpreted first for the primary EndorseRank–AWP comparison; supplementary GF-PR notes appear where they clarify construct overlap or ceiling effects. All proxies are oriented so that higher values imply higher expected reputation before correlation, following Do et al.^[10] and the operational definitions in Chapter 3.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

4.5.1 Transfer family (Do et al. baseline)

Table 4.8 reports alignment with transfer in-degree and inbound transfer value—the baseline validation axis in Do et al.^[10].

Table 4.8: Domain alignment with transfer proxies (Do et al. 2023 baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings.

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
In-degree	0.422	0.344
In-value (sum inbound)	0.398	0.321
<i>AWP</i>		
In-degree	0.791	0.617
In-value (sum inbound)	0.610	0.450
<i>GF-PR</i>		
In-degree	0.396	0.275
In-value (sum inbound)	0.314	0.216

AWP achieves the highest social-method alignment on both transfer proxies: in-degree $\tau = 0.617$ ($\rho = 0.791$) and in-value $\tau = 0.450$ ($\rho = 0.610$), for a family mean of 0.534. This result is expected by construction: AWP ranks on the time-decayed transfer graph, so correlation with transfer centrality is a primary-construct check rather than an independent external validation. The magnitude is nonetheless informative. AWP does not collapse to a trivial identity with in-degree; logistic decay and value weighting produce rankings that agree strongly but not perfectly with simple degree and value summaries.

EndorseRank shows moderate secondary-axis alignment (in-degree $\tau = 0.344$; in-value $\tau = 0.321$; family mean 0.333). Wallets that receive substantial transfer activity are somewhat more likely to appear as endorsed spenders, but the relationship is far from deterministic. Allowance delegation and transfer receipt are related economic behaviors—active protocols and market makers tend to attract both—yet they are not interchangeable. The moderate EndorseRank–transfer τ therefore supports Claim C2 (credible secondary alignment) without undermining Claim C3 (AWP remains the transfer-family winner among social methods).

GF-PR is weak on transfer proxies (family mean 0.246), as expected for a method with no wallet-to-wallet transfer edges. Any residual correlation likely reflects indirect association between profitable GMX traders and transfer activity, not transfer-graph semantics.

For RQ4, the transfer family is the Do et al. baseline validation frame. Both social methods clear a non-negligible bar on this frame, but AWP’s lead is large. If a protocol’s notion of reputation is “wallets that receive economically meaningful transfers,” AWP is the appropriate social method on this evidence. EndorseRank remains relevant as a secondary check: moderate transfer alignment suggests that endorsed spenders are not an entirely disjoint population from transfer recipients, so allowance-based scores are not orthogonal to economic flow—only imperfectly aligned with it.

4.5.2 Allowance family

Table 4.9 reports alignment with spender-side in-approve degree and in-approve value—EndorseRank’s primary construct.

Table 4.9: Domain alignment with allowance proxies (spender-side in-approve degree and value).

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
In-approve degree	0.360	0.356
In-approve value (sum)	0.360	0.354
<i>AWP</i>		
In-approve degree	-0.037	-0.030
In-approve value (sum)	-0.037	-0.030
<i>GF-PR</i>		
In-approve degree	-0.051	-0.042
In-approve value (sum)	-0.050	-0.041

EndorseRank is strongly aligned with both allowance proxies (in-approve degree $\tau = 0.356$; in-approve value $\tau = 0.354$; family mean 0.355). As with AWP on transfer, this is a primary-construct check: PageRank on G_{approve} should correlate with spender-side approval centrality. The near-equality of degree and value correlations suggests that, on this cohort, being approved by many owners and being approved for large aggregate value convey similar ranking information under

latest-allowance semantics.

AWP alignment on allowance is near zero and slightly negative ($\tau = -0.030$ for both proxies). Transfer centrality does not identify wallets that receive spending authorization. This dissociation is central to the dissertation’s contribution: if transfer-based reputation were a sufficient proxy for endorsement, AWP would show at least moderate allowance alignment. It does not. Protocols that care about explicit spending authorization—for example, which addresses are trusted to move tokens on behalf of others—cannot substitute AWP for an allowance-aware method on this evidence. GF-PR is likewise near zero on allowance (family mean ≈ -0.041), confirming that outcome-native PnL flow does not encode endorsement structure. The allowance family therefore isolates EndorseRank’s distinctive signal among both the social baseline and the supplementary ceiling reference.

The allowance result is the empirical core of EndorseRank’s contribution. Prior transfer-based reputation work could not have produced Table 4.9’s pattern, because transfer graphs do not encode spending authorization. By building G_{approve} from latest allowances and ranking with PageRank, EndorseRank recovers a signal that AWP misses entirely. Combined with the runtime results, the allowance family supports the efficiency–alignment trade-off in Claim C4: the cheaper method is also the only social method that aligns with authorization proxies.

4.5.3 Inverse-risk family (GMX PnL)

Table 4.10 reports alignment with loss-avoidance, non-loss close rate, and worst-close PnL score. Among social methods, alignment is weak overall. EndorseRank family mean τ is 0.035, with individual proxies ranging from -0.031 (non-loss close rate) to 0.079 (loss avoidance). AWP is low-to-moderate (family mean 0.107), driven by loss avoidance ($\tau = 0.165$) and worst-close PnL score ($\tau = 0.137$), while non-loss close rate remains negligible ($\tau = 0.019$). Transfer activity is modestly associated with avoiding large losses, but neither social method ranks wallets in close agreement with inverse-risk summaries.

GF-PR achieves the highest family mean (0.376), with loss avoidance $\tau = 0.545$ and worst-close

Table 4.10: Domain alignment with inverse-risk proxies from GMX realized PnL (higher = lower loss).

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Loss avoidance ($-\sum \min(\text{PnL}, 0)$)	0.097	0.079
Non-loss close rate	-0.037	-0.031
Worst-close PnL score ($-\min \text{PnL}$)	0.070	0.057
<i>AWP</i>		
Loss avoidance ($-\sum \min(\text{PnL}, 0)$)	0.243	0.165
Non-loss close rate	0.028	0.019
Worst-close PnL score ($-\min \text{PnL}$)	0.204	0.137
<i>GF-PR</i>		
Loss avoidance ($-\sum \min(\text{PnL}, 0)$)	0.675	0.545
Non-loss close rate	0.150	0.104
Worst-close PnL score ($-\min \text{PnL}$)	0.618	0.478

PnL score $\tau = 0.478$. This elevation is expected: GF-PR’s POOL/SINK star graph is built from the same GMX PnL stream that defines these proxies. High GF-PR τ on inverse risk is therefore a construct-overlap diagnostic, not evidence that outcome-native ranking measures social reputation. Section 4.7 develops this interpretation formally as claim SC1.

For the primary comparison, the inverse-risk family supports a cautious reading of RQ4: social graphs are not strong predictors of loss-avoidance rankings on this six-month GMX window. That finding does not invalidate EndorseRank’s allowance alignment; it delimits what social reputation can be expected to explain. Wallets that are central in approval or transfer networks need not be the same wallets that avoid large perpetual losses, and the data confirm that separation.

4.5.4 Sybil-adjusted stability family

Table 4.11 reports alignment with inbound counterparty ratio, transfer tenure, and active months. AWP exceeds EndorseRank among social methods (family means 0.286 vs. 0.233). The gap is concentrated in tenure and activity proxies: AWP achieves $\tau = 0.436$ on transfer tenure and $\tau = 0.482$ on active months, versus 0.297 and 0.330 for EndorseRank. Because AWP’s edge weights incor-

Table 4.11: Domain alignment with Sybil-adjusted stability proxies from transfer activity.

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Inbound counterparty ratio	0.093	0.073
Transfer tenure (days)	0.366	0.297
Active months	0.381	0.330
<i>AWP</i>		
Inbound counterparty ratio	-0.063	-0.061
Transfer tenure (days)	0.602	0.436
Active months	0.622	0.482
<i>GF-PR</i>		
Inbound counterparty ratio	-0.339	-0.235
Transfer tenure (days)	0.359	0.244
Active months	0.372	0.270

porate logistic time-decay over transfer history, wallets with sustained transfer activity naturally receive higher scores, producing alignment with tenure and active-month summaries.

Inbound counterparty ratio—a proxy intended to favor diverse counterparties over concentrated Sybil-like structures—is weak or negative for both methods (EndorseRank $\tau = 0.073$; AWP $\tau = -0.061$). Neither social ranking strongly tracks counterparty diversity on this cohort. GF-PR is weak overall (family mean 0.093) and strongly negative on inbound counterparty ratio ($\tau = -0.235$), consistent with the absence of transfer-activity semantics in the PnL star graph.

The Sybil-adjusted family therefore reinforces AWP’s advantage on transfer-derived activity constructs, while showing that neither method currently provides a strong counterparty-diversity signal. Future work could incorporate explicit diversity penalties into G_{approve} or G_{transfer} ; the present results establish the baseline against which such extensions would be judged.

4.5.5 Trading-success family

Table 4.12 reports alignment with close-success count, realized gain proxy, and close success rate. Among social methods, AWP exceeds EndorseRank (family means 0.179 vs. 0.096). AWP’s strongest trading proxies are realized gain ($\tau = 0.239$) and close-success count ($\tau = 0.224$); close

Table 4.12: Domain alignment with GMX V2 margin-trading success proxies on Arbitrum One.

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Close-success count	0.180	0.149
Realized gain proxy	0.132	0.107
Close success rate	0.041	0.033
<i>AWP</i>		
Close-success count	0.319	0.224
Realized gain proxy	0.349	0.239
Close success rate	0.110	0.075
<i>GF-PR</i>		
Close-success count	0.690	0.522
Realized gain proxy	1.000	0.999
Close success rate	0.329	0.229

success rate remains weak ($\tau = 0.075$). EndorseRank shows the same ordering at lower magnitude (count $\tau = 0.149$; realized gain $\tau = 0.107$; rate $\tau = 0.033$). Transfer-based social propagation therefore aligns more strongly with profitable perpetual closes than allowance-based propagation on this cohort, but both social-method means remain in the weak-to-moderate band.

GF-PR achieves the highest family mean (0.583), including near-perfect alignment with the realized gain proxy ($\rho = 1.000$, $\tau = 0.999$). That near-identity is the clearest empirical demonstration of construct overlap: when the ranking input and the validation proxy share the same PnL stream, alignment ceases to be an independent test of reputation quality. GF-PR’s close-success count ($\tau = 0.522$) and success rate ($\tau = 0.229$) are also elevated relative to social methods, but the realized-gain cell is decisive for interpretation.

The trading-success family thus plays a dual role. For Claims C2–C3, it shows that AWP is the stronger social method on outcome proxies, while EndorseRank retains only weak outcome alignment. For SC1, it shows that GF-PR’s ceiling must not be misread as a deployable reputation result. Protocols seeking social endorsement signals should not adopt GF-PR merely because it maximizes trading-family τ .

Comparing trading-success means to transfer means for the same social methods answers RQ4

directly. Transfer proxies yield higher mean τ than trading-success proxies for both EndorseRank (0.333 vs. 0.096) and AWP (0.534 vs. 0.179). On this chain and window, the Do et al. transfer validation frame is the stronger external check for social reputation methods. That finding does not diminish EndorseRank’s allowance-specific validity; it situates trading outcomes as a harder, more weakly related target for social-graph scores. Protocols that require trading-skill discrimination may need features beyond either social graph—or may need to accept that social reputation and trading performance are only loosely coupled in perpetual markets.

4.6 Rank divergence (H1 / RQ1)

Hypothesis H1 and research question RQ1 ask whether EndorseRank and AWP yield meaningfully different wallet orderings on an identical sample. Inter-method Kendall τ between the two score vectors is 0.316 on the matched cohort ($n = 5,521$). In the alignment bands of Chapter 2, this value falls in the weak-to-moderate range: the rankings are related but far from interchangeable. Correlation is present because economically active wallets tend to participate in both transfer and approval markets, yet allowance and transfer graphs emphasize different relationships, so many pairwise orderings disagree.

Figure 4.7 shows the marginal distributions of EndorseRank and AWP scores. Both are right-skewed, as is typical for PageRank on sparse directed graphs, but they are not identically shaped. AWP’s mass reflects the denser transfer operator; EndorseRank’s mass reflects the sparser endorsement operator. Skewness alone does not prove divergence—two methods could share the same ordering with different score scales—but it motivates examining rank-rank relationships directly. Figure 4.8 plots EndorseRank ranks against AWP ranks. Points along the diagonal would indicate identical orderings; the observed cloud shows substantial off-diagonal mass. Wallets that rank highly under transfer centrality are not reliably the same wallets that rank highly under allowance endorsement, and vice versa. The inter-method $\tau = 0.316$ summarizes this scatter: agreement exceeds chance, but a protocol that substitutes one ranking for the other would mis-order a large

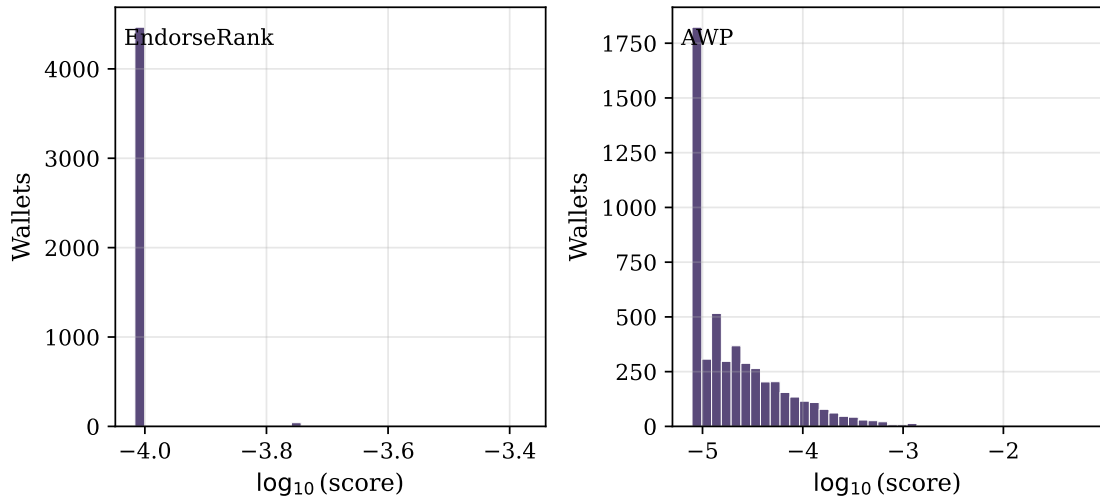


Figure 4.7: Marginal distributions of EndorseRank and AWP scores on the matched cohort ($n = 5,521$), shown on a $\log_{10}$ scale for wallets with positive score (EndorseRank excludes 966 zero-score wallets; AWP excludes 381). Both are right-skewed PageRank distributions; differences in shape reflect the distinct endorsement and transfer operators.

fraction of wallet pairs.

Rank divergence is the structural counterpart of construct-specific alignment. If the two methods produced nearly identical orderings, family-level winners in Table 4.7 would be difficult to interpret as construct effects; they could be noise around a shared ranking. The observed $\tau = 0.316$, together with EndorseRank’s allowance lead and AWP’s transfer lead, supports reading the methods as related but distinct reputation instruments. H1 is supported: EndorseRank and AWP yield partially overlapping yet construct-distinct orderings on the matched cohort.

An operational reading of $\tau = 0.316$ is that for a randomly chosen pair of wallets, the probability that EndorseRank and AWP agree on which wallet ranks higher is only moderately above chance. Protocols that currently use transfer-based scores cannot assume that switching to allowance-based scores—or blending the two—will preserve existing wallet orderings. Migration would re-rank a substantial fraction of pairs, with the largest disagreements likely concentrated among wallets that are transfer hubs but rarely endorsed as spenders, or endorsed spenders with limited transfer footprints. Figure 4.8 is consistent with that interpretation: the cloud is positively sloped but wide.

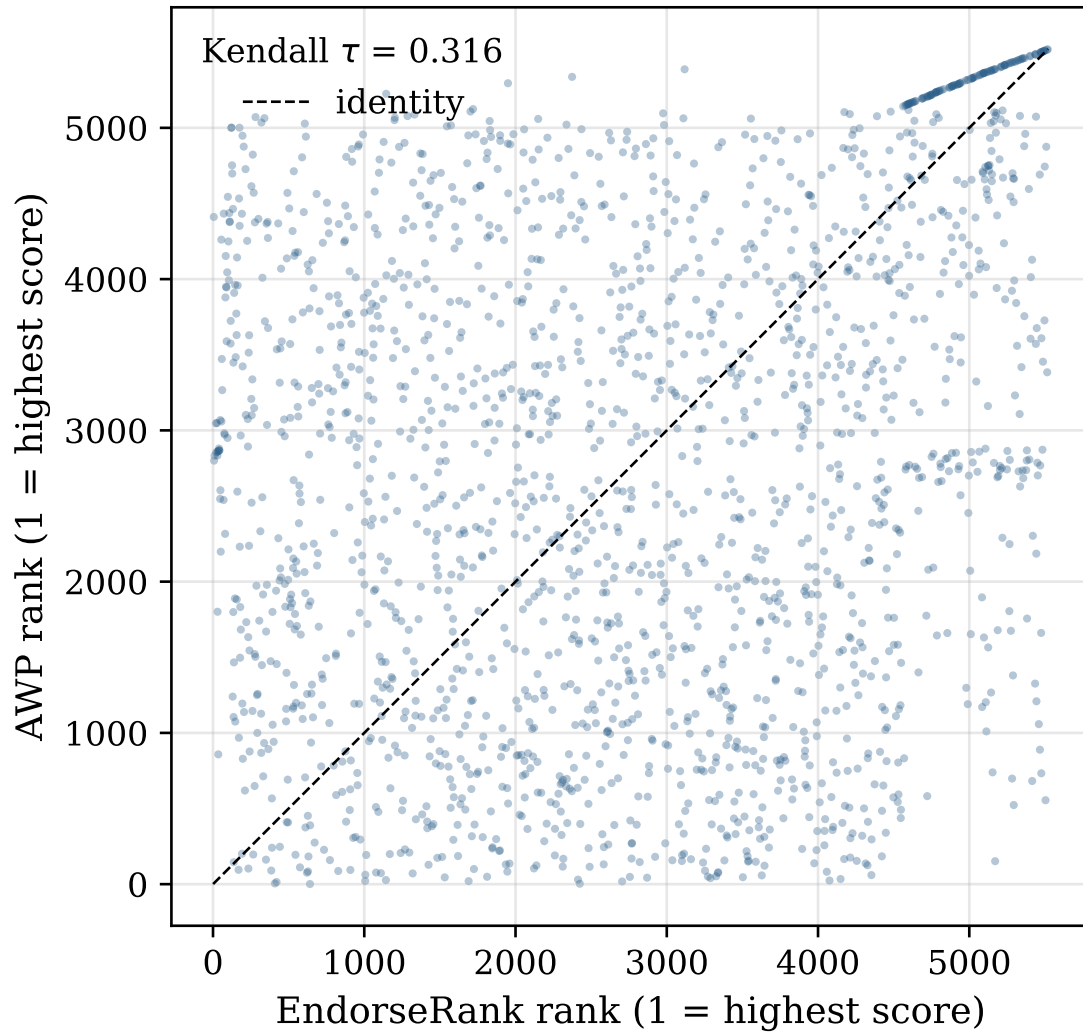


Figure 4.8: Rank–rank scatter of EndorseRank versus AWP on the matched cohort. Inter-method Kendall $\tau = 0.316$ indicates partial overlap with substantial pairwise disagreement, supporting H1/RQ1 (construct-distinct orderings).

The divergence result also disciplines rhetoric about “on-chain reputation” as a singular object. On this cohort, reputation-from-authorization and reputation-from-transfer are empirically distinct enough that treating them as interchangeable would misstate the evidence. Chapter 5 returns to the practical implication: method choice should follow the trust construct a protocol needs, not a generic preference for any PageRank variant.

4.7 Supplementary GF-PR ceiling analysis (SRQ1 / SC1)

GF-PR was added during empirical work to bound trading-outcome proxy alignment when ranking directly on outcome flow rather than social edges. Table 4.7 (row 3) shows GF-PR achieving the highest mean τ on inverse risk (0.376) and trading success (0.583) among all methods. This pattern is expected by construction: GF-PR’s POOL/SINK star graph is built from the same GMX PositionDecrease PnL stream used to define trading-success and inverse-risk proxies. The result illustrates *construct overlap*—shared input semantics inflate alignment without implying that GF-PR measures deployable social reputation.

Two further observations clarify the ceiling’s role. First, GF-PR is weak on allowance and only modest on transfer, confirming that outcome-native flow does not recover social-graph constructs. Second, among social methods on trading-success proxies, AWP ($\tau = 0.179$) still exceeds EndorseRank (0.096), confirming that transfer-based social propagation aligns more strongly with trading outcomes on this cohort than allowance-based propagation, independent of the GF-PR ceiling.

The ceiling is therefore a diagnostic for evaluation design, not a competitor in the primary comparison. When a validation proxy shares inputs with a ranking method, high τ is uninformative about social reputation quality. Researchers comparing EndorseRank and AWP on trading-outcome families should treat GF-PR’s τ as an upper reference that flags how much alignment is achievable by construction, then interpret social-method τ values relative to that reference and to each other.

The near-perfect GF-PR alignment with the realized-gain proxy ($\tau \approx 0.999$ in Table 4.12) is espe-

cially instructive. It shows that the evaluation pipeline can recover extremely high τ when inputs and proxies coincide, so the moderate social-method values are not an artifact of a broken correlation implementation. Low-to-moderate social-method τ on trading families reflects limited shared structure between social graphs and GMX outcomes, not an inability of Kendall τ to detect strong relationships when they exist.

Supplementary claim SC1: GF-PR sets an outcome-native upper bound on trading-outcome proxy alignment; high GF-PR τ on the trading-success and inverse-risk families must not be read as evidence that outcome-native ranking replaces social endorsement graphs for reputation deployment.

4.8 Primary claims (C1–C4) and supplementary claim (SC1)

Chapter 4 empirical results support the following claims, aligned with Table 1.3.

Primary claims (EndorseRank vs. AWP):

- C1.** EndorseRank runtime is lower than AWP runtime on identical n . On the matched cohort ($n = 5,521$), EndorseRank completes PageRank in 2.105 s versus 18.711 s for AWP (speedup $\approx 8.9\times$), with 14,727 vs. 137,087 edges (Table 4.2). On the expanded wallet pool, speedups remain $\approx 10.4\times$ at $n = 10,000$ (1.18 s vs. 12.28 s) and $\approx 9.4\times$ at $N = 31,612$ (2.29 s vs. 21.40 s) (Table 4.3).
- C2.** EndorseRank and AWP achieve credible Spearman ρ /Kendall τ on selected proxies under Do et al.’s framework; family-level alignment varies by construct (Tables 4.8–4.12). EndorseRank’s allowance mean $\tau = 0.355$ and AWP’s transfer mean $\tau = 0.534$ are the primary-construct anchors; secondary axes are weaker but non-negligible where reported.
- C3. Construct-specific alignment:** EndorseRank exceeds AWP on allowance proxies ($\tau = 0.355$ vs. -0.030); AWP exceeds EndorseRank on transfer ($\tau = 0.534$ vs. 0.333) and on most outcome-based families among social methods, including trading success (0.179 vs. 0.096) (Table 4.7). Inter-method rank agreement is only moderate ($\tau = 0.316$), consistent

with construct-distinct orderings (Section 4.6).

C4. Efficiency–alignment trade-off: EndorseRank offers $\approx 8.9\times$ lower PageRank runtime than AWP on the matched cohort and $\approx 9\text{--}10\times$ on expanded-pool scaling (Tables 4.2, 4.3) while maintaining moderate transfer alignment ($\tau \approx 0.33$) and stronger allowance alignment ($\tau \approx 0.35$). Damping sweeps confirm allowance-family stability at ≈ 0.355 across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.4); top-token and sample-size checks preserve the construct-specific pattern (Tables 4.5, 4.6).

Supplementary claim:

SC1. GF-PR’s highest trading-success mean τ (0.583) and inverse-risk mean τ (0.376) (Table 4.7, Section 4.7) reflect construct overlap with outcome proxies, not superior social reputation. GF-PR bounds interpretable alignment for EndorseRank and AWP on trading-outcome families (SRQ1) and must not be treated as a deployable substitute for social endorsement graphs.

In summary, the matched-cohort evaluation supports a coherent empirical narrative. EndorseRank is substantially faster than AWP because G_{approve} is an order of magnitude sparser than G_{transfer} . The two methods produce related but distinct rankings (inter-method $\tau = 0.316$) and win on different proxy families: EndorseRank on allowance, AWP on transfer and several social-outcome axes. Robustness checks do not overturn these patterns. GF-PR clarifies the ceiling on trading-outcome alignment under construct overlap, protecting the primary comparison from misinterpretation.

The claims can also be read as answers to the primary research questions. RQ1 is addressed by inter-method $\tau = 0.316$ and Figures 4.7–4.8: orderings diverge enough to treat the methods as distinct instruments. RQ2 is addressed by Table 4.7 and the per-family tables: winners are construct-specific. RQ3 is addressed by Tables 4.2–4.3 and Figure 4.4: EndorseRank is approximately $8.9\times\text{--}10.4\times$ faster depending on sample definition. RQ4 is addressed by comparing transfer-family means to trading-success means for social methods: transfer proxies yield stronger alignment than trading-success proxies on this cohort (AWP 0.534 vs. 0.179; EndorseRank 0.333 vs. 0.096). SRQ1 is addressed by the GF-PR ceiling and SC1.

What the chapter does *not* claim is equally important. It does not claim that EndorseRank predicts leveraged trading profits better than AWP; the opposite is true among social methods on the trading-success family. It does not claim that AWP is obsolete; AWP remains the stronger method for transfer-hub detection and for several activity- and outcome-adjacent proxies. It does not claim that GF-PR is a production reputation score. The contribution is a precise, construct-aware comparison: EndorseRank offers a sparse, authorization-native ranking that is much cheaper to compute, aligns strongly with allowance proxies, diverges moderately from AWP, and remains stable under the reported robustness checks.

Chapter 5 interprets these findings against RQ1–RQ4 and SRQ1 in greater depth and discusses implications, threats to validity, and limitations.

5 Discussion

This chapter interprets the Chapter 4 findings against RQ1–RQ4 and SRQ1, discusses theoretical and practical implications for DeFi risk assessment, formulates a Sybil attack cost model on allowance graphs, addresses regulatory and ethical considerations, evaluates threats to validity, and states limitations of the primary EndorseRank–AWP comparison. Throughout, empirical quantities are those reported for the matched wallet cohort ($n = 5,521$) unless otherwise noted: inter-method Kendall $\tau = 0.316$, EndorseRank allowance $\tau = 0.355$, AWP transfer $\tau = 0.534$, trading-success $\tau = 0.096$ (EndorseRank) and 0.179 (AWP), GF-PR trading-success $\tau = 0.583$ and inverse-risk $\tau = 0.376$, and PageRank runtimes of 2.105 s versus 18.711 s ($\approx 8.9\times$), with pool scaling of $\approx 10.4\times$ at $n = 10,000$ and $\approx 9.4\times$ at $N = 31,612$.

5.1 Findings by research question

The primary empirical comparison concerns EndorseRank and AWP as social reputation methods on wallet-to-wallet graphs. The following subsections interpret each research question in light of the five-proxy alignment design and the computational benchmarks.

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP produce meaningfully different wallet orderings on an identical Arbitrum sample. Inter-method Kendall τ between the two score vectors is 0.316 on the matched cohort ($n = 5,521$), within the weak-to-moderate alignment band defined in Chapter 2. Correlation is present—both methods operate on overlapping wallet activity in the same observation window—but the magnitude is far from unity. Allowance and transfer graphs therefore emphasize different wallet relationships.

This divergence is structurally expected. The endorsement graph G_{approve} contains 14,727 edges encoding current spending authorization; the transfer graph G_{transfer} contains 137,087 edges encoding time-decayed economic flow ^[10]. PageRank propagates influence along these edges under the same damping convention ^{[30],[25]}, so differences in edge sets and weights produce differences in stationary mass. Allowances encode authorization intent; transfers encode movement of value. A wallet that receives large transfers without granting allowances can rank highly under AWP and modestly under EndorseRank, and conversely for a wallet that authorizes spenders without being a transfer hub.

H1 is therefore supported: EndorseRank and AWP yield partially overlapping yet construct-distinct orderings. The practical implication is that the two scores are not interchangeable substitutes. Deployments that adopt EndorseRank for efficiency must accept a different ranking geometry, not merely a faster approximation of AWP.

5.1.2 RQ2: Construct-specific social-method alignment

RQ2 asks which social method aligns more strongly with each of the five proxy families. Table 4.7 and the per-family tables (Tables 4.8–4.12) show clear construct-specific winners rather than a universal ranking.

On the **allowance** family, EndorseRank achieves mean Kendall $\tau = 0.355$, whereas AWP is near zero (-0.030). This pattern is the strongest evidence that EndorseRank measures its intended construct: latest-allowance centrality aligns with allowance-based proxies, while transfer-graph PageRank does not. In measurement terms, the indicator tracks the construct it was designed to represent ^[8].

On the **transfer** family, AWP achieves $\tau = 0.534$ and EndorseRank a moderate secondary-axis

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google's PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

$\tau = 0.333$. AWP’s advantage is expected because transfer proxies (in-degree and in-value) are the validation frame of Do et al.^[10] and share input semantics with G_{transfer} . EndorseRank’s moderate transfer alignment indicates partial overlap—wallets active in approvals often participate in transfers—but not equivalence.

On **trading-success** proxies derived from GMX V2 perpetual closes, AWP exceeds EndorseRank among social methods ($\tau = 0.179$ vs. 0.096). Allowance-based propagation does not align more strongly with profitable perpetual closes than transfer-based AWP on this cohort. Both values lie in the weak-to-moderate band; neither social method approaches the outcome-native GF-PR ceiling discussed under SRQ1. On **inverse-risk** and **Sybil-adjusted stability** families, AWP likewise leads among social methods, consistent with transfer activity capturing more of the variance in leveraged outcomes and activity-adjusted stability than allowance snapshots alone.

H2 is supported in its construct-specific form: EndorseRank wins on allowance; AWP wins on transfer and most social-outcome families. The result argues against treating reputation as a single latent trait measured equally well by any graph. Instead, graph semantics determine which external proxies a score can credibly track.

5.1.3 RQ3: Computational efficiency

RQ3 asks whether EndorseRank is computationally more efficient than AWP on the same matched sample. On the full cohort ($n = 5,521$), EndorseRank completes PageRank in 2.105 s versus 18.711 s for AWP—approximately $8.9\times$ faster—with a sparser graph (14,727 vs. 137,087 edges). Both methods converge within tolerance 10^{-8} ; EndorseRank requires fewer iterations on average, consistent with a smaller effective edge set^[30].

Scaling on the expanded wallet pool ($N = 31,612$) preserves the advantage: $\approx 10.4\times$ at $n = 10,000$ and $\approx 9.4\times$ at full pool size. Robustness checks on damping and top-token subgraphs do

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

not alter the allowance–transfer trade-off pattern: EndorseRank’s allowance τ remains stable near 0.355 across damping $d \in \{0.75, 0.85, 0.95\}$, while transfer-family τ for both methods varies by at most approximately 0.005.

H3 is supported. The efficiency gain follows from latest-allowance snapshots instead of aggregating full transfer history with logistic decay^[10]. For protocols that must refresh reputation frequently, this difference is operationally material: an order-of-magnitude reduction in PageRank wall time enables tighter update loops without proportional infrastructure cost.

5.1.4 RQ4: Transfer versus trading-success validation frames

RQ4 asks which validation frame yields stronger rank alignment for social reputation methods on this chain: Do et al. transfer proxies or GMX V2 trading-success proxies. Transfer proxies yield higher mean τ than trading success for both EndorseRank and AWP (Table 4.7). AWP’s transfer $\tau = 0.534$ substantially exceeds its trading-success $\tau = 0.179$; EndorseRank’s transfer $\tau = 0.333$ exceeds its trading-success $\tau = 0.096$.

Two interpretations follow. First, social graphs built from wallet-to-wallet economic relationships are more tightly coupled to transfer centrality than to perpetual trading profitability. Trading success depends on market timing, leverage choice, and liquidation dynamics that are only partially reflected in approval or transfer topology^{[15],[26]}. Second, outcome-based proxies show weak-to-modest alignment overall for social graphs, suggesting that no single proxy family fully captures “reputation” on this sample. Domain alignment remains a relative, multi-construct exercise^{[8],[33],[22]}, not a search for one ground-truth ranking.

RQ4 therefore favors transfer proxies as the stronger validation frame for social methods on Arbi-

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[15]GMX. (n.d.). GMX V2 documentation.

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

trum One in this window, while leaving room for trading-success proxies as a secondary, economically motivated axis—especially for protocols whose risk is defined by leveraged outcomes rather than transfer hubs.

5.2 Supplementary finding (SRQ1)

GF-PR achieves the highest trading-success mean τ (0.583) and inverse-risk mean τ (0.376; Table 4.7), as expected for an outcome-native graph built from the same PnL stream used to define those proxies. The POOL/SINK star construction ranks wallets by synthetic flow of realized gains and losses; it shares input semantics with trading-success and inverse-risk proxies by design. High alignment is therefore evidence of *construct overlap*, not of superior social-graph semantics^[8].

This ceiling clarifies interpretation of social-method results. EndorseRank’s trading-success $\tau = 0.096$ and AWP’s $\tau = 0.179$ should be read against an outcome-native upper bound of 0.583, not against an implicit expectation of near-perfect prediction. Among deployable social methods, AWP still exceeds EndorseRank on trading-success proxies, confirming that transfer-based propagation aligns more strongly with trading outcomes on this cohort than allowance-based propagation, independent of the GF-PR ceiling. EndorseRank and AWP remain the deployable social reputation methods; GF-PR is a diagnostic reference only (Claim SC1).

5.3 Theoretical implications

The findings extend three strands of theory: hyperlink-style propagation on financial graphs, construct validity for on-chain scores, and the economics of trust under pseudonymity.

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

5.3.1 Propagation semantics beyond transfers

PageRank’s original formulation treats inbound hyperlinks as citations ^{[30],[5]}. Do et al.^[10] map that metaphor to Ethereum transfers. EndorseRank maps it to ERC-20 allowances: an owner who authorizes a spender issues a directed endorsement of spending authority ^{[12],[13]}. The inter-method $\tau = 0.316$ demonstrates that changing the edge semantics changes the ranking geometry even when the node set is held fixed. Graph-based reputation is therefore not a single algorithm applied to “the” transaction graph; it is a family of methods whose validity depends on which relationships are encoded as edges ^{[25],[26]}.

Related trust-propagation systems in peer-to-peer and web settings—EigenTrust and TrustRank—similarly show that seed sets and edge definitions dominate outcomes ^{[21],[17]}. EndorseRank contributes an edge definition native to token standards rather than to message-passing overlays or hyperlink graphs.

5.3.2 Multi-construct domain alignment

The five-proxy framework extends Do et al.’s validation logic by separating transfer centrality, allowance delegation, leveraged outcomes, and Sybil-adjusted activity. EndorseRank aligns with its primary allowance construct (G_{approve} , $\tau = 0.355$); transfer proxies serve as a secondary valida-

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[5]Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. [Computer Networks and ISDN Systems]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[21]Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. [Proceedings of the 12th International Conference on World Wide Web]

^[17]Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. [Proceedings of the Thirtieth International Conference on Very Large Data Bases]

tion axis (moderate $\tau = 0.333$, below AWP 0.534). AWP aligns with transfer—its primary graph construct—and several outcome proxies derived from the same wallet activity. The supplementary GF-PR analysis demonstrates the upper bound of trading-outcome proxy alignment achievable when ranking directly on outcome flow rather than social edges (SRQ1).

This pattern is consistent with classical construct-validity reasoning: indicators should correlate with measures of the same construct and diverge from measures of different constructs ^[8]. Rank correlations ^{[33],[22]} operationalize that reasoning for ordinal reputation scores without imposing classification thresholds.

5.3.3 Reputation under cheap pseudonyms

Public ledgers permit cheap identity creation ^{[27],[36],[11]}. Reputation systems that reward inbound transfers or allowances are therefore exposed to Sybil strategies in which an attacker manufactures edges among controlled addresses. The empirical results do not eliminate that threat; they show that allowance and transfer graphs respond differently to the activity patterns present in the observational sample. Section 5.5 formalizes attack cost on allowance graphs, arguing that non-zero allowances to external spenders can impose opportunity and revocation costs that pure transfer wash trading does not.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers face a design choice between transfer-hub detection and authorization-aware reputation. The empirical trade-off supports scenario-dependent adoption rather than a single recommended score.

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

^[27]Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.

^[36]Wood, G. (2014). Ethereum: A secure decentralised generalised transaction ledger. [Ethereum Project]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

5.4.1 Transfer-hub and flow monitoring

Protocols that prioritize detection of high-flow wallets—for example, for market-maker identification, routing analysis, or transfer-based compliance heuristics—may prefer AWP despite higher cost. AWP’s transfer $\tau = 0.534$ and stronger social-outcome alignment make it the better match when the operational target is economic flow centrality^[10]. The runtime of 18.711 s on $n = 5,521$ wallets is acceptable for batch analytics; it is less attractive for sub-second or near-real-time refresh.

5.4.2 Authorization-aware and low-latency refresh

Protocols that emphasize explicit spending authorization—token vaults, allowance-gated integrations, or risk engines that treat approvals as trust acts—may prefer EndorseRank. The method’s allowance $\tau = 0.355$, sparse graph (14,727 edges), and 2.105 s PageRank time support frequent recomputation. Pool scaling ($\approx 10.4\times$ at $n = 10,000$; $\approx 9.4\times$ at $N = 31,612$) indicates that the efficiency advantage persists beyond the matched cohort. Operators accept lower transfer-proxy alignment ($\tau = 0.333$) and weaker trading-success alignment ($\tau = 0.096$) in exchange for authorization semantics and speed.

5.4.3 Evaluation design and hybrid use

GF-PR is suitable only as a diagnostic ceiling for outcome-proxy overlap during evaluation design, not as a deployable social reputation score. Evaluation pipelines should report social-method τ alongside an outcome-native reference when trading-success proxies are used, to avoid mistaking construct overlap for social validity (Claim SC1).

Hybrid deployments are feasible: EndorseRank for continuous authorization-aware monitoring, AWP for periodic transfer-hub audits, and GMX-linked outcome metrics as separate risk dashboards rather than as substitutes for social scores^{[15],[32]}. The open pipeline enables cross-protocol

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[15]GMX. (n.d.). GMX V2 documentation.

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal

replication on any ERC-20 chain with allowance logs (Appendix 8.1). Integration with live risk oracles and governance tooling remains future work (Chapter 6), but the efficiency profile of EndorseRank is compatible with oracle update cadences that transfer-graph AWP may not sustain at equal cost.

5.5 Sybil attack cost model on allowance graphs

Sybil attacks exploit cheap pseudonyms to manufacture reputation ^[11]. On transfer graphs, an attacker can wash-trade value among controlled wallets, inflating in-degree and in-value at the cost of fees and inventory risk. On allowance graphs, the attack surface differs because edges represent authorization rather than completed transfers. This section formalizes a minimal cost model for inflating EndorseRank via fabricated allowances.

5.5.1 Attack setup

Let $\mathcal{A}$ be the set of addresses controlled by an attacker and $\mathcal{H}$ the set of honest wallets in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) with weight $w_{uv} > 0$ denotes that u currently authorizes v to spend decimal-adjusted token units totaling w_{uv} . EndorseRank computes the PageRank vector $\mathbf{r}$ on G_{approve} with damping $d \in (0, 1)$ ^{[30],[25]}:

$$\mathbf{r} = d P^\top \mathbf{r} + \frac{1-d}{|V|} \mathbf{1}, \quad (5.1)$$

where P is the row-stochastic matrix obtained by normalizing outgoing allowance weights (with a standard handling of dangling nodes).

Reserve Bank of St. Louis Review]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google's PageRank and beyond: The science of search engine rankings. [Princeton University Press]

The attacker seeks to maximize the mean EndorseRank of a target subset $T \subseteq \mathcal{A}$,

$$J(T) = \frac{1}{|T|} \sum_{i \in T} r_i, \quad (5.2)$$

subject to a budget on attack cost defined below.

5.5.2 Cost components

Fabricating an allowance edge (u, v) with weight w_{uv} incurs at least three cost components:

1. **Transaction cost.** Submitting approve or permit consumes gas. Let c_{gas} be the expected fee per allowance update. For m updates,

$$C_{\text{gas}} = m c_{\text{gas}}. \quad (5.3)$$

2. **Capital lock / opportunity cost.** A non-zero allowance to an external spender $v \notin \mathcal{A}$ exposes tokens owned by u to transfer by v . Even if $v \in \mathcal{A}$, maintaining large allowances to contracts or routers may be required to mimic honest behavior. Let $\ell(w_{uv})$ be the opportunity cost of capital associated with weight w_{uv} over the reputation refresh horizon (for example, forgone yield). Then

$$C_{\text{cap}} = \sum_{(u,v) \in E_{\mathcal{A}}} \ell(w_{uv}), \quad (5.4)$$

where $E_{\mathcal{A}}$ denotes attacker-created edges.

3. **Revocation and detection cost.** Honest users revoke stale allowances; attackers who never revoke may be distinguishable. Let c_{rev} be the expected cost of periodic revocation and re-approval needed to maintain a target edge set over k refresh epochs:

$$C_{\text{rev}} = k c_{\text{rev}}. \quad (5.5)$$

Total attack cost is

$$C = C_{\text{gas}} + C_{\text{cap}} + C_{\text{rev}}. \quad (5.6)$$

5.5.3 Influence and a cost-effectiveness bound

Under standard PageRank sensitivity results, the influence of an edge on $\mathbf{r}$ is mediated by the stationary mass at the source and the normalized outgoing weight ^[25]. A coarse upper bound on the gain in $J(T)$ from adding edges among $\mathcal{A}$ alone—a closed Sybil cluster with no honest inbound endorsements—is limited by the teleportation mass:

$$\sum_{i \in \mathcal{A}} r_i \leq |\mathcal{A}| \cdot \frac{1-d}{|V|} + d \cdot \sum_{i \in \mathcal{A}} r_i, \quad (5.7)$$

which rearranges to

$$\sum_{i \in \mathcal{A}} r_i \leq \frac{|\mathcal{A}|}{|V|}. \quad (5.8)$$

Thus a closed attacker cluster cannot capture more than its population share of total rank mass. To exceed that share, the attacker needs inbound endorsements from $\mathcal{H}$, which requires either compromising honest wallets or inducing honest owners to approve attacker-controlled spenders—actions whose cost is typically dominated by C_{cap} and social-engineering effort rather than gas alone.

When the attacker purchases inbound allowances from honest owners (for example, by operating a service that requests approvals), the marginal cost per unit of honest-sourced weight w_{ha} can be written

$$c_{\text{in}}(w_{ha}) = c_{\text{gas}} + \ell_{\text{honest}}(w_{ha}) + \pi(w_{ha}), \quad (5.9)$$

where ℓ_{honest} is compensation paid to the honest owner and π is a risk premium for potential loss from malicious spending. Cost-effectiveness of the attack is then

$$\eta = \frac{\Delta J(T)}{C}, \quad (5.10)$$

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

with $\Delta J(T)$ the increase in mean target rank attributable to the attack edges.

Equations (5.7)–(5.10) imply that EndorseRank is not Sybil-proof, but that closed-cluster inflation is population-share limited, and that cross-cluster inflation requires costly honest-sourced allowances. Transfer wash trading faces different costs (fees and inventory) but does not require exposing spending authority to external addresses. Sybil-adjusted stability proxies in Chapter 4 partially address observational manipulation; they do not replace adversarial evaluation, which remains future work.

5.6 Regulatory and ethical considerations

On-chain reputation scores occupy a contested space between transparency and fairness. Packin and Lev-Aretz^[29] caution that decentralized credit scoring can reproduce black-box harms if models are opaque, unappealable, or coupled to identity systems without due process. EndorseRank and AWP are fully determined by public ledger events and published algorithms; any party can recompute scores from the open pipeline (Appendix 8.1). That auditability is a partial response to black-box concerns, but it does not resolve distributional effects: wallets with limited approval or transfer history receive low scores regardless of off-chain creditworthiness.

Ethically, scores must not be presented as identity, creditworthiness in the consumer-lending sense, or proof of unique personhood^[11]. They are ordinal indicators of on-chain graph position under specified edge semantics. Deployments that gate access to financial services should disclose the graph definition, observation window, and known failure modes—including Sybil exposure and weak trading-success alignment for social methods ($\tau = 0.096$ and 0.179). Regulatory regimes that require explainability favor graph propagation over opaque machine-learning scores^{[24],[29]}, provided operators resist overclaiming predictive power for loan default.

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]

Privacy is structurally limited on public ledgers ^{[27],[36]}: reputation computation uses data already broadcast. The ethical obligation is therefore not to avoid analysis but to avoid deanonymization beyond address-level aggregation and to prevent misuse of scores as persistent biometric-like identifiers. This dissertation analyzes addresses only; no off-chain identity linkage is performed (Chapter 3).

5.7 Threats to validity

Interpretation of the empirical claims should account for the following threats, organized by construct, internal, external, and adversarial validity.

5.7.1 Construct validity

Proxy not ground truth. No under-collateralized lending default labels are available at sufficient density. Perpetual-trading liquidation and PnL on GMX V2 are domain-intuition proxies, not loan repayment ^{[15],[26]}. Alignment with trading-success proxies ($\tau = 0.096$ for EndorseRank; 0.179 for AWP) must not be read as credit-default prediction.

GF-PR construct overlap. GF-PR shares input semantics with trading-success and inverse-risk proxies; its high τ on those families (0.583 and 0.376) must not be interpreted as superior social reputation ^[8].

Secondary-axis interpretation. EndorseRank’s moderate transfer $\tau = 0.333$ is a secondary validation axis, not evidence that allowance graphs measure transfer centrality as well as AWP ($\tau = 0.534$).

^[27]Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.

^[36]Wood, G. (2014). Ethereum: A secure decentralised generalised transaction ledger. [Ethereum Project]

^[15]GMX. (n.d.). GMX V2 documentation.

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

5.7.2 Internal validity

Shared observation window. All methods and proxies are computed over 2025-12-01 to 2026-05-31. Market regimes specific to that window—volatility, incentive programs, or protocol changes—may inflate or deflate alignment. Robustness checks on damping and top-token subgraphs reduce sensitivity to PageRank hyperparameters and token concentration but do not vary the calendar window.

Cohort selection. The matched wallet cohort requires at least three GMX closes and non-zero reputation subgraph connectivity ($n = 5,521$; 365,488 `PositionDecrease` events). Results may not extend to wallets with thinner perpetual histories or to addresses absent from allowance and transfer subgraphs.

Edge-weight measurement. Token-decimal normalization, USD conversion choices, and permit-versus-approve coverage can affect w_{uv} . Measurement error in edge weights propagates to $\mathbf{r}$ and to rank correlations ^[25].

5.7.3 External validity

Single protocol and chain. Evidence is limited to Arbitrum One and GMX V2. Generalization to other perpetual markets, lending protocols, or Layer-1 chains requires cross-protocol validation ^{[32],[35],[18]}.

Pool scaling versus alignment scaling. Runtime scaling on the expanded wallet pool ($N = 31,612$) supports efficiency claims at larger n ; alignment results remain anchored to the matched cohort with GMX outcomes. Efficiency gains should not be assumed to preserve the same τ profile at arbitrary population sizes without further alignment studies.

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

5.7.4 Adversarial validity

Sybil and wash trading. Adversarial allowance or transfer patterns may manipulate rankings ^[11]. Sybil-adjusted proxies partially address but do not eliminate this risk. The cost model in Section 5.5 is analytical; it is not an empirical red-team evaluation.

Strategic approval behavior. Once EndorseRank is deployed, rational actors may grant allowances strategically to farm rank. Observational τ estimated before widespread deployment may overstate robustness to strategic behavior.

5.8 Limitations

Results reflect one six-month window on Arbitrum One for a matched wallet cohort of $n = 5,521$ addresses. The endorsement subgraph comprises 14,727 edges; the transfer subgraph comprises 137,087 edges. Trading-success alignment for social methods is weak to modest (0.096 and 0.179), and transfer proxies remain the stronger validation frame (RQ4). EndorseRank’s primary demonstrated strengths are allowance-construct alignment ($\tau = 0.355$), rank distinctness from AWP (inter-method $\tau = 0.316$), and computational efficiency (2.105 s vs. 18.711 s; pool scaling $\approx 10.4\times$ and $\approx 9.4\times$).

Token-decimal normalization, USD price oracles, and permit-versus-approve coverage may affect edge weights. Extended baselines (alternative PageRank variants and lending-oriented graphs) are archived for future comparison (Appendix 8.1) but are excluded from the main dissertation narrative. Unsecured lending default labels are out of scope. Regulatory and ethical deployment constraints (Section 5.6) limit how scores should be used even when alignment is favorable.

These limitations bound the claims: EndorseRank is a credible, efficient, allowance-specific social reputation method on the studied cohort and window; it is not a universal credit score, a Sybil-proof

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

identity system, or a replacement for protocol-level collateralization ^{[45],[29]}.

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

6 Conclusion and Future Work

This chapter synthesizes the dissertation, states the original contribution to knowledge, revisits the research questions, and outlines future work on cross-protocol validation, adversarial manipulation, hybrid methods, and live oracle integration.

6.1 Summary of findings

This dissertation formulated EndorseRank, a PageRank method on a latest-allowance endorsement graph that produces an on-chain reputation score on Arbitrum One, and evaluated it against Adaptive Weighted PageRank (AWP), the transfer-graph baseline of Do et al.^[10], on an identical matched wallet cohort ($n = 5,521$). Validation used five proxy families and GMX V2 perpetual outcomes over the observation window 2025-12-01 to 2026-05-31, comprising 365,488 decoded `Position Decrease` events, 14,727 EndorseRank edges, and 137,087 AWP edges.

The primary findings for the EndorseRank–AWP comparison are as follows.

On reputation structure (RQ1), EndorseRank and AWP yield partially overlapping yet construct-distinct orderings. Inter-method Kendall τ between the two score vectors is 0.316, within the weak-to-moderate band: correlation is present, but allowance and transfer graphs emphasize different wallet relationships. H1 is supported.

On social-method alignment (RQ2), construct-specific winners emerge. EndorseRank leads on allowance proxies ($\tau = 0.355$ versus AWP -0.030); AWP leads on transfer proxies ($\tau = 0.534$ versus EndorseRank 0.333) and on most social-outcome families. On trading-success proxies specifically, AWP exceeds EndorseRank (0.179 versus 0.096). H2 is supported in its construct-specific form: neither social method universally dominates.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

On computational efficiency (RQ3), EndorseRank requires less PageRank wall-clock time than AWP on the matched cohort and on the expanded wallet pool. At $n = 5,521$, runtimes are 2.105 s versus 18.711 s ($\approx 8.9\times$). Pool scaling preserves the advantage at $n = 10,000$ ($\approx 10.4\times$) and at $N = 31,612$ ($\approx 9.4\times$). Robustness checks on damping and top-token subgraphs do not overturn the efficiency–alignment pattern. H3 is supported (Claims C1 and C4).

On validation frames (RQ4), transfer proxies yield stronger rank alignment than trading-success proxies for both social methods. Outcome-based proxies show weak-to-modest alignment overall, indicating that social graphs track transfer centrality more tightly than perpetual trading profitability on this sample.

As a supplementary finding (SRQ1), GF-PR achieves the highest trading-success mean τ (0.583) and inverse-risk mean τ (0.376) by construction, illustrating construct overlap when ranking on outcome flow. GF-PR bounds trading-outcome proxy interpretation for social methods and is not a deployable social reputation score (Claim SC1).

Collectively, Claims C1–C4 establish an efficiency–alignment trade-off: EndorseRank offers substantially lower PageRank cost and stronger allowance-construct validity, while AWP offers stronger transfer and social-outcome alignment. The appropriate choice depends on whether operators prioritize authorization semantics and refresh latency or transfer-hub detection and outcome co-movement.

6.2 Original contribution to knowledge

This dissertation contributes to on-chain reputation research and DeFi risk assessment in four respects.

First, it introduces **EndorseRank** as an allowance-based endorsement graph model. By treating latest ERC-20 approve and EIP-2612 permit authorizations as directed endorsement edges ^{[12],[13]},

^[12]Ethereum Improvement Proposals (2015). ERC-20 token standard (EIP-20). [Ethereum Improvement Proposals]

^[13]Ethereum Improvement Proposals (2020). ERC-20 permit extension (EIP-2612). [Ethereum Improvement Proposals]

it extends PageRank’s citation metaphor ^{[30],[5]} from transfer graphs ^[10] to explicit spending delegation. The model eliminates temporal decay through allowance override semantics and yields a sparse graph (14,727 edges on the matched cohort) suitable for frequent recomputation.

Second, it provides a **five-proxy domain alignment evaluation** that compares EndorseRank and AWP on leveraged DeFi activity under a shared cohort and window. Building on construct-validity reasoning ^[8] and rank correlation measures ^{[33],[22]}, the evaluation separates transfer, allowance, inverse-risk, Sybil-adjusted stability, and trading-success families. The result is a construct-specific map of strengths rather than a single global ranking—a methodological template for future on-chain score comparisons.

Third, it releases a **reproducible open-source evaluation pipeline** linking extraction, graph construction, scoring, proxy computation, and LaTeX-ready tables to a machine-readable summary (Appendix 8.1). Auditability of decentralized credit-related scores is a recognized concern ^[29]; the pipeline makes the empirical claims independently verifiable.

Fourth, as a supplementary contribution, it introduces a **GF-PR ceiling analysis** that clarifies construct overlap on trading-outcome proxies (SRQ1). By ranking directly on GMX PnL flow, GF-PR sets an outcome-native upper bound ($\tau = 0.583$ on trading success; $\tau = 0.376$ on inverse risk) against which social-method alignment can be interpreted without mistaking shared inputs for social validity.

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[5]Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. [Computer Networks and ISDN Systems]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

Relative to prior graph-reputation and DeFi credit work ^{[21],[17],[19],[10],[24],[26]}, the original knowledge contribution is the demonstration that allowance-graph PageRank is a distinct, efficient, and allowance-valid social reputation construct on Arbitrum One leveraged trading data—not a drop-in approximation of transfer-graph AWP, and not a substitute for collateralization ^{[45],[32],[18],[35]}.

Authorization graphs and payment graphs are therefore not interchangeable reputation substrates. They differ in sparsity (14,727 versus 137,087 edges on the matched cohort), runtime (2.105 s versus 18.711 s), and alignment profiles (allowance $\tau = 0.355$ for EndorseRank; transfer $\tau = 0.534$ for AWP), and they should be validated against proxies that match their intended semantics ^[8].

6.3 Future work

Several extensions follow from the limitations in Chapter 5 and the scope of this study.

Cross-protocol and cross-chain validation. Replication on other perpetual markets, lending protocols, and Layer-2 or Layer-1 networks would test whether the allowance–transfer trade-off generalizes beyond Arbitrum One and GMX V2 ^{[32],[35]}. Where lending default or liquidation labels are

^[21]Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. [Proceedings of the 12th International Conference on World Wide Web]

^[17]Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. [Proceedings of the Thirtieth International Conference on Very Large Data Bases]

^[19]Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. [IEEE Transactions on Network Science and Engineering]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[18]Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). DeFi and the future of finance. [Wiley]

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

^[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

^[35]Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized

denser, credit-ground-truth validation can complement the trading-success frame used here ^{[45],[26]}. Cross-chain GMX deployments and alternative L2s offer natural replication sites with comparable event schemas.

Adversarial evaluation. The Sybil cost model in Chapter 5 is analytical. Empirical red-teaming—including wash approvals, coordinated spender farms, and strategic revocation—would quantify realized attack cost-effectiveness under live fee markets and detection heuristics ^[11]. Sybil clustering on G_{approve} and comparison with transfer-graph manipulation costs are natural next experiments. Such work would also stress-test whether Sybil-adjusted stability proxies remain informative under active adversaries.

Hybrid social and outcome-aware methods. Operators may combine endorsement-graph features with transfer-graph features, or with DeFi-informed representation learning ^[24], while preserving the auditability emphasized by Packin and Lev-Aretz^[29]. Outcome-native diagnostics in the spirit of GF-PR can remain on a supplementary track for bounding trading-outcome proxy evaluation without being deployed as social scores. Hybrid designs should report construct-specific alignment rather than a single aggregate metric, following the five-proxy template of this dissertation.

Live oracles and governance tooling. Integration with risk oracles, liquidation prioritization, and governance dashboards would test whether EndorseRank’s runtime profile (2.105 s on the matched cohort; order-of-magnitude gains on the expanded pool) sustains production refresh cadences. Careful interface design is required so that ordinal graph scores are not over-interpreted as consumer credit ratings. Human-in-the-loop safeguards remain appropriate wherever scores gate access to

finance (DeFi). [Proceedings of the 4th ACM Conference on Advances in Financial Technologies]

^[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

^[11]Douceur, J. R. (2002). The Sybil attack. [Peer-to-Peer Systems]

^[24]Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. [2023 Intelligent Methods, Systems, and Applications (IMSA)]

^[29]Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. [American Business Law Journal]

financial services.

Archived baselines and richer authorization semantics. Extended baselines archived in Appendix 8.1 can be revisited when additional social-graph semantics are needed. Richer authorization events—including protocol-specific delegation logs where available—may refine endorsement edge definitions beyond ERC-20 allowances alone. Longer observation windows and additional market regimes would further stress-test stability of the reported τ profiles.

6.4 Closing reflection

On-chain reputation will not dissolve the need for collateral in adversarial, pseudonymous markets [32],[45]. It can, however, make behavioral structure legible. This dissertation shows that legibility depends on which edges we choose to read. Transfers tell a story about economic flow; allowances tell a story about delegated authority. PageRank can narrate either story, but not both at once with the same ranking. EndorseRank commits to the authorization narrative, documents its costs and benefits, and refuses to confuse outcome-native ceilings with social proof.

The practical upshot is modest and actionable. If a protocol or researcher needs a fast, auditable score of who is trusted to spend, EndorseRank is a credible instrument on Arbitrum One data of the kind studied here. If the goal is to find transfer hubs or to track activity-correlated outcomes, AWP remains stronger among social methods. If the goal is to predict trading profits from social graphs alone, neither method is sufficient—and GF-PR warns against evaluations that bake the answer into the graph.

This dissertation reports primary empirical claims C1–C4 (RQ1–RQ4) and supplementary claim SC1 (SRQ1) on a matched Arbitrum One wallet cohort ($n = 5,521$). EndorseRank stands as an efficient, allowance-specific alternative to transfer-based AWP for on-chain reputation under pseudonymity: distinct in ranking geometry (inter-method $\tau = 0.316$), aligned with its primary

[32]Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. [Federal Reserve Bank of St. Louis Review]

[45]Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. [2020 Crypto Valley Conference on Blockchain Technology]

construct (allowance $\tau = 0.355$), and approximately $8.9\times$ faster to compute on the studied sample, with sustained advantages on the expanded wallet pool ($\approx 10.4\times$ at $n = 10,000$; $\approx 9.4\times$ at $N = 31,612$). Those properties motivate continued research on authorization-aware trust signals in decentralized markets.

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the open evaluation pipeline used to reproduce Chapter 4 numerics, including repository layout, offline synthetic evaluation, real BigQuery extraction procedures, parameter summaries, proxy formulas, and a glossary.

8.1.1 Repository layout

The evaluation pipeline lives in `A-Skill-Programs/margin_rank/`. It ingests (1) latest ERC-20 allowances, (2) time-stamped transfers, and (3) decoded GMX `PositionDecrease` events. `EndorseRank`, `AWP`, and `GF-PR` scores are merged into `wallet_rankings.parquet` via `compute_reputation_ranks.py`. Five proxy families are computed in `proxy_metrics.py`; alignment and benchmarks run through `run_dissertation_eval.py`. Figures are produced by `generate_dissertation_figures.py`.

Key outputs: `data/processed/eval_summary.json` (numerics), LaTeX fragments under `2-Dissertation-Draft/results/tables/`, and PDFs under `2-Dissertation-Draft/Figures/generated/`. Extended baselines (7-method, 6-Aave) export to `2-Dissertation-Draft/archive/extended-baselines/`.

8.1.2 Offline synthetic evaluation (no Google Cloud)

```
cd A-Skill-Programs/margin_rank
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python scripts/run_dissertation_eval.py --fixtures --n-wallets 571 --export-latex
python scripts/generate_dissertation_figures.py
```

```
cd ../../2-Dissertation-Draft && ./build.ps1
```

8.1.3 Real BigQuery extraction

Requires Application Default Credentials and `GOOGLE_CLOUD_PROJECT`. See `docs/bigquery_data_plan.md` for cost guardrails.

```
export GOOGLE_CLOUD_PROJECT=your-project-id
```

```
# Phase 1 - trading-success label extraction (GMX V2) (6-month window)
```

```
python scripts/extract_margin_week.py --dry-run
```

```
python scripts/extract_margin_week.py --extract --yes
```

```
python scripts/decode_gmx_events.py
```

```
python scripts/compute_rankings.py
```

```
# Phase 2 - Reputation subgraph (monthly batches)
```

```
python scripts/extract_reputation_data.py --dry-run
```

```
python scripts/extract_reputation_data.py --extract --yes
```

```
python scripts/preprocess_arbitrum_allowances.py
```

```
python scripts/preprocess_arbitrum_transfers.py
```

```
python scripts/compute_reputation_ranks.py
```

```
# Phase 3 - Five-proxy evaluation + robustness (local)
```

```
python scripts/sample_active_wallets.py --from-parquet --extract
```

```
python scripts/run_dissertation_eval.py --real --robustness --export-latex
```

```
python scripts/generate_dissertation_figures.py
```

Extraction metadata (bytes processed, row counts) is written to `data/processed/extraction_manifest.json`.

8.1.4 Scaling tables

Table 8.1 retains in-cohort runtime scaling ($n \leq 5,521$). Robustness tables appear in Chapter 4; they are regenerated with `run_dissertation_eval.py--real--robustness--export-late x`.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort).

n	ER (s)	AWP (s)	Speedup	ER $ E $	AWP $ E $
1000	0.393	3.215	$8.2\times$	2584	23058
2000	0.729	6.794	$9.3\times$	5147	47031
4000	2.075	18.031	$8.7\times$	10557	97105
5521	2.105	18.711	$8.9\times$	14727	137087

8.1.5 Credit Delegation extract (deferred)

Aave V3 BorrowAllowanceDelegated logs are configured in `margin_config.yaml` but were not extracted in this study (`aave_delegation_events.parquet` absent). The archived six-Aave baseline (Delegation-PR) therefore reports degenerate alignment ($\tau = \text{—}$). Re-enable via a dedicated delegation SQL template and `extract_aave_lending.py--include-delegation` in a follow-on extract pass.

8.1.6 Supplementary alignment summary table

Table 8.2 duplicates family-level mean τ from Table 4.7 in columnar form (EndorseRank / AWP / GF-PR / highest-alignment label). It is retained for CSV-oriented reproducibility; the main text uses Table 4.7 only.

8.1.7 Parameter summary

8.1.8 Proxy operational definitions (reference)

The following formulas match Chapter 3 and `proxy_metrics.py`.

Table 8.2: Mean Kendall τ by proxy family: EndorseRank, AWP, and GF-PR ($n = 5521$).

Family	EndorseRank	AWP	GF-PR	Highest
Transfer	0.333	0.534	0.246	AWP
Allowance	0.355	-0.030	-0.041	EndorseRank
Liq.	0.093	0.113	0.132	GF-PR
Inv.risk	0.035	0.107	0.376	GF-PR
Sybil	0.233	0.286	0.093	AWP
GMX	0.096	0.179	0.583	GF-PR

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations	300	Power-iteration cap
Logistic k	0.01	AWP decay slope
Logistic t_0	180 days	AWP decay midpoint
Min. GMX closes	3	Cohort eligibility
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
GF-PR loss $\varepsilon_{\text{loss}}$	0.1	Ordinary loss edge boost
GF-PR liquidation ε_{liq}	1.0	Liquidation edge boost
Expanded pool seed	benchmark-tier2-v1	SHA256 subsample seed

Transfer. $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.

Allowance. $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_o w_a(o, w)$ with w as spender.

Inverse risk. $\text{loss_avoidance}(w) = -\sum_c \min(\text{PnL}_c, 0)$; $\text{non_loss_close_rate}(w) = 1 - |\{c : \text{PnL}_c < 0 \wedge \neg \text{liq}_c\}|/|\mathcal{C}(w)|$; $\text{worst_close_pnl_score}(w) = -\min_c \text{PnL}_c$.

Sybil-adjusted stability. $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}|/|T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.

Trading success. $\text{close_success_count}(w) = |\{c : \text{PnL}_c > 0 \wedge \neg \text{liq}_c\}|$; $\text{realized_gain_proxy}(w) = \sum_c \max(\text{PnL}_c, 0)$; $\text{close_success_rate}(w) = \text{close_success_count}(w)/|\mathcal{C}(w)|$.

8.1.9 Glossary

Allowance ERC-20 authorization for a spender to transfer tokens on behalf of an owner.

Alignment Rank agreement between reputation scores and a validation proxy (ρ, τ) .

Construct overlap Shared data source between a ranking method and a proxy, inflating alignment.

Endorsement graph Directed graph of latest allowances (EndorseRank input).

Matched cohort Primary sample of $n = 5,521$ wallets meeting eligibility rules.

Expanded wallet pool Larger address set ($N = 31,612$) for runtime scaling.

PageRank Spectral ranking by stationary distribution of a damped random walk.

Proxy Observable wallet metric used for domain-intuition validation.

Teleport Uniform jump in PageRank with probability $1 - d$.

8.2 SQL Templates, Configuration, and Extended Notes

This appendix collects representative SQL patterns, configuration keys, and extended methodological notes that support reproduction without cluttering the main chapters.

8.2.1 Approval log extraction pattern

ERC-20 Approval events are identified by topic 0 equal to the keccak256 hash of Approval(address, address, uint256). A simplified monthly query pattern is:

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS owner_topic,
    topics[SAFE_OFFSET(2)] AS spender_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
      AND block_timestamp <  TIMESTAMP('2026-01-01')
      AND topics[SAFE_OFFSET(0)] =
      ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
      AND (
          LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
          ↪ UNNEST(@wallet_list)
          OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
          ↪ UNNEST(@wallet_list)
      );
```

Wallet filtering via @wallet_list keeps scans proportional to the cohort rather than the full chain. Monthly batching respects per-query byte caps documented in the pipeline manifest.

8.2.2 Transfer log extraction pattern

Transfer events use topic 0 for Transfer(address,address,uint256):

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS from_topic,
    topics[SAFE_OFFSET(2)] AS to_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
        ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
            ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
            ↪ UNNEST(@wallet_list)
    );
```

Post-processing applies token decimals, builds time-decayed edge weights, and restricts nodes to the evaluation sample before PageRank.

8.2.3 GMX PositionDecrease extraction pattern

GMX V2 events are read from the EventEmitter contract. The pipeline filters on the emitter address `0xC8ee91A54287DB53897056e12D9819156D3822Fb` and decodes key-value fields to recover account, basePnlUsd, sizeDeltaUsd, and liquidation flags. Only PositionDecrease rows enter proxy computation. Wallets with fewer than three closes are excluded from the matched cohort.

8.2.4 Configuration keys

Primary configuration lives in `config/margin_config.yaml`. Salient keys include:

- observation window start and end timestamps;
- PageRank damping, tolerance, and maximum iterations;
- logistic decay k and t_0 ;
- GMX emitter address and topic filters;
- minimum closes for cohort eligibility ($m = 3$);
- benchmark repetition count and subsample seeds;
- output paths for parquet and LaTeX fragments.

Changing damping or decay parameters without re-exporting tables will desynchronize the dissertation numerics from `eval_summary.json`. The recommended workflow is always `run_dissertation_eval.py--real--robustness--export-latex` followed by `generate_dissertation_figures.py` and `build.ps1`.

8.2.5 Complexity notes

Let n be the number of nodes and $m = |E|$ the number of edges. Building latest-allowance edges is $O(R_a \log R_a)$ if R_a approval rows are sorted by timestamp, or $O(R_a)$ expected time with a hash map keyed by (token, owner, spender). Building transfer edges is $O(R_t)$ over transfer rows. PageRank is $O(mI)$ for I iterations. Empirically $I \approx 37$ for EndorseRank and $I \approx 100$ for AWP

on the matched cohort, and $m_{\text{approve}} \ll m_{\text{transfer}}$, so both factors favor EndorseRank.

Memory is dominated by edge lists and score vectors. Peak solve memory of 4.6 MB versus 38.2 MB (Table 4.2) is consistent with the edge-count ratio 14,727 : 137,087.

8.2.6 Interpretation checklist for readers

When reading Chapter 4 tables, the following checklist reduces common misreadings:

1. Identify whether a row is a *social* method (EndorseRank, AWP) or the supplementary GF-PR reference.
2. Match each family to its intended construct (allowance vs. transfer vs. outcome).
3. Prefer relative comparisons on the same cohort over absolute τ thresholds.
4. Treat GF-PR highs on trading-success and inverse-risk as construct-overlap diagnostics (SRQ1).
5. Read runtime tables as PageRank solve time on pre-built edges, not end-to-end ETL time.
6. Confirm robustness tables before generalizing a point estimate.

8.2.7 Archived extended baselines

Seven-method and six-Aave alignment matrices are stored under 2-Dissertation-Draft/archive/extended-baselines/ and corresponding CSV matrices under A-Skill-Programs/margin_rank/data/archive/extended-baselines/. They are excluded from the main narrative because several variants require additional event types (e.g., credit delegation) or introduce constructs outside the primary EndorseRank–AWP question. Researchers extending this work may re-enable exports via `export_method_matrix.py` presets documented in the archive README.

8.2.8 Limitations reprise

Appendix material does not expand the empirical claims beyond Chapter 4. In particular, SQL templates do not imply that other chains were queried, and archived baselines do not constitute

peer comparisons in this dissertation. The authoritative numerics remain those in `eval_summary.json` for the matched cohort ($n = 5,521$) and expanded wallet pool ($N = 31,612$).

8.3 Elementary Properties of PageRank and Rank Correlation

This appendix records elementary properties used implicitly in Chapters 2–4. The material is standard ^{[30],[25],[33],[22]} and is included for self-contained reading.

8.3.1 Existence and uniqueness of PageRank

Let P be a row-stochastic matrix on n nodes, possibly after dangling-node repair as in Equation (2.1). For $d \in (0, 1)$, the Google matrix

$$G_{\text{PR}} = dP^\top + \frac{1-d}{n}\mathbf{1}\mathbf{1}^\top$$

is positive and column-stochastic (up to the transpose convention in Equation (2.2)). By the Perron–Frobenius theorem for positive matrices, G_{PR} has a unique eigenvalue of modulus one with a strictly positive eigenvector π normalized so that $\sum_i \pi_i = 1$. Hence PageRank scores exist and are unique for any directed graph once damping and dangling-node rules are fixed ^[25].

This uniqueness underpins comparisons between EndorseRank and AWP: given fixed parameters, each method induces a single score vector on a given edge set. Differences between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ are therefore attributable to edge semantics and weights, not to solver multiplicity.

^[30]Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). The PageRank citation ranking: Bringing order to the web. [Stanford Digital Library Technologies Project]

^[25]Langville, A. N., & Meyer, C. D. (2006). Google’s PageRank and beyond: The science of search engine rankings. [Princeton University Press]

^[33]Spearman, C. (1904). The proof and measurement of association between two things. [The American Journal of Psychology]

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

8.3.2 Continuity in edge weights

PageRank is a continuous function of the entries of P for fixed $d \in (0, 1)$. Small perturbations of allowance or transfer weights therefore induce small perturbations of scores in the $\|\cdot\|_1$ norm, with Lipschitz constants depending on d and n ^[25]. Continuity justifies robustness checks that vary damping or restrict to top tokens: if alignment were knife-edge fragile, small hyperparameter changes would scramble family-level τ . Empirically, EndorseRank allowance τ remains ≈ 0.355 across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.4), consistent with a stable ranking geometry.

8.3.3 Sparsity and per-iteration cost

A sparse matrix–vector product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations for m nonzero edges. Power iteration therefore costs $O(mI)$ for I iterations. On the matched cohort, $m_{\text{ER}} = 14,727$ and $m_{\text{AWP}} = 137,087$, so even at identical I the transfer method would be roughly $9.3\times$ more expensive per iteration. Observed iteration counts ($I_{\text{ER}} \approx 37$, $I_{\text{AWP}} \approx 100$) widen the gap, matching the measured wall-time ratio $\approx 8.9\times$ (Table 4.2).

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation applied to ranks. It is invariant to strictly increasing transformations of either variable and lies in $[-1, 1]$. Perfect monotone agreement yields $\rho = 1$; perfect reverse agreement yields $\rho = -1$. Ties require mid-rank adjustments; the evaluation pipeline uses standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association is more appropriate than Pearson correlation on raw values ^[10].

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

8.3.5 Properties of Kendall's τ

Kendall's τ equals the normalized difference between concordant and discordant pairwise orderings. It admits a probabilistic reading: for two random distinct wallets, τ is the probability their relative order agrees under both rankings, minus the probability it disagrees^[22]. This reading is useful when communicating results to practitioners: $\tau = 0.355$ means a clear but imperfect tendency for allowance-proxy orderings to agree with EndorseRank orderings.

For large n , τ and ρ are often similar in sign and magnitude but not identical. This dissertation reports both, following Do et al.^[10], and uses family-mean τ as the primary summary.

8.3.6 Construct validity note

Cronbach and Meehl^[8] distinguish criterion-related validity from construct validity. In the absence of a gold-standard default label, this dissertation pursues a construct-oriented strategy: each method is tested against proxies that operationalize its intended meaning (allowance receipt for EndorseRank; inbound transfer activity for AWP) and against secondary families that probe neighboring constructs (risk, stability, trading success). High within-construct τ supports interpretation; low cross-construct τ is not automatically failure—it may indicate discriminant validity. GF-PR's extreme trading-success τ is a cautionary case of *contaminated* criteria, where the criterion shares inputs with the measure (SRQ1).

8.3.7 Numerical summary sheet

For quick reference, the primary matched-cohort numerics are:

^[22]Kendall, M. G. (1938). A new measure of rank correlation. [Biometrika]

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

^[8]Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. [Psychological Bulletin]

Quantity	Value
Matched wallets n	5,521
GMX closes (decoded)	365,488
EndorseRank edges	14,727
AWP edges	137,087
EndorseRank runtime	2.105 s
AWP runtime	18.711 s
Runtime ratio (AWP/ER)	$\approx 8.9\times$
Inter-method τ	0.316
ER allowance mean τ	0.355
AWP transfer mean τ	0.534
ER trading-success mean τ	0.096
AWP trading-success mean τ	0.179
GF-PR trading-success mean τ	0.583
GF-PR inverse-risk mean τ	0.376
Expanded pool N	31,612
Speedup at $n = 10,000$	$\approx 10.4\times$
Speedup at $N = 31,612$	$\approx 9.4\times$
Observation window	2025-12-01 – 2026-05-31

These values are authoritative only insofar as they match `eval_summary.json` and the exported LaTeX tables; if a regenerated export differs, the JSON artifact prevails.

8.4 Archived Extended Baselines

During exploratory analysis, additional PageRank variants were scored on the same matched cohort and proxy families. They are archived here for completeness and future work. They are *not* part of the primary EndorseRank–AWP research questions and should not be read as peer claims in the

main narrative.

8.4.1 Seven-method alignment matrix

Table 8.4 reports family-level mean Kendall τ for seven methods, including LP-PR, CW-AWP, LF-PR, and RiskProp-style variants implemented in the evaluation codebase. EndorseRank remains the allowance-family leader; AWP (and closely related transfer variants) lead on transfer; GF-PR leads on inverse-risk and GMX success by construction.

Table 8.4: Mean Kendall τ by proxy family across 7 PageRank methods ($n = 5521$). Bold = column maximum. Runtime = mean PageRank wall time (s, 3 repeats). Edge counts: liq-pr=45, borrow-pr=484, repay-pr=787, borrow-pr-pool=262, repay-pr-pool=232, delegation-pr=0.

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX	Runtime (s)
AWP	0.534	-0.030	0.113	0.107	0.286	0.179	22.757
EndorseRank	0.333	0.355	0.093	0.035	0.233	0.096	2.544
GF-PR	0.246	-0.041	0.132	0.376	0.093	0.583	1.879
LP-PR	0.529	-0.027	0.136	0.098	0.280	0.196	27.630
CW-AWP	0.504	-0.013	0.128	0.114	0.253	0.204	20.364
LF-PR	0.197	-0.031	0.045	0.017	0.102	0.032	0.113
RiskProp	0.534	-0.030	0.113	0.107	0.286	0.179	24.301

Interpretation guidelines:

- Methods built on transfer-like edges cluster near AWP on transfer and Sybil families.
- Methods with very small edge counts (e.g., liquidation-oriented graphs with dozens of edges) can produce unstable or degenerate rankings; their runtimes may be low simply because m is tiny.
- RiskProp-style rows that mirror AWP numerically indicate shared transfer structure rather than independent discovery.
- GF-PR’s dominance on outcome families remains a construct-overlap warning, identical to Chapter 4.

8.4.2 Six-method Aave-oriented matrix

Table 8.5 reports an Aave-oriented preset. Several lending-event graphs are sparse on Arbitrum within the study window; delegation-based rows may be degenerate when BorrowAllowanceDelegated events were not extracted (Appendix 8.1).

Table 8.5: Mean Kendall τ by proxy family across 6 PageRank methods ($n = 5521$). Bold = column maximum. Runtime = mean PageRank wall time (s, 3 repeats). Aave W→W edges: LiqCall (liquidator→user), Borrow (initiator→onBehalfOf), Repay (repayer→user), Delegation (delegator→delegatee). Credit Delegation extract deferred when parquet absent— τ may be degenerate. Edge counts: liq-pr=45, borrow-pr=484, repay-pr=787, borrow-pr-pool=262, repay-pr-pool=232, delegation-pr=0.

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX	Runtime (s)
AWP	0.534	-0.030	0.113	0.107	0.286	0.179	22.757
EndorseRank	0.333	0.355	0.093	0.035	0.233	0.096	2.544
LiqCall-PR	0.053	-0.012	-0.008	0.026	0.025	0.030	0.011
Borrow-PR	0.113	-0.019	0.020	0.014	0.055	0.023	0.083
Repay-PR	0.143	-0.024	0.016	0.022	0.068	0.026	0.131
Delegation-PR	—	—	—	—	—	—	0.000

These matrices motivate future work on credit-delegation edges and cross-protocol lending labels, but they do not alter Claims C1–C4 or SC1.

8.4.3 Why the main narrative uses three methods

Focusing the dissertation on EndorseRank, AWP, and supplementary GF-PR serves three goals:

1. **Clarity of contribution.** The novel construct is allowance-based endorsement. Comparing it to the established transfer baseline ^[10] isolates the research question.
2. **Avoiding baseline sprawl.** Additional variants that share transfer edges with AWP add rows without adding independent constructs.
3. **Honest use of GF-PR.** Keeping GF-PR supplementary prevents readers from treating an outcome-native ceiling as a social reputation competitor.

^[10]Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. [International Journal of Software Innovation]

Researchers who need broader baselines can regenerate archive tables with `export_method_matrix.py` presets documented in `archive/extended-baselines/README.md`.

8.4.4 Reproducing archive tables

```
cd A-Skill-Programs/margin_rank
python scripts/run_dissertation_eval.py --real --export-latex
# or explicitly:
python scripts/export_method_matrix.py --preset seven \
    --out-dir ../../2-Dissertation-Draft/archive/extended-baselines/tables
python scripts/export_method_matrix.py --preset six-aave \
    --out-dir ../../2-Dissertation-Draft/archive/extended-baselines/tables
```

CSV copies are written under `data/archive/extended-baselines/` for spreadsheet inspection.

8.4.5 Notes on archived method identifiers

The following identifiers appear in archive tables and CSV exports. They document exploratory implementations in `pagerank_variants.py` and related scripts.

LP-PR A transfer-graph variant with alternative loss-penalized or liquidity-oriented weighting explored during development. Empirically it tracks AWP closely on transfer families (Table 8.4), indicating shared payment-flow structure.

CW-AWP A counterparty-weighted AWP variant that down-weights repeated transfers between the same pair. It slightly reduces transfer τ relative to AWP while remaining in the same construct family.

LF-PR A low-frequency or lightly filtered graph with very small edge counts and near-zero run-times; rankings are not comparable to full AWP and are retained only to document a failed sparse baseline.

RiskProp A risk-propagation style ranking on transfer-like edges inspired by Lin et al.^[26]. On this cohort it numerically mirrors AWP on several families, suggesting that the implemented risk signal did not diverge strongly from transfer centrality under the study’s parameterization.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs built from liquidation calls, borrows, and repays when those events are present. On Arbitrum in the study window, edge counts are small (tens to hundreds), so alignment is weak and runtimes are negligible.

Delegation-PR Intended to use credit-delegation allowances; with zero delegation edges extracted, scores are degenerate and τ is not interpretable.

These notes exist so that future researchers can decide which variants deserve full re-implementation rather than treating archive rows as peer-reviewed baselines.

8.4.6 Decision log: what stayed in the main text

The decision to keep only EndorseRank, AWP, and supplementary GF-PR in Chapters 4–5 was made after inspecting the seven-method and six-Aave matrices. Criteria were:

1. **Construct independence.** A baseline must implement a distinct edge semantics, not a minor reweighting of transfers.
2. **Data adequacy.** Edge counts must be large enough for stable PageRank on the matched cohort.
3. **Narrative focus.** The dissertation’s contribution is allowance-based endorsement, not a survey of all possible PageRank variants.
4. **Evaluation honesty.** Outcome-native methods remain supplementary diagnostics.

Under these criteria, EndorseRank (allowance), AWP (transfer), and GF-PR (outcome-native ceiling) form a minimal sufficient set. Archive tables remain available for readers who want the broader

^[26]Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. [IEEE Transactions on Dependable and Secure Computing]

landscape without diluting the primary claims.

8.5 Claim-to-Evidence Map

This appendix maps each hypothesis, research question, and empirical claim to the specific evidence in Chapter 4. It is intended as a navigation aid for examiners and replicators.

8.5.1 H1 / RQ1 — Rank divergence

Statement. EndorseRank and AWP yield distinct wallet orderings on the same cohort.

Evidence. Inter-method Kendall $\tau = 0.316$ between score vectors (Section 4.6); rank scatter in Figure 4.8; score distributions in Figure 4.7.

Interpretation band. Weak-to-moderate agreement: related but not interchangeable rankings.

8.5.2 H2 / RQ2 — Construct-specific alignment

Statement. Social methods differ in which proxy families they align with.

Evidence. Table 4.7; per-family Tables 4.8–4.12; heatmap Figure 4.6.

Winners. EndorseRank on allowance ($\tau = 0.355$ vs. -0.030); AWP on transfer (0.534 vs. 0.333) and trading success among social methods (0.179 vs. 0.096).

8.5.3 H3 / RQ3 — Computational efficiency

Statement. EndorseRank is faster than AWP at equal n .

Evidence. Table 4.2 (2.105 s vs. 18.711 s, $\approx 8.9\times$); Table 4.3 ($\approx 10.4\times$ at $n = 10,000$; $\approx 9.4\times$ at $N = 31,612$); Figure 4.4; edge counts $14,727$ vs. $137,087$.

8.5.4 RQ4 — Validation frame

Statement. For social methods, transfer proxies align more strongly than trading-success proxies on this chain.

Evidence. Table 4.7: transfer means 0.333 (ER) and 0.534 (AWP) exceed trading-success means 0.096 (ER) and 0.179 (AWP).

8.5.5 SRQ1 / SC1 — GF-PR ceiling

Statement. GF-PR bounds trading-outcome alignment via construct overlap.

Evidence. Table 4.7 row GF-PR: inverse-risk 0.376, trading success 0.583; Section 4.7.

8.5.6 Claims C1–C4

- **C1:** runtime evidence as in H3/RQ3.
- **C2:** credible alignment on selected proxies—allowance for ER, transfer for AWP—in Tables 4.9 and 4.8.
- **C3:** construct-specific winners as in H2/RQ2.
- **C4:** efficiency–alignment trade-off combining C1 with allowance $\tau \approx 0.355$ and transfer $\tau \approx 0.33$ for EndorseRank; robustness in Tables 4.4–4.6 and Figure 4.5.

8.5.7 Explicit non-claims

The dissertation does *not* claim:

- that EndorseRank predicts trading profits better than AWP;
- that AWP is obsolete;
- that GF-PR is a deployable social reputation score;
- that results generalize beyond Arbitrum One, GMX V2, and the study window without replication;
- that on-chain scores replace collateral or legal identity systems.

8.5.8 Artifact checklist for replication

1. `data/processed/wallet_rankings.parquet` — scores for ER, AWP, GF-PR.

2. `data/processed/gmx_decoded_events.parquet` — closes and PnL.
3. `data/processed/eval_summary.json` — authoritative numerics.
4. `results/tables/*.tex` — exported tables.
5. `Figures/generated/*.pdf` — exported figures.
6. `archive/extended-baselines/` — optional broader baselines.

Regenerate with `run_dissertation_eval.py--real--robustness--export-latex` and `generate_dissertation_figures.py`, then build the dissertation with `build.ps1`.